Interactive Systems Properties with a Focus on Learning and Training

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Échauffement

- Que savez-vous de l'utilisabilité ?
- Que savez-vous de UX ?
- Que savez-vous de la différence entre les 2 ?
- Quand doit-on travailler sur UX versus usability ?
- Quand ne doit-on pas travailler sur UX ?
- Quand ne doit-on pas travailler sur usability ?

- Et la formation des opérateurs/utilisateurs dans tout ça?

Outline of the course/seminar

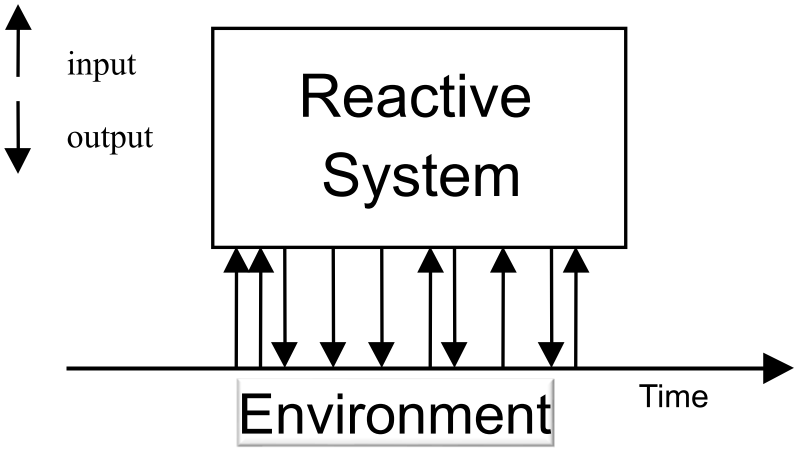
- Context and definitions
  - Usability
  - User Experience
- Designing for Usability
- Designing for User Experience
- Measuring Usability and User Experience
- Going further
  - Acceptance
  - Learnability
- Specificities of command and control systems
- Take away message

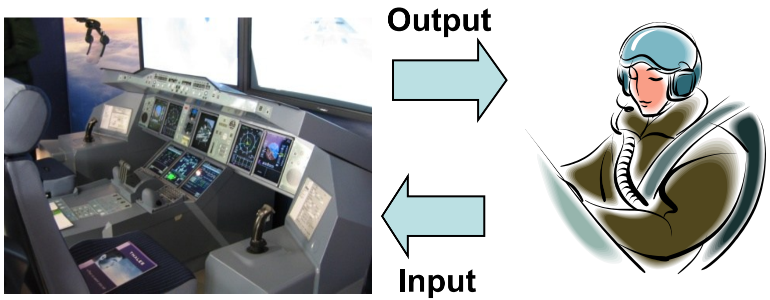
Interaction rules - please

- I'll try to speak slowly and to repeat the information in different places
  - Presentation of what will be presented next
  - Presentation of content
  - Summary of what was presented
- Questions are welcome at any time
- Prepare questions for the end of the course too (we will save some time for it)

Specificities of Interactive Systems?

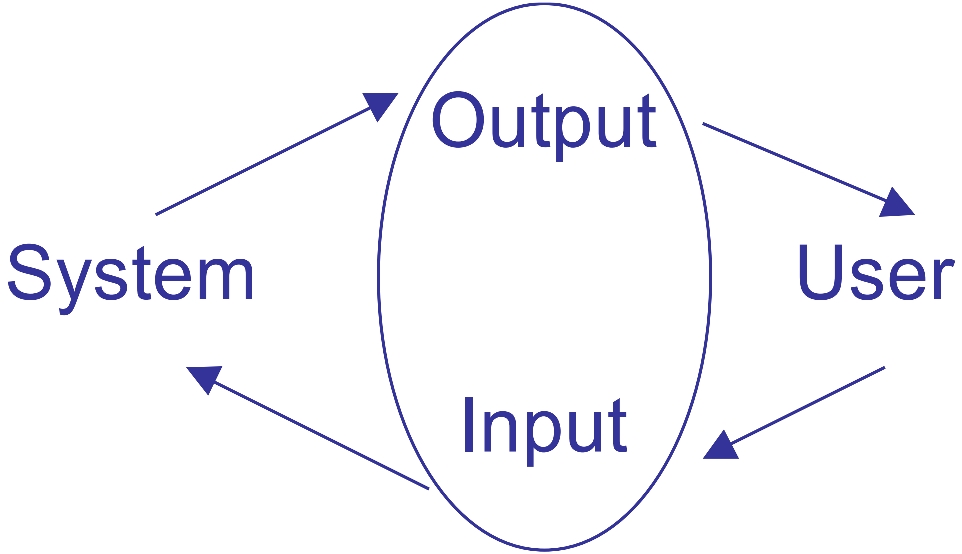
- One principle: Reactive Systems and Users
- One challenge: Variability (context of uses, of users, …)
- One philosophy: Tool approach (user **IN THE LOOP**)





Foundations of Interactive Systems

- Not a transformational system
- Not a reactive system
- An interactive system
  - Unpredictability
  - Perception
  - Understanding



Interactive Systems

- Everywhere
- Command and control systems
- Web applications
- Entertainment (games)
- Home appliances (dishwasher, coffee machine, …)

Each time you use a computer (or something with a computer inside e.g. dishwasher), you use an interactive system

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Definition of Usability

- Usability is concerned with making systems **easy to learn** and **use**
- A usable system is one which enables users to perform their job **effectively** and **efficiently**

- Usability is the „**effectiveness**, **efficiency** and **satisfaction** with which a specified set of users can achieve a specified set of tasks in a particular environment“

Standard ISO 9241- part 11c

Usability and usability functions

- Usability is a **non functional requirement** for the system
- However, there are **usability functions**
  - Undo/redo
  - Macro-commands (styles in MS Word)
  - Cancelation of commands
  - …
- Other properties
  - **Safety function**: safety belt
  - **Security function**: authentication mechanism
  - **Dependability function**: redundant systems

Usability?

Participant ID: _____ Site: ____________________________ Date: ___/___/___

System Usability Scale

Instructions: For each of the following statements, mark <u>one</u> box that best describes your reactions to the website *today*.

		Strongly Disagree				Strongly Agree
1.	I think that I would like to use this website frequently.	☐	☐	☐	☐	☐
2.	I found this website unnecessarily complex.	☐	☐	☐	☐	☐
3.	I thought this website was easy to use.	☐	☐	☐	☐	☐
4.	I think that I would need assistance to be able to use this website.	☐	☐	☐	☐	☐
5.	I found the various functions in this website were well integrated.	☐	☐	☐	☐	☐
6.	I thought there was too much inconsistency in this website.	☐	☐	☐	☐	☐
7.	I would imagine that most people would learn to use this website very quickly.	☐	☐	☐	☐	☐
8.	I found this website very cumbersome/awkward to use.	☐	☐	☐	☐	☐
9.	I felt very confident using this website.	☐	☐	☐	☐	☐
10.	I needed to learn a lot of things before I could get going with this website.	☐	☐	☐	☐	☐

Please provide any comments about this website:

Principes d’un questionnaire ?

- À quoi sert un questionnaire ?
- Comment fabrique-t-on un questionnaire ?
- Comment valide-t-on un questionnaire ?
- Quels sont les risques associés avec un questionnaire ?
- Quelles sont les limitations d’un questionnaire?
  - Domaine d’application
  - Type d’information recueillies
  - Type de personnes consultées

Effectiveness

CS 25
Amendment 19
1094 pages

European Aviation Safety Agency

Certification Specifications
for
Large Aeroplanes

CS-25

Amendment 4
27 December 2007

CS 25.1302

GENERAL

CS 25.1301 Function and installation
(See AMC 25.1301)

Each item of installed equipment must –

(a) Be of a kind and design appropriate to its intended function;

(b) Be labelled as to its identification, function, or operating limitations, or any applicable combination of these factors. (See AMC 25.1301(b).)

(c) Be installed according to limitations specified for that equipment.

[Amdt. No.:25/2]

CS 25.1302 Installed systems and equipment for use by the flight crew
(See AMC 25.1302)

This paragraph applies to installed equipment intended for flight-crew members' use in the operation of the aeroplane from their normally seated positions on the flight deck. This installed equipment must be shown, individually and in combination with other such equipment, to be designed so that qualified flight-crew members trained in its use can safely perform their tasks associated with its intended function by meeting the following requirements:

(a) Flight deck controls must be installed to allow accomplishment of these tasks and information necessary to accomplish these tasks must be provided.

(b) Flight deck controls and information intended for flight crew use must:

(1) Be presented in a clear and unambiguous form, at resolution and precision appropriate to the task.

(2) Be accessible and usable by the flight crew in a manner consistent with the urgency, frequency, and duration of their tasks, and

(3) Enable flight crew awareness, if awareness is required for safe operation, of the effects on the aeroplane or systems resulting from flight crew actions.

(c) Operationally-relevant behaviour of the installed equipment must be:

(1) Predictable and unambiguous, and

(2) Designed to enable the flight crew to intervene in a manner appropriate to the task.

(d) To the extent practicable, installed equipment must enable the flight crew to manage errors resulting from the kinds of flight crew interactions with the equipment that can be reasonably expected in service, assuming the flight crew is acting in good faith. This sub-paragraph (d) does not apply to skill-related errors associated with manual control of the aeroplane.

[Amdt. No.:25/3]

CS 25.1303 Flight and navigation instruments

(a) The following flight and navigation instruments must be installed so that the instrument is visible from each pilot station:

(1) A free-air temperature indicator or an air-temperature indicator which provides indications that are convertible to free-air temperature.

(2) A clock displaying hours, minutes, and seconds with a sweep-second pointer or digital presentation.

(3) A direction indicator (non-stabilised magnetic compass).

(b) The following flight and navigation instruments must be installed at each pilot station:

(1) An airspeed indicator. If airspeed limitations vary with altitude, the indicator must have a maximum allowable airspeed indicator showing the variation of V_{MO} with altitude.

(2) An altimeter (sensitive).

(3) A rate-of-climb indicator (vertical speed).

(4) A gyroscopic rate of turn indicator combined with an integral slip-skid indicator (turn-and-bank indicator) except that only a slip-skid indicator is required on aeroplanes with a third attitude instrument system usable through flight attitudes of 360° of pitch and roll, which is powered from a source independent of the electrical generating system and continues reliable operation for a minimum of 30 minutes after total failure of the electrical generating system, and is installed in accordance with CS 25.1321 (a).

GENERAL

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Efficiency/performance

- Time and number of actions to perform a task
- Number of information to recall
- Number of errors
- **Time to select a target (Fitt’s law)**
- Number of items to look at and to select (Hick-Hyman Law)
- …

Fitts' Law for Design

$$MT = a + b \times ID, \quad \text{where} \quad ID = \log_2\left(\frac{D}{W} + 1\right)$$

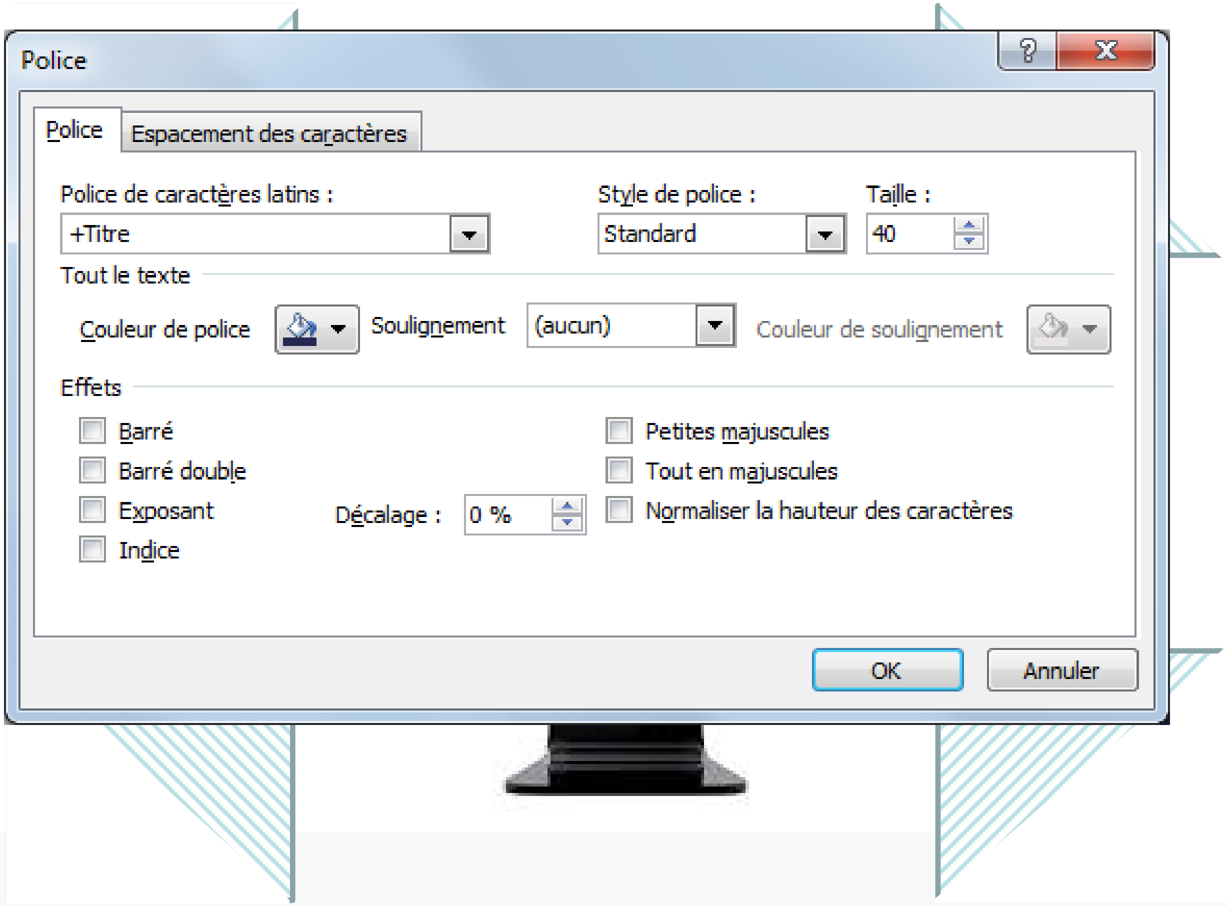
MT: Mouvement Time – ID: Index of Difficulty

Paul M. Fitts (1954). **The information capacity of the human motor system in controlling the amplitude of movement**. *Journal of Experimental Psychology*, volume 47, number 6, June 1954, pp. 381–391. (Reprinted in *Journal of Experimental Psychology: General*, 121(3):262–269, 1992)

Scott MacKenzie. **Fitts' law as a research and design tool in human-computer interaction**. Hum.-Comput. Interact., 7:91–139, 1992.


QUIZ
TIME

Fitts Law for Design


Police
Police
Espacement des caractères
Police de caractères latins :
+Titre
Style de police :
Standard
Taille :
40
Tout le texte
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Soulignement
(aucun)
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Effets
Barré
Barré double
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Petites majuscules
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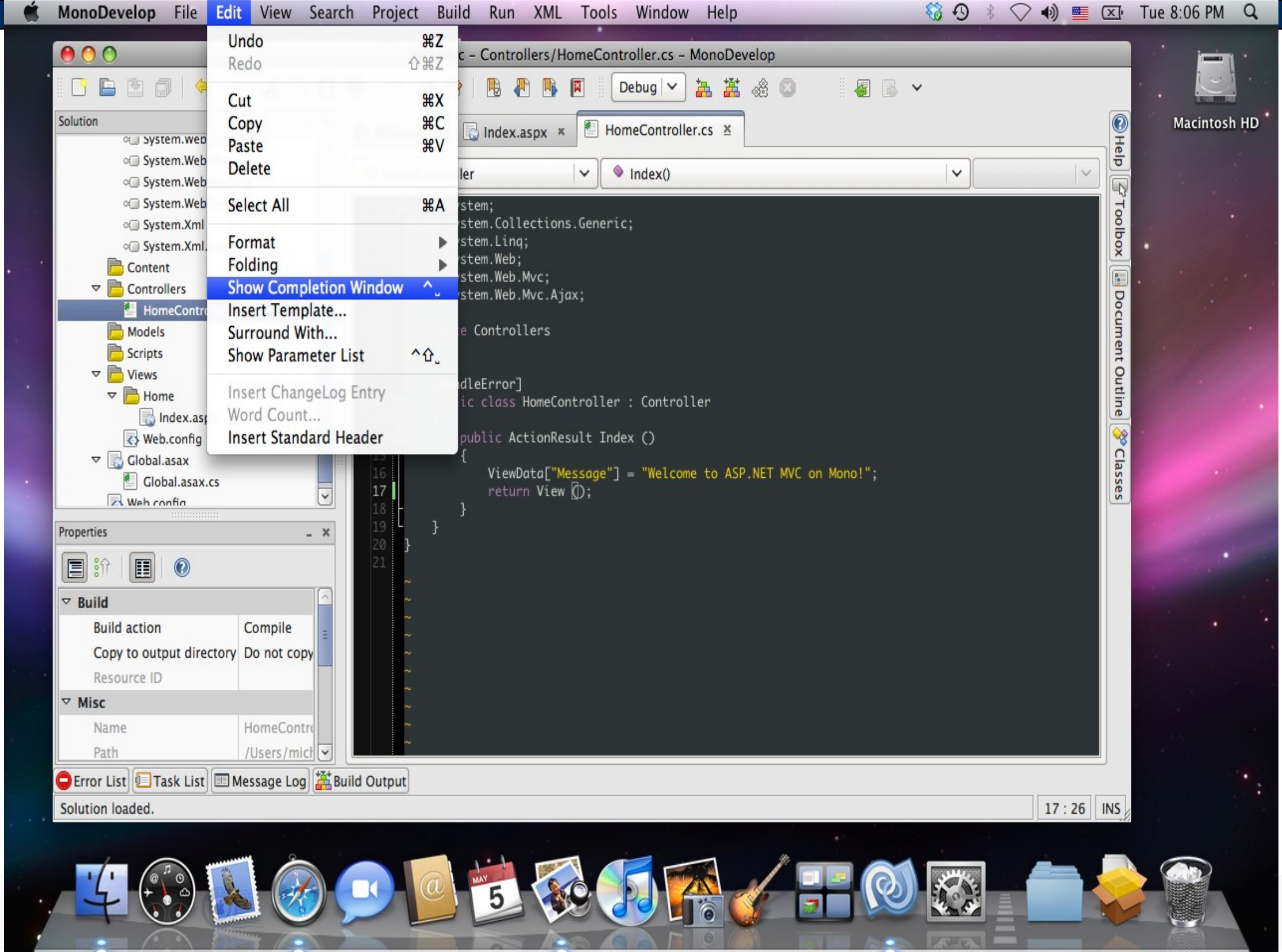
Normaliser la hauteur des caractères
OK
Annuler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Solution
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Models
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Views
Home
Web.config
Global.asax
Global.asax.cs
Properties
Build
Build action
Compile
Copy to output directory
Do not copy
Resource ID
Misc
Name
Path
public ActionResult Index ()
ViewData["Message"] = "Welcome to ASP.NET MVC on Mono!";
return View ();
Error List
Task List
Message Log
Build Output
Solution loaded.
17 : 26
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Macintosh HD
Help
Toolbox
Document Outline
Classes

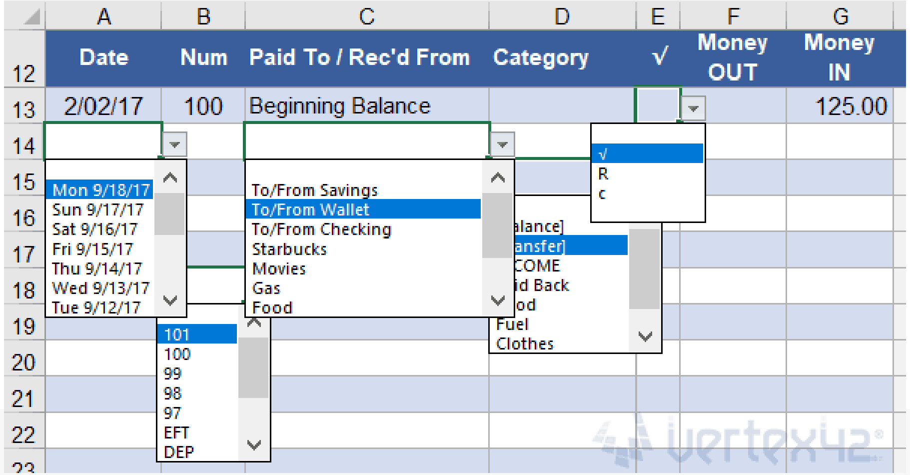
Efficiency

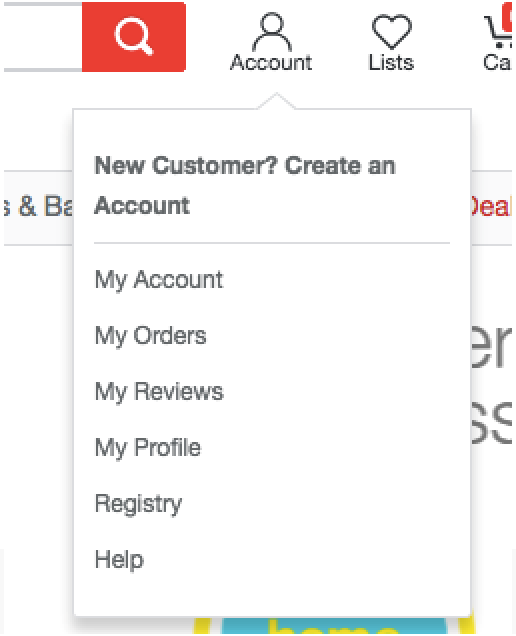
- Time and number of actions to perform a task
- Number of information to recall
- Number of errors
- Time to select a target (Fitt’s law)
- **Number of items to look at and to select (Hick-Hyman Law)**
- …

Hick-Hyman Law

- Given *n* equally probable choices, the average reaction time *T* required to choose among the choices is approximately

$$T = b \cdot \log_2(n + 1)$$





Is Usability important?

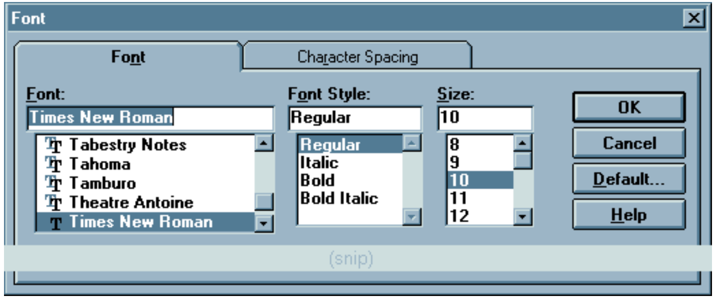
- Following the EC Directive 90/270/EEC countries legislations require employers to ensure that when designing, selecting, commissioning or modifying software:
  - That it is suitable for the task
  - That it is easy to use and, where appropriate, adaptable to the user‘s knowledge and experience
  - That it provides feedback on performance
  - That it displays information in a format and at a pace that is adapted to the user
  - That it conforms to the principles of software ergonomics

Can we tell if a system is usable?

Can we tell if a system is usable?

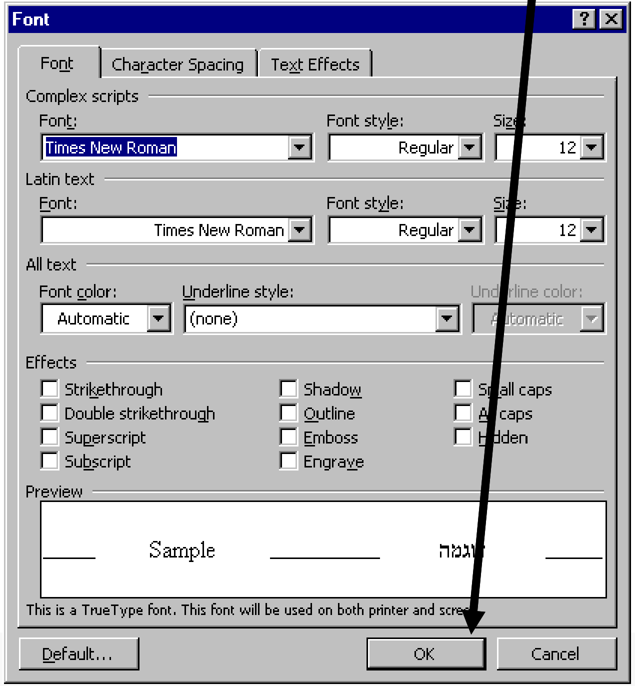
Can we tell if a system is usable?

USABILITY PROBLEMS WITH INTERACTIVE SYSTEMS

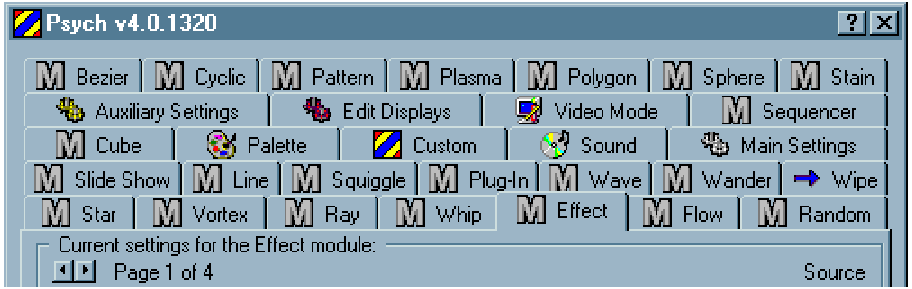


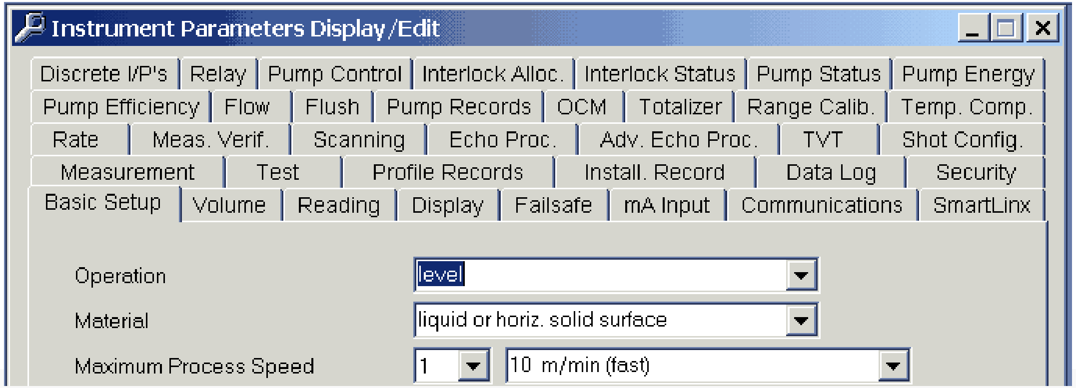
Does Ok closes the window?

Should buttons be inside a tab or outside the tab?



Complexity of user interfaces

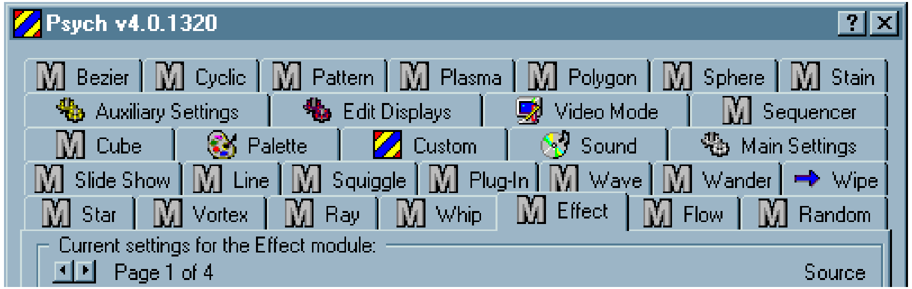


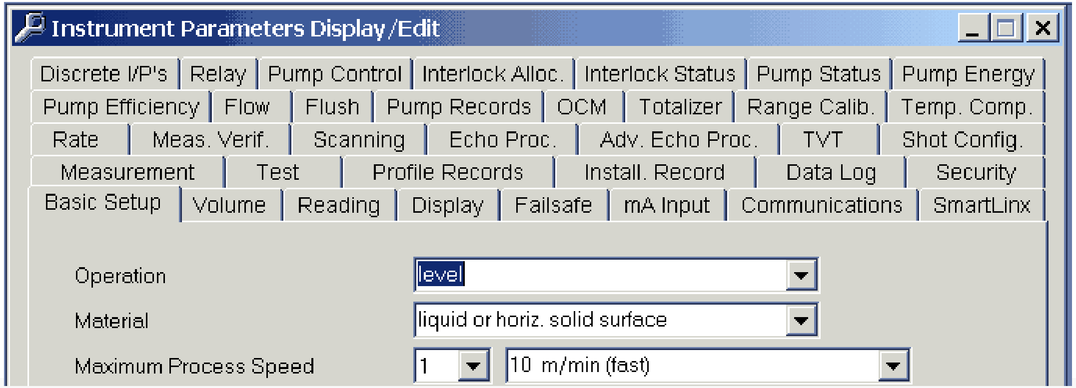


Complexity of user interfaces

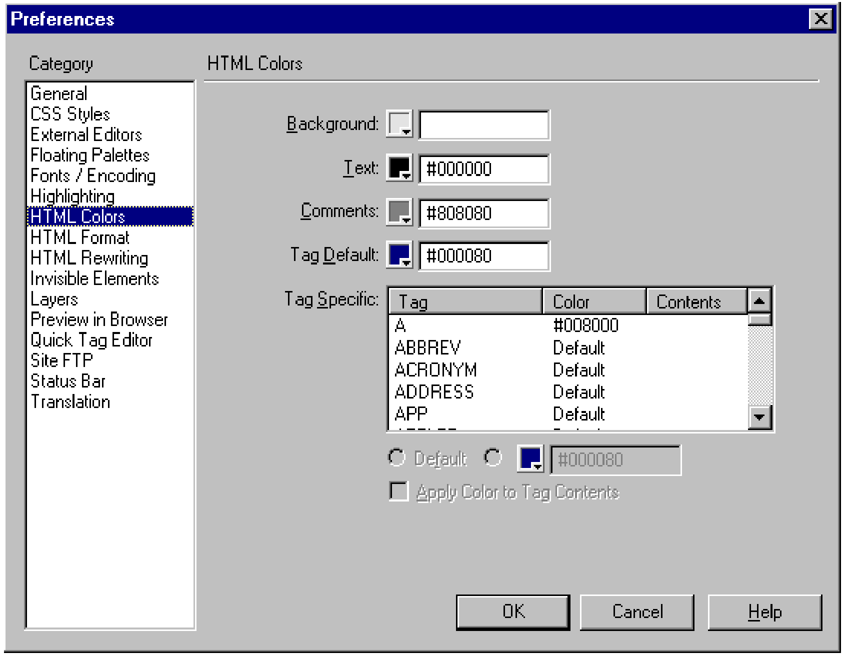
People will say: “if the system is complex, the user interface must be complex”

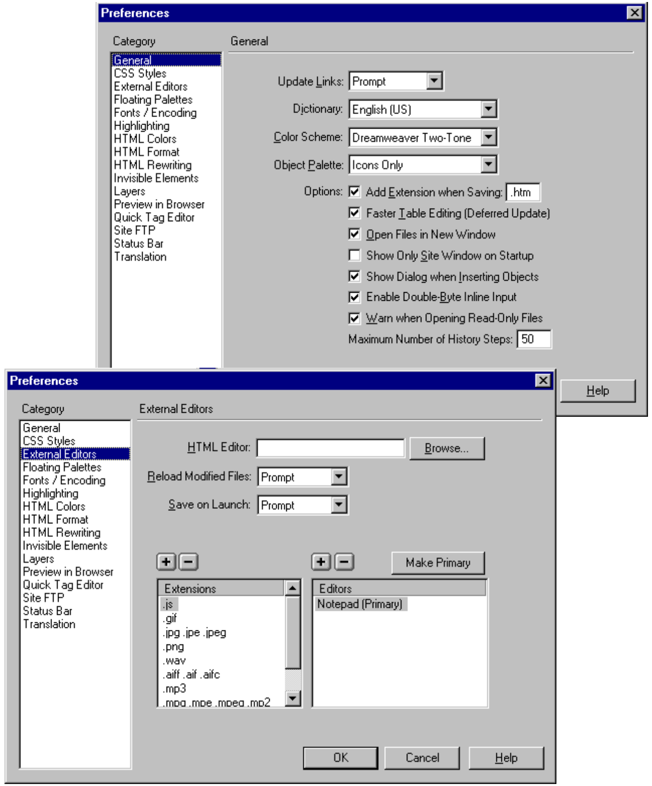
Complexity is “usually” a property of bad designs



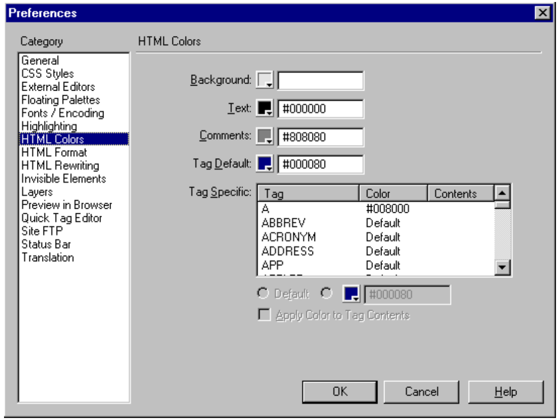


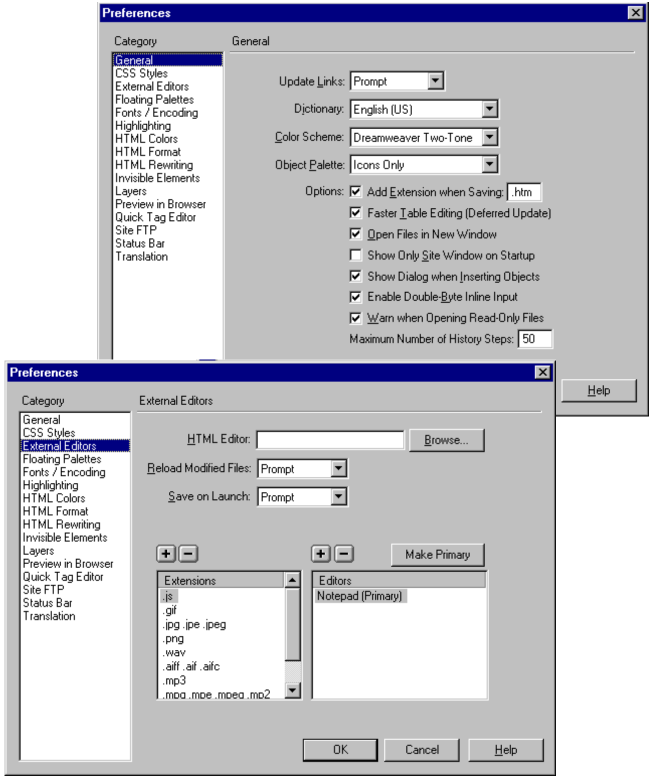
Is there a solution? …yes





Alternative designs …





The desire for creativity is not an excuse to ignorance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Support mebottles.com Buy Your T-Shirt Today
01/18/2011 The Lauber auction was a once in a lifetime treat.
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Yvette's Fashion Models!!
Мы любим Россия.
Yvette's is here for YOU !! ~*~*~*
Celebrate! Prom 2011 !! ~*~*~*
Apple Pie
Nous aimons la France.
please click here to
Invitations !!
Established 1980
우리는 한국을 사랑한다.
我們愛中國。
ette's habla poquito espanola !! ~*
私達は日本を愛する。
~barbara ann~
IT'S CUTE HOW JEALOUZ YOU HATERZ ARE OF WHAT WE HAVE!
HotFreeLayouts.com
message
friendz
chat
join
forward
favorites
haterz
rate
music
Pop Out Player
Clara Ft Justin Love Sex An...
Lady Gaga -Love...
STARSTRUKK
Cyclone
Lollipop
2010 ANTH...

Usability problem: do I have enough time for a coffee?

EXERCICES

Bubble cursor

- Tovi Grossman and Ravin Balakrishnan. 2005. The bubble cursor: enhancing target acquisition by dynamic resizing of the cursor's activation area. Proceedings of the SIGCHI Conference on Human Factors in Computing Systems. Association for Computing Machinery, New York, NY, USA, 281–290. DOI:https://doi.org/10.1145/1054972.1055012

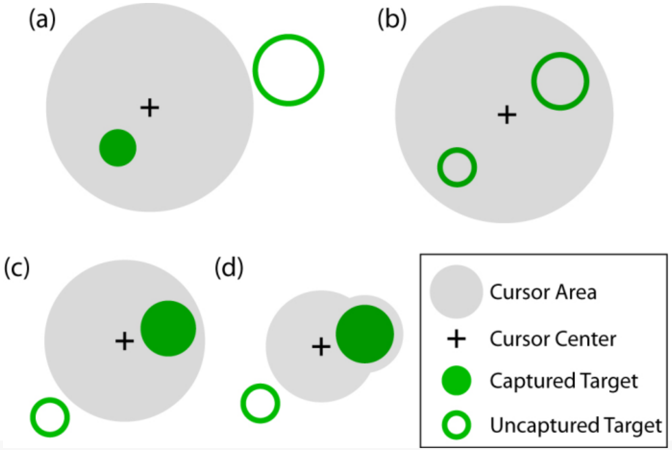

Figure 1. (a) Area cursors ease selection with larger hotspots than point cursors. (b) Isolating the intended target is difficult when the area cursor encompasses multiple possible targets. (c) The bubble cursor solves the problem in (b) by changing its size dynamically such that only the target closest to the cursor centre is selected. (d) The bubble cursor morphs to encompass a target when the basic circular cursor cannot completely do so without intersecting a neighboring target.

Bubble Cursor (voir article sur Moodle)

- Which property?
- How much?

- Have you used it?
- Why is it not deployed?

Semantic Pointing (voir article sur Moodle)

- Renaud Blanch, Yves Guiard, and Michel Beaudouin-Lafon. 2004. Semantic pointing: improving target acquisition with control-display ratio adaptation. Proceedings of the SIGCHI Conference on Human Factors in Computing Systems. Association for Computing Machinery, New York, NY, USA, 519–526. DOI:https://doi.org/10.1145/985692.985758

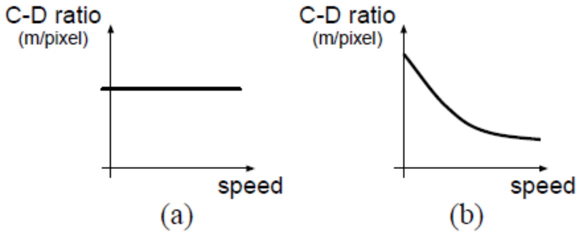

Figure 1: C-D ratio as a function of mouse speed
(a) constant C-D ratio (b) mouse acceleration

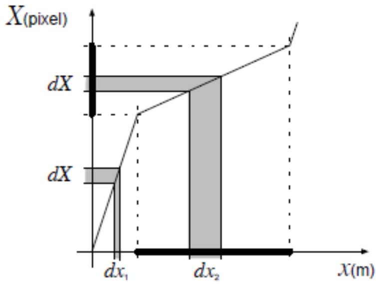

Figure 2: C-D ratio adapted to a target

Figure 14: Button redesign
(a) unchanged visual version (b) motor space version

Semantic Pointing

- Which property?
- How much?

- Have you used it?
- Why is it not deployed?

Pie menus (voir article sur Moodle)

- Don Hopkins. 1991. The design and implementation of pie menus. Dr. Dobb's J. 16, 12 (Dec. 1991), 16–26.
- J. Callahan, D. Hopkins, M. Weiser, and B. Shneiderman. 1988. An empirical comparison of pie vs. linear menus. In Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '88). Association for Computing Machinery, New York, NY, USA, 95–100. DOI:https://doi.org/10.1145/57167.57182

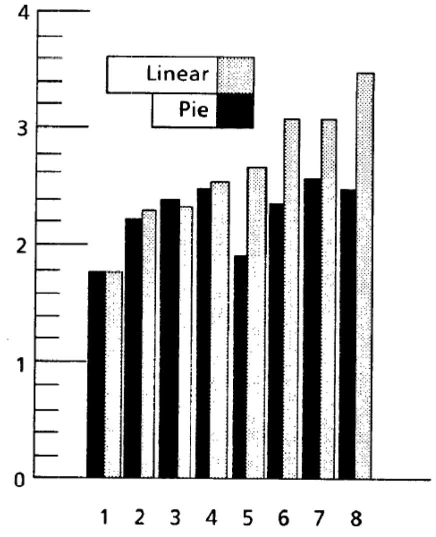

Figure 4: Target location (x) vs. seek time (y) in seconds

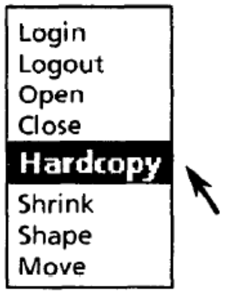

Figure 1: A typical linear menu

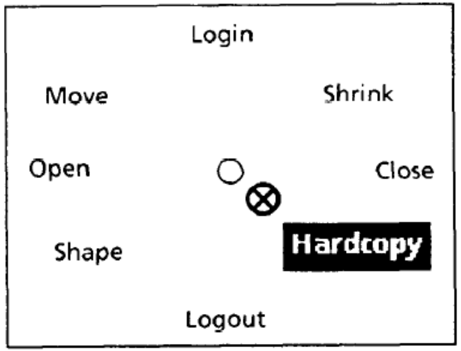

Figure 2: A crude pie menu

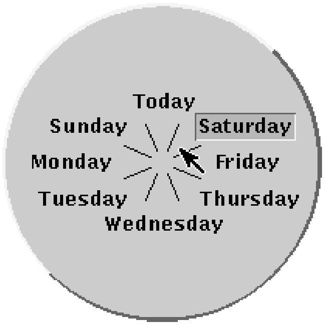

Figure 1: Eight Days a Week Pie Menu

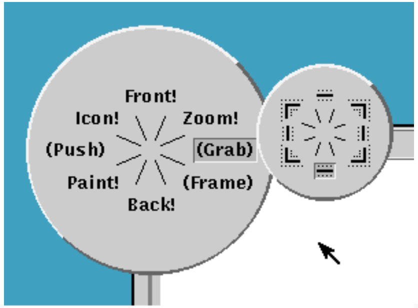

Figure 2: Window Management Pie Menu

Pie Menu

- Which property?
- How much?

- Have you used it?
- Why is it not deployed?

PLUS DIFFICILE

Distributeur de billets

- Retrait rapide 40€

Distributeur de billet

- 1) retrait rapide
- 2) bouton rouge annulation

- Which property?
- How much?

- Have you used it?
- Why is it deployed?

Outline of the course/seminar

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  - User Experience
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- Going further
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  - Learnability
- Specificities of command and control systems
- Take away message

Definition

- Currently there is no unique and agreed upon definition
- UX is not only “playability” or “fun”
- Examples of definitions:
  - The entire set of affects that is elicited by the interaction between a user and a product including the degree to which all our senses are gratified (aesthetic experience) the meanings we attach to the product (experience of meaning) and the feelings and emotions that are elicited (emotional experience). [Desmet & Hekkert, 2007]
  - A person’s perceptions and responses that result from the use and/or anticipated use of a product, system or service.

    [ISO 9241- part 210]

ISO 9241 – Part 210

arch | ISO 9241-210:2010(en) ×

Ergonomics of human-system interaction — Part 210: Human-centred design for interactive systems

FOLLOW

This standard has been withdrawn

Available in: EN FR

Redlines ▼

2.15

user experience

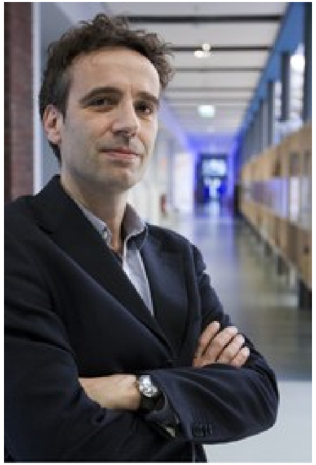
person's perceptions and responses resulting from the use and/or anticipated use of a product, system or service

Note 1 to entry: User experience includes all the users' emotions, beliefs, preferences, perceptions, physical and psychological responses, behaviours and accomplishments that occur before, during and after use.

Note 2 to entry: User experience is a consequence of brand image, presentation, functionality, system performance, interactive behaviour and assistive capabilities of the interactive system, the user's internal and physical state resulting from prior experiences, attitudes, skills and personality, and the context of use.

Note 3 to entry: Usability, when interpreted from the perspective of the users' personal goals, can include the kind of perceptual and emotional aspects typically associated with user experience. Usability criteria can be used to assess aspects of user experience.

Definition



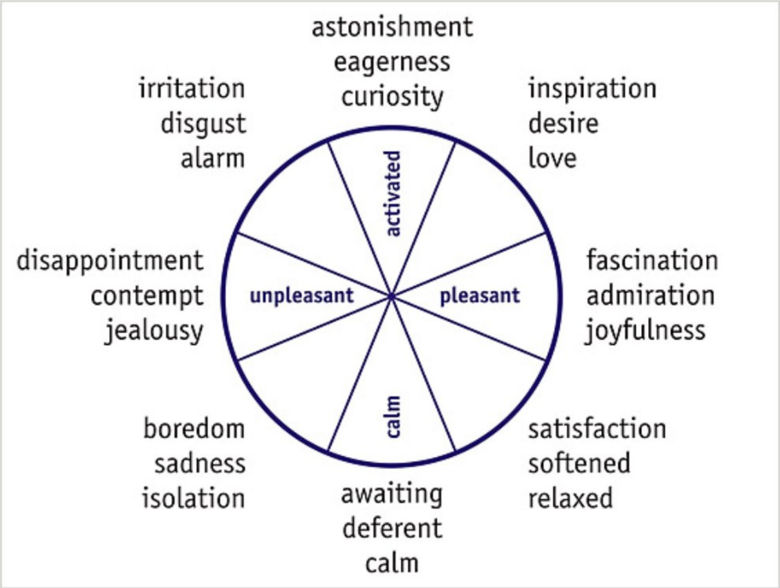
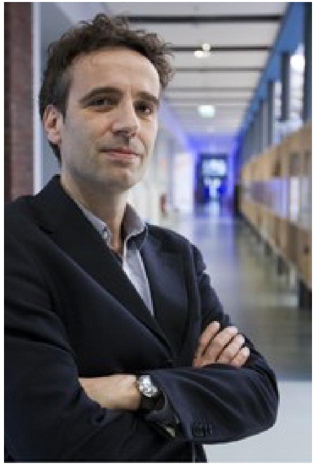


Figure 1. Circumplex model of core affect with product relevant emotions (Desmet, 2007; adapted from Russell, 1980).

Pieter Desmet & Paul Hekkert (both TU Delft)

http://www.ijdesign.org/index.php/IJDesign/article/view/66/15

Definition



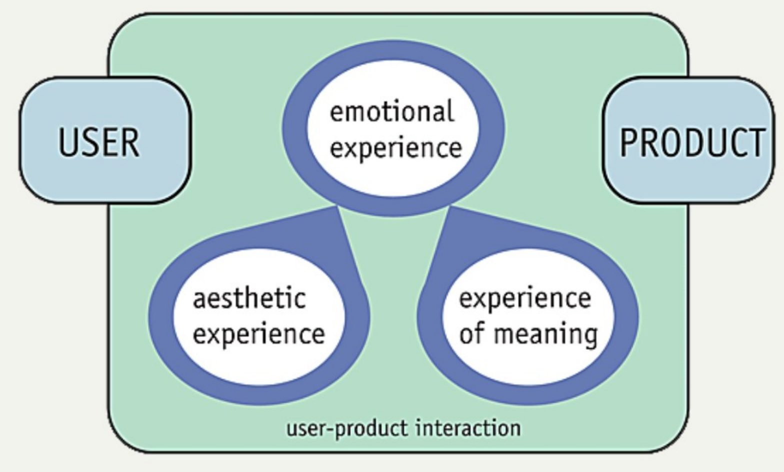


Figure 2. Framework of product experience.

Pieter Desmet & Paul Hekkert (both TU Delft)

http://www.ijdesign.org/index.php/IJDesign/article/view/66/15

DEFINITION OF USER EXPERIENCE

- A consequence of
  - a user’s internal state,
  - the characteristics of the designed system
  - and the context
- within which the interaction occurs.”

[Hassenzahl & Tractinsky 2006]

- The value derived from interaction(s) [or anticipated interaction(s)] with a product or service and the supporting cast in the context of use.”

[Sward & MacArthur 2007]

User Experience: Dimensions/Factors

UX Dimension	Explanation
Aesthetics	The aesthetic level involves a product's capacity to delight one or more of our sensory modalities. [Hekkert, 2007]
Emotion	The emotional level involves those experiences that are typically considered in emotion psychology and in everyday language about emotions, such as love and anger, which are elicited by the appraised relational meaning of products.
Meaning & Value	The meaning level involves our ability to assign personality or other expressive characteristics and to assess the personal or symbolic significance of products. [Desmet & Hekkert, 2007]. Value refers to the concept that products are sometimes chosen because they reflect or represent values that are important to the person.

User Experience: Dimensions/Factors

UX Dimension	Explanation
Stimulation	The stimulation dimension describes to what extent a product can support the human need for innovative and interesting functions, interactions and contents. (innovativeness, novelty)
Identification	Identification; a hedonic attribute group, which captures the product's ability to communicate important personal values to relevant others [Hassenzahl, 2004]
Social Connectedness	Socio pleasure deals with interaction with others. Thus, products that facilitate communication as well as those that serve as conversation pieces contribute to socio pleasure.

Aesthetics

Peugeot Onyx Nicest concept car of the year 2013

Emotion

Line to buy iPhones before opening hours

Emotion

Line to buy iPhones before opening hours:

Happy??

Positive or negative UX

Meaning and Value

Electric cars with green color, leaves, … symbols

Meaning and Value

Samsung batteries – green light

Stimulation

Stimulation

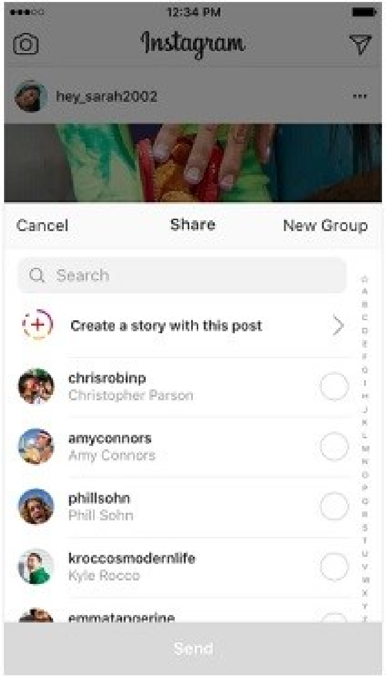
Stimulation

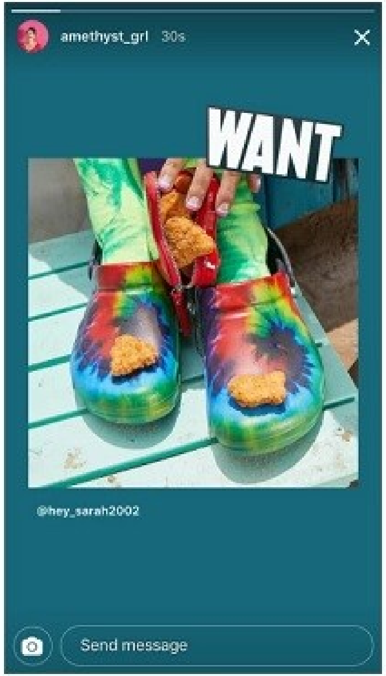
Identification

Identification

Social Connectedness

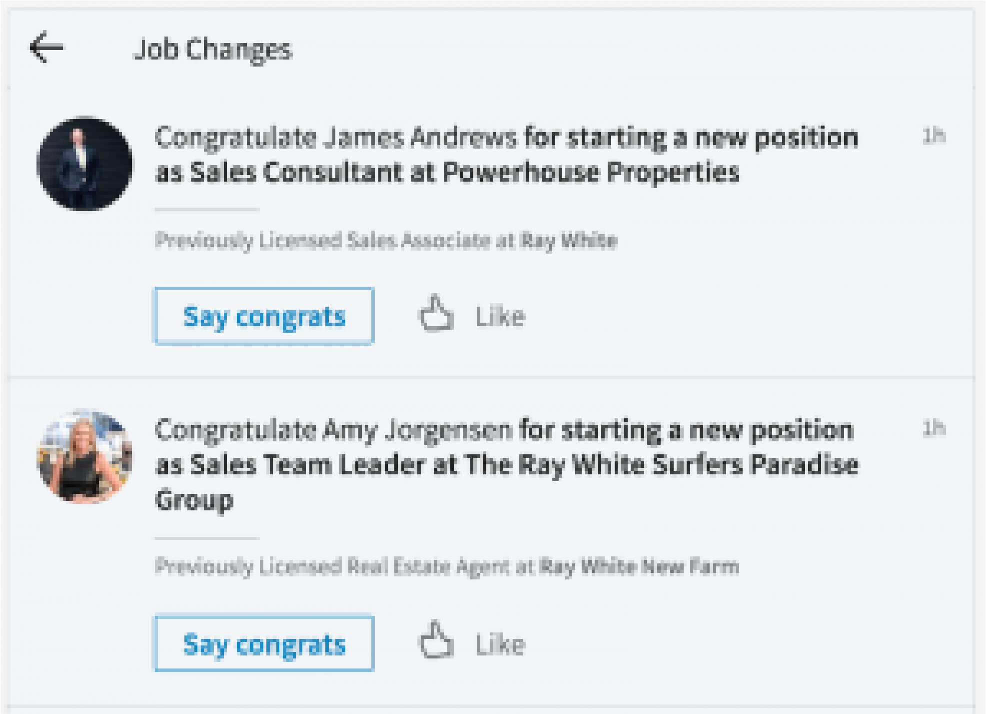
Create stories in Instagram





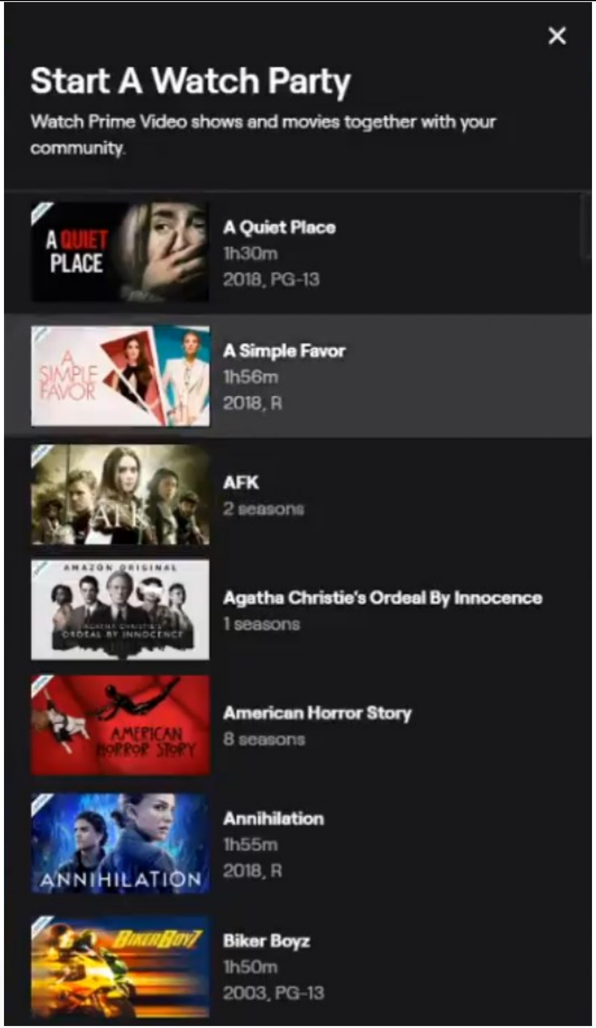
Social Connectedness

Build a professional network



Social Connectedness

Amazon Prime Watch Party (activity together)

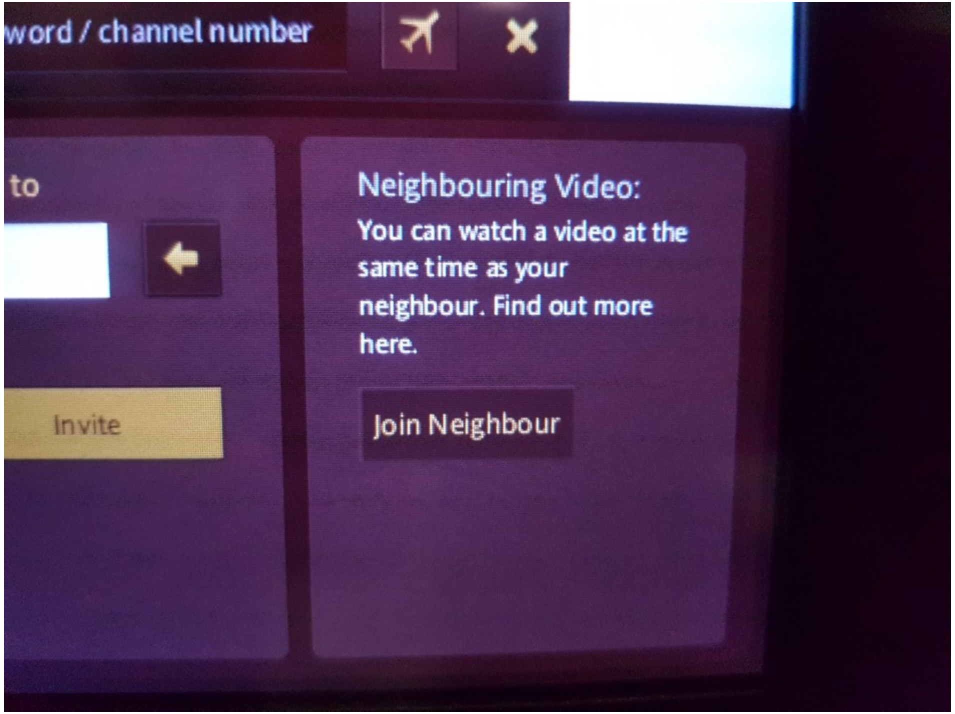


Social Connectedness

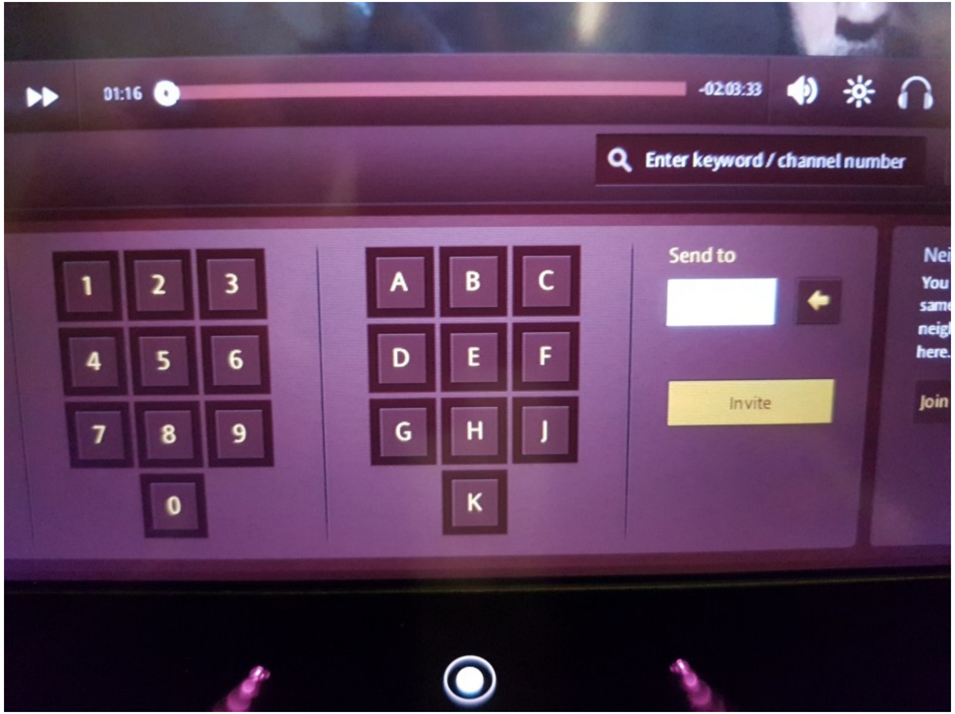
Ethiad Airways Flight Entertainment System

Ethiad Airways FES

Ethiad Airways FES



Ethiad Airways FES

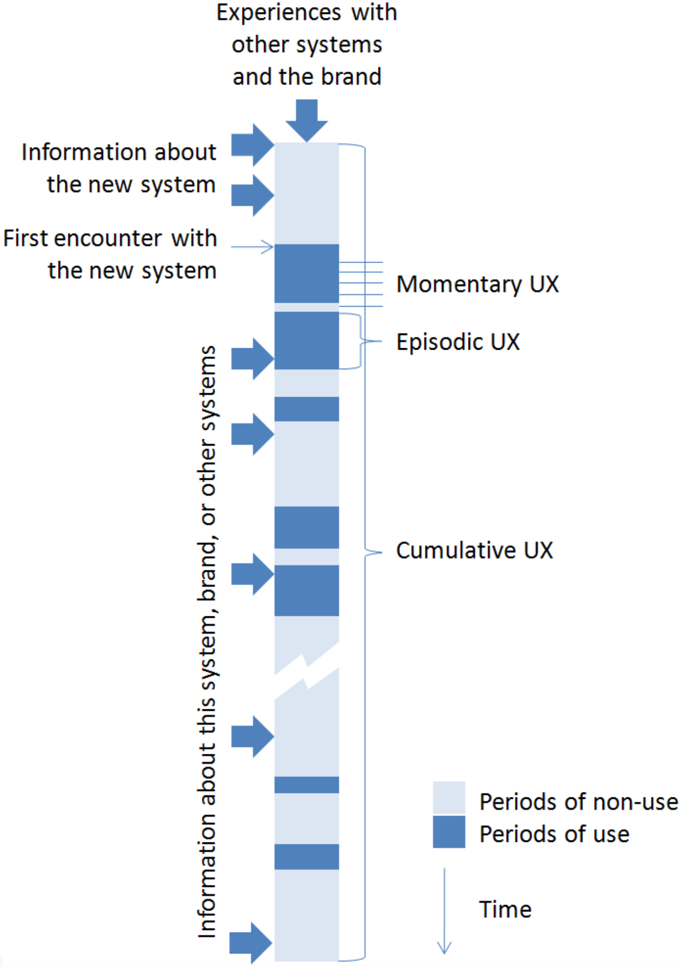


Defining Ux: User Experience

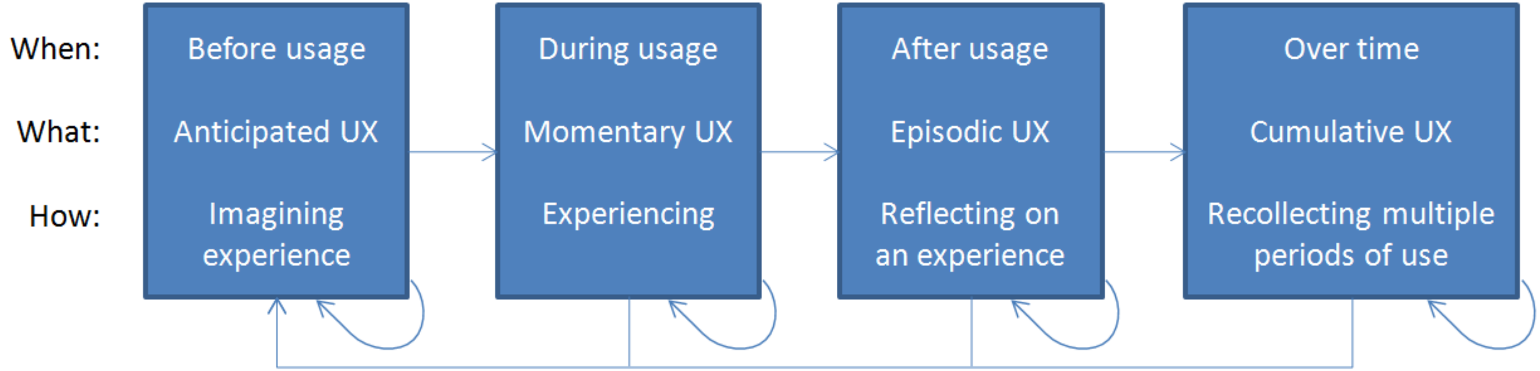
When looking at UX we have to identify the appropriate time span

- a specific change of feeling during interaction (**momentary UX**),
- appraisal of a specific usage episode (episodic UX),
- views on a system as a whole, after having used it for while (**cumulative UX**).
- **Anticipated UX** may relate to the period before first use, or any of the three other time spans of UX, since person may imagine a specific moment during interaction, a usage episode, or life after taking a system into use.

[allaboutUX. White paper]



Defining Ux: User Experience



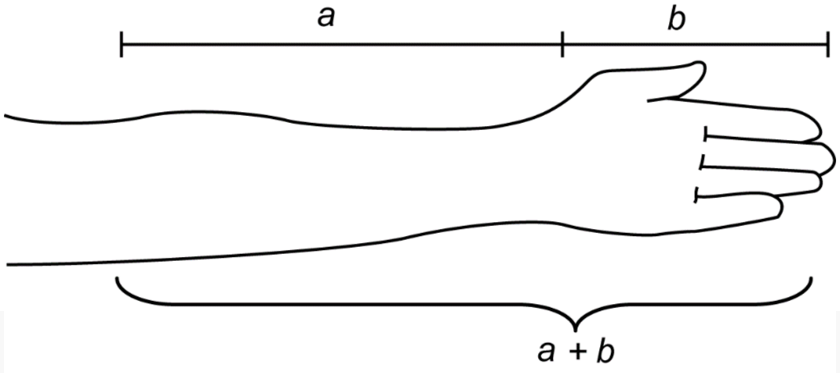
[allaboutUX. White paper]

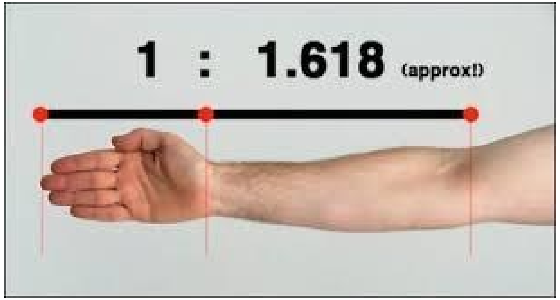

hello! studio · hellostudio.eu

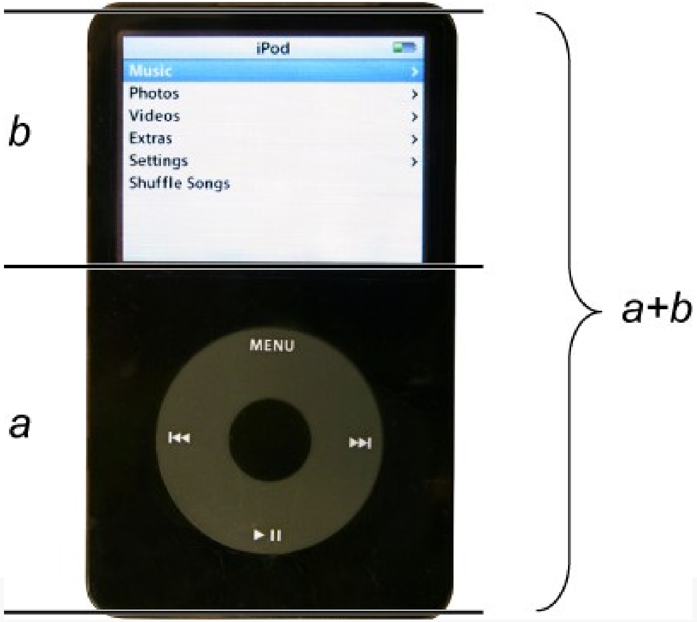
Aesthetics

The Golden ratio: The golden ratio expresses the relationship between two aspects of a form such as height to width and must equal 0.618

$$\frac{a+b}{a} = \frac{a}{b} \qquad R = \frac{1+\sqrt{5}}{2}$$

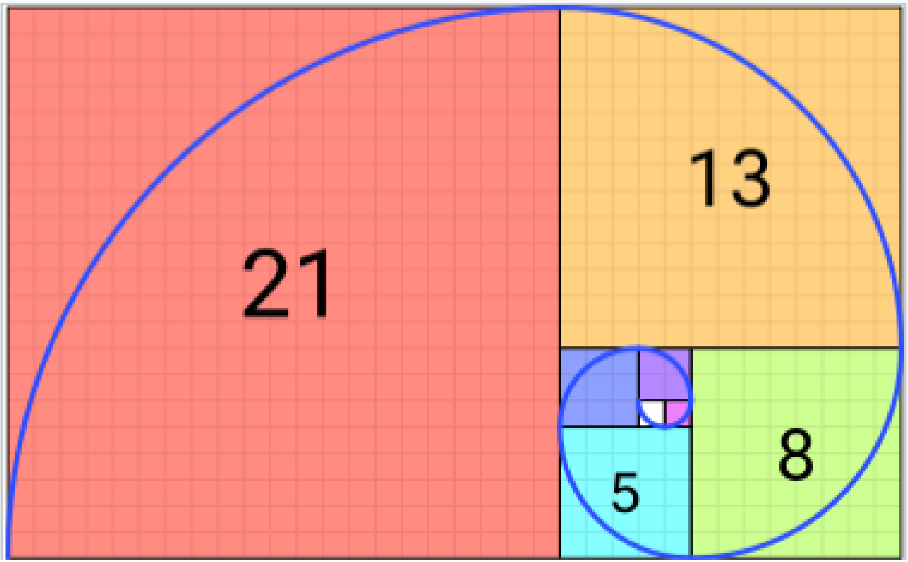




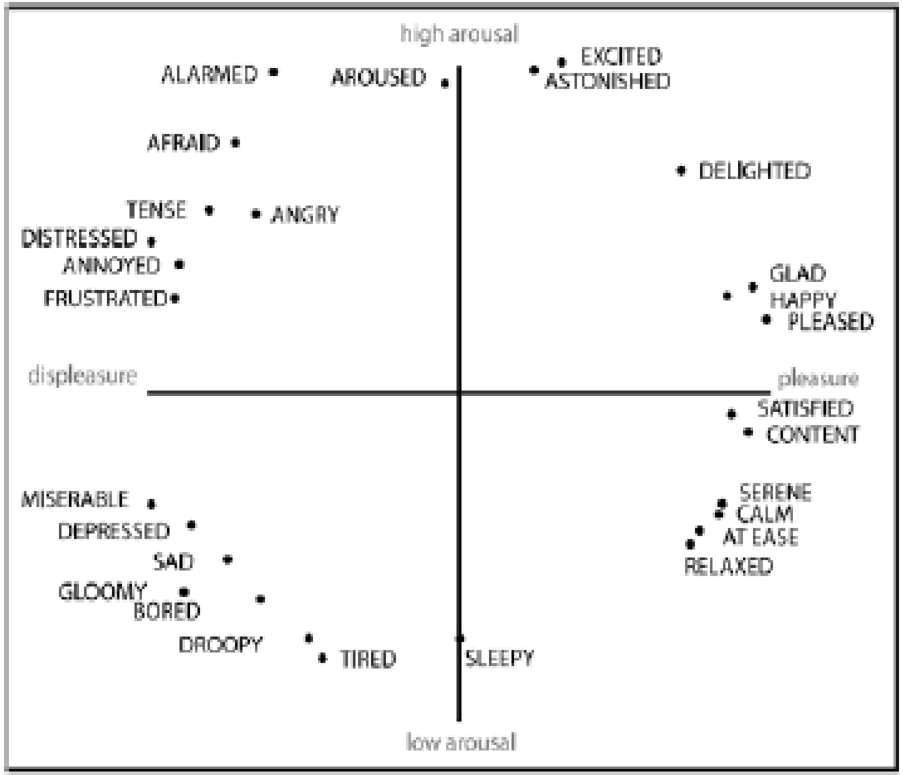


Aesthetics

The Golden ratio (pour les courbes) :
Spirale de Fibonacci

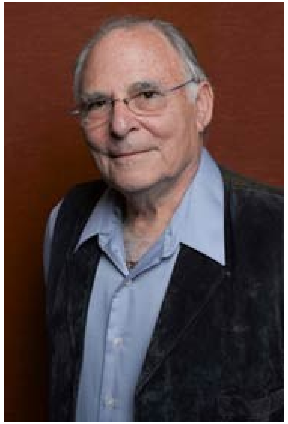


Emotions

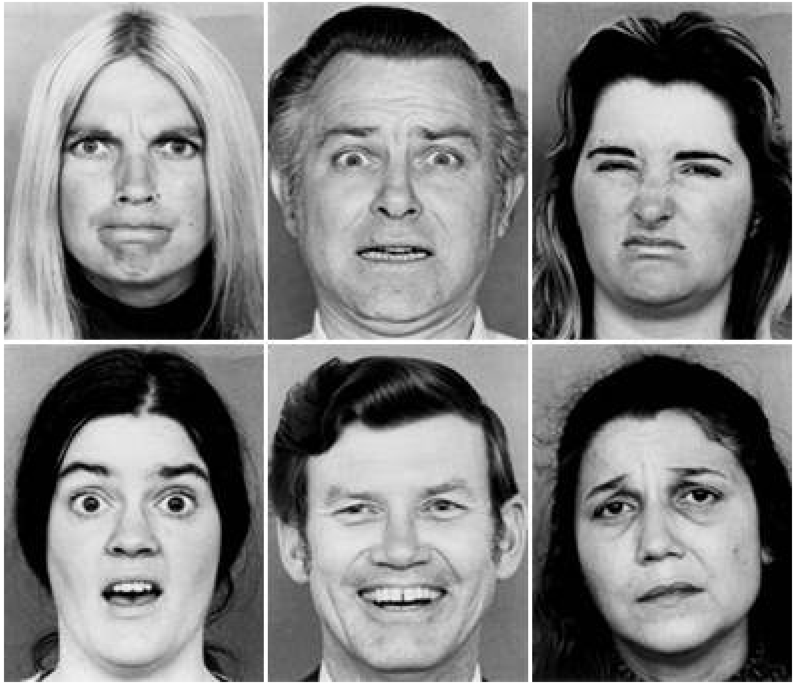


Two-dimensional affective space defined by valence and arousal: The circumplex model of affect (Russell, 1980).

Emotions



Paul Ekman

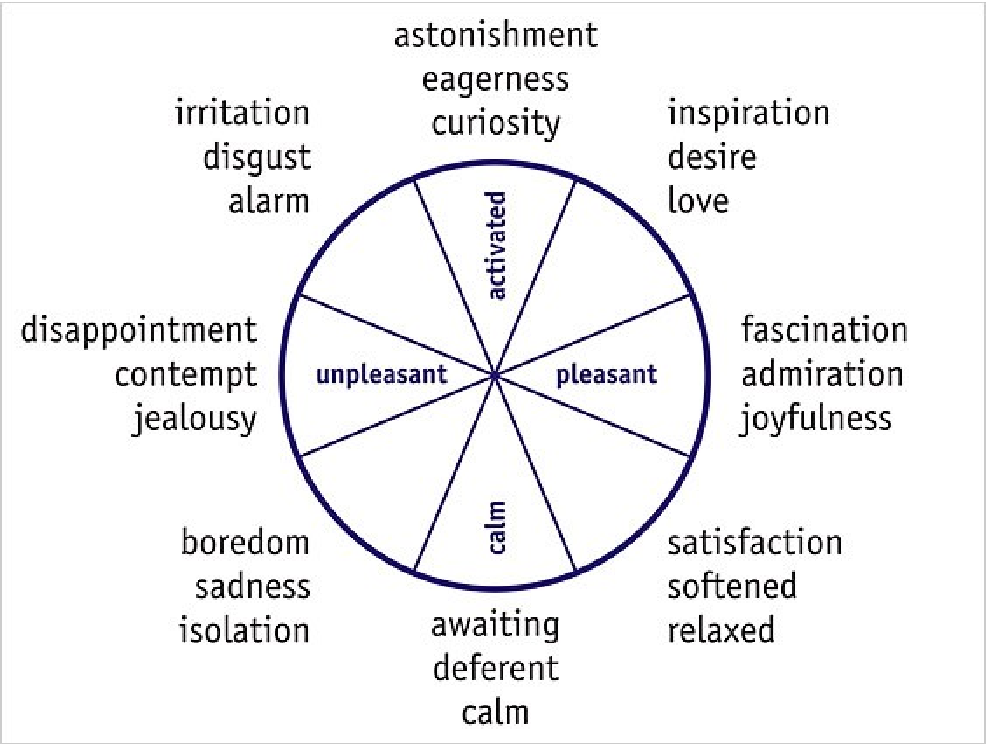


Ekman, Paul; Friesen, Wallace V.
Journal of Personality and Social Psychology, Vol 17(2), Feb 1971, 124-129. doi:10.1037/h0030377

Emotions

- Feelings – **subjective** representation private to individual
- Moods – **diffuse states** that **last longer** and are less intense than emotions
- Affect – term to describe the topics of emotion, feeling and moods (encompasses all, interchangeable with emotion)
- **Emotions:** Six Basic Emotions (Ekman & Friesen, 1971) anger, disgust, fear, happiness, sadness and surprise

Emotions: quantitative



Desmet, P. & Hekkert, P. (2007) Framework of Product Experience, in Int. Journal of Design, Vol, 1, Ed. 1, online

Other factors and dimensions of emotion

Affection
Anger
Angst
Anguish
Annoyance
Anxiety
Apathy
Arousal
Awe
Boldness
Boredom
Contempt
Contentment
Curiosity
Depression
Desire
Despair
Disappointment

Disgust
Distrust
Dread
Ecstasy
Embarrassment
Envy
Euphoria
Excitement
Fear
Fearlessness
Frustration
Gratitude
Grief
Guilt
Happiness
Hatred
Hope
Horror

Hostility
Hurt
Hysteria
Indifference
Indulgance
Interest
Jealousy
Joy
Loathing
Loneliness
Love
Lust
Outrage
Panic
Passion
Pity
Pleasure
Pride
Rage
Regret

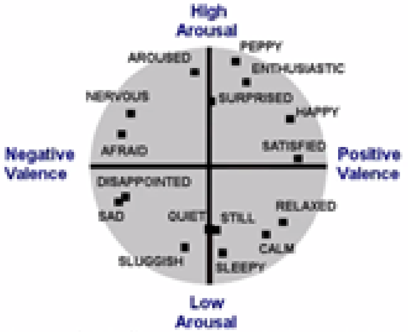
Remorse
Sadness
Satisfaction
Shame
Shock
Shyness
Sorrow
Suffering
Surprise
Terror
Trust
Wonder
Worry
Zest

Emotion Representation

- Qualitative model
  - Six basic emotions:
    Anger, happiness, surprise, disgust, sadness, fear
  - <u>Understandable</u>

- Quantitative model
  - Valence-Arousal model
  - Valence: pleased -> unpleased
  - Arousal: excited -> bored
  - <u>Flexible and accurate</u>

https://managementmania.com/en/six-basic-emotions



3

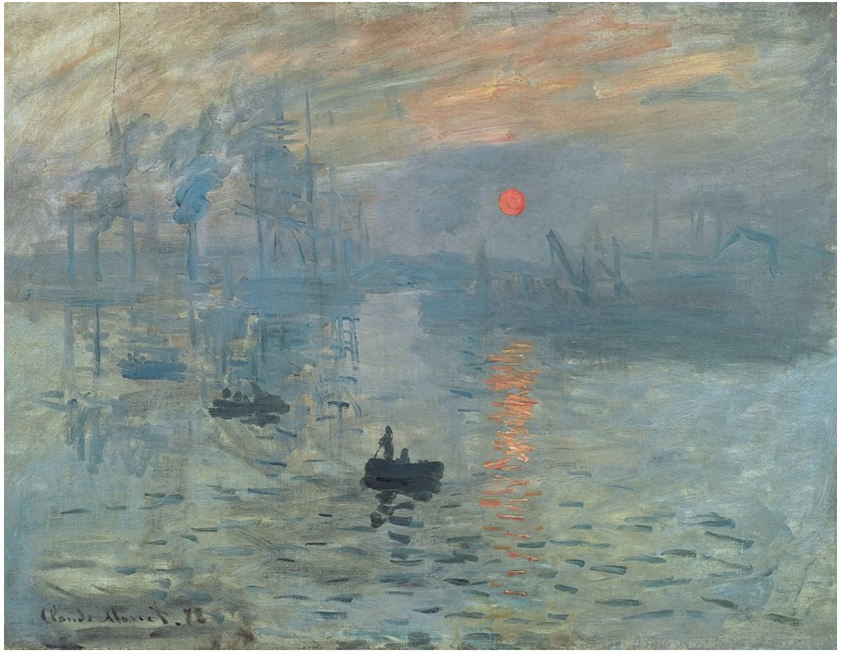
https://plos.figshare.com/articles/figure/Two_dimensional_arousal-valence_model_/4027827

UX and Usability in arts – trigger emotions

Expressionisme

Impressionnisme

Edvard Munch – le cri, 1893



Claude Monet, Impression, soleil levant (Impression, Sunrise), 1872

Evolution of emotions – feel what you feel

UX versus Usability (conclusion)

	Usability-focus	UX focus
Holistic	**Do-goals** (action theory – goal oriented) **Instrumental** (chairs to sit on, to be used for working long hours) Chairs that can stack for reducing storage space and ease storing activities Chairs under which you can stack your bag **Efficiency** I don’t fall from the chair **Performance** (we can sit 4 people on that chair)	**be-goals** (be competent, be happy) **hedonic** (more balanced toward non pragmatic and pragmatic aspects – massage while seated) **Enjoyment** (this chair rolls very fast) **Beautiful** (this chair is not comfortable at all but I’ll buy it) **Trust** (Caterpillar chair) **Pride** (I am proud of my Apple chair)
Subjective	**Objective** (don’t want to ask the user but to observe the user)	**Subjective** (ask/interpret how the user feels)
Positive	**Avoid the negative** (no pain on the back) **Hygiene** (washable chair) **Prevention** / reduction of error rate (hard to fall from) and error impact (small carpet to fall on)	**Seek the positive** (fun all day chair) **Motivators** (the chair displays the number of hours you have been seating on it) **Promotion** (everybody can see your chair)

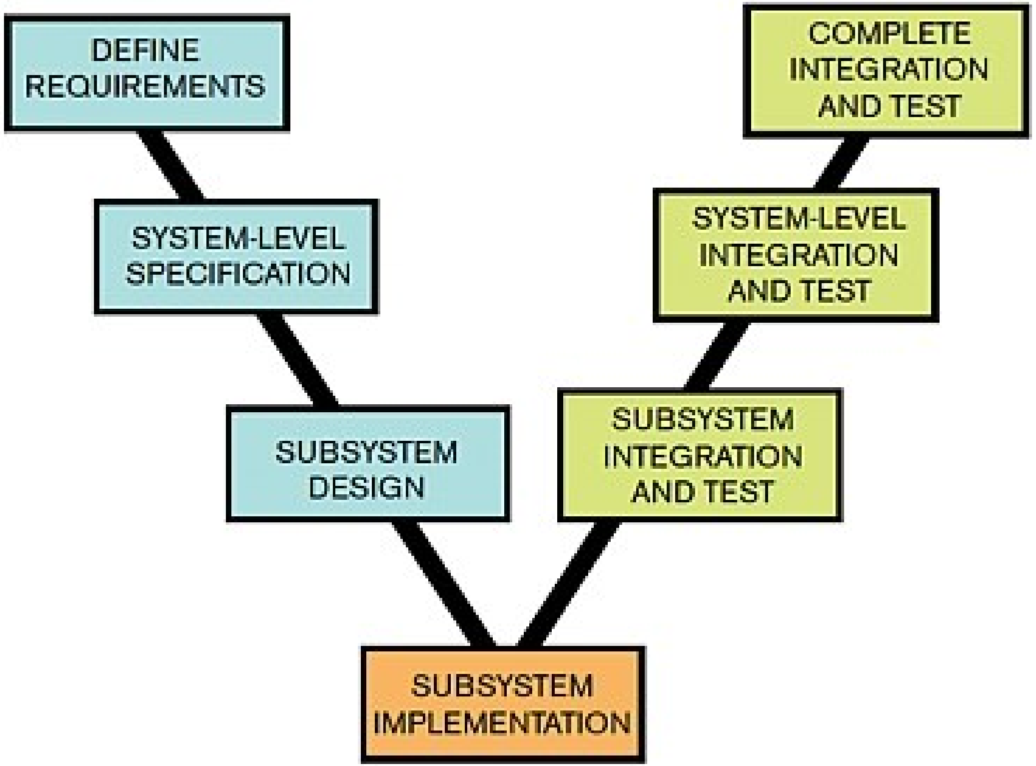
Outline of the course

- Context and definitions
  - Usability
  - User Experience
- Designing for Usability
- Designing for User Experience
- Measuring Usability and User Experience
- Going further
  - Acceptance
  - Learnability
- Specificities of command and control systems
- Take away message

Designing for usability

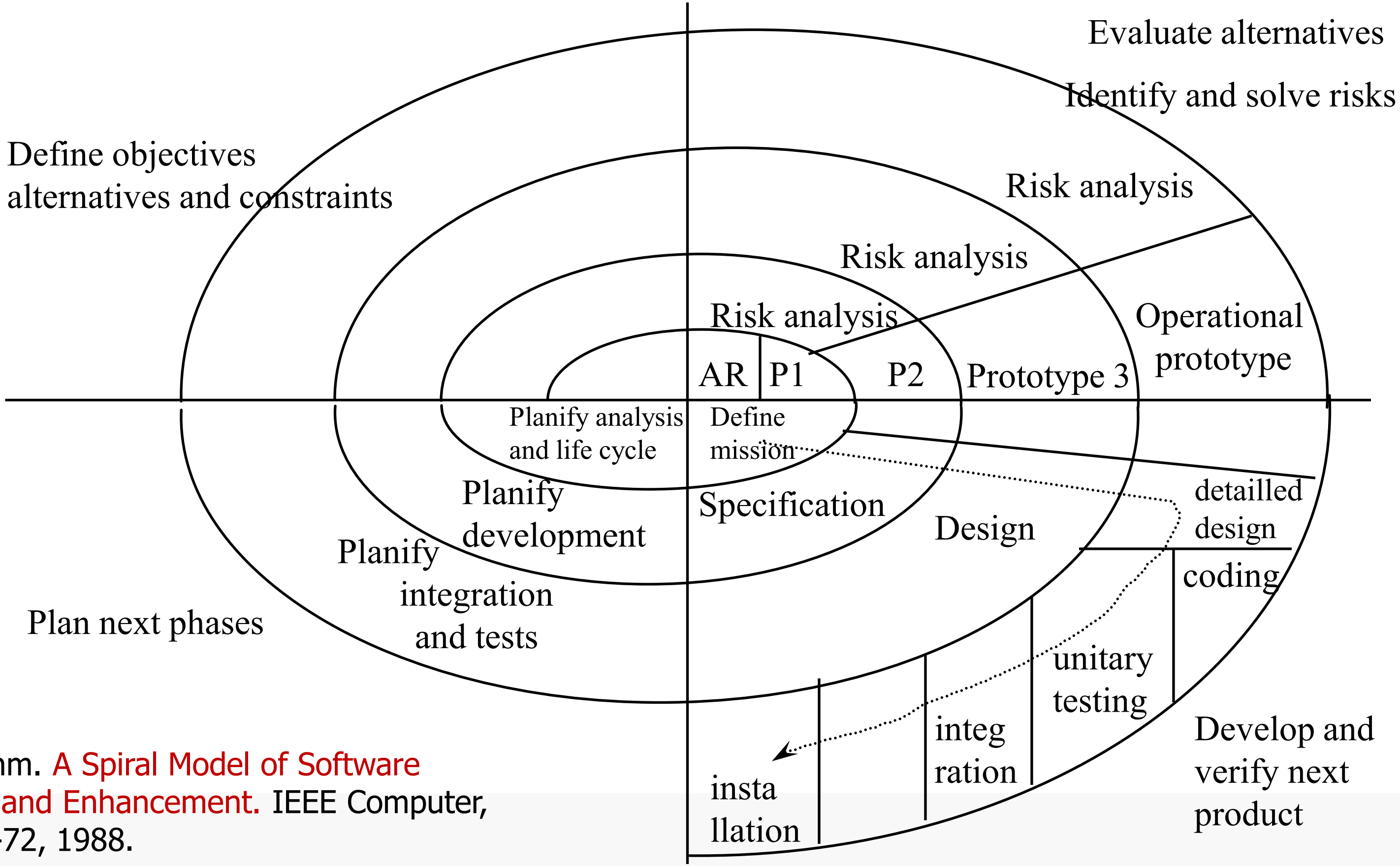
- Software development processes do not bring usability
- Software development processes do not bring user experience
- Software development processes focus on other properties
  - Reliability: V-cycle and Waterfall cycle
  - Deliverability: Spiral model
  - Iterability and fast deliverability: Agile cycles

V life-cycle



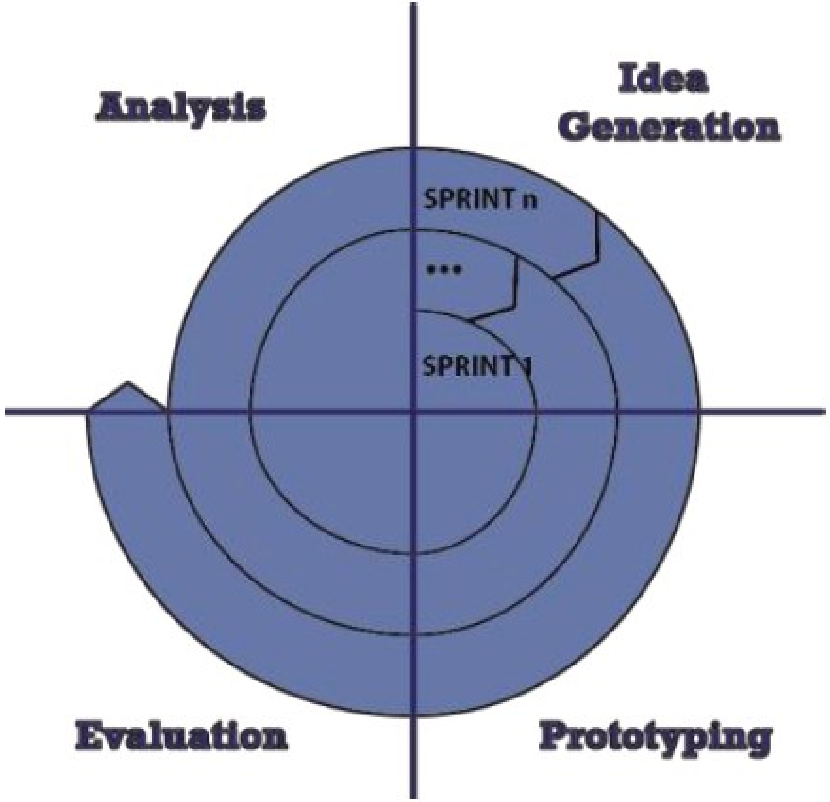
John McDermid et Knut Ripken. Life cycle support in the ADA environment. University Press, 1984.

Spiral life cycle



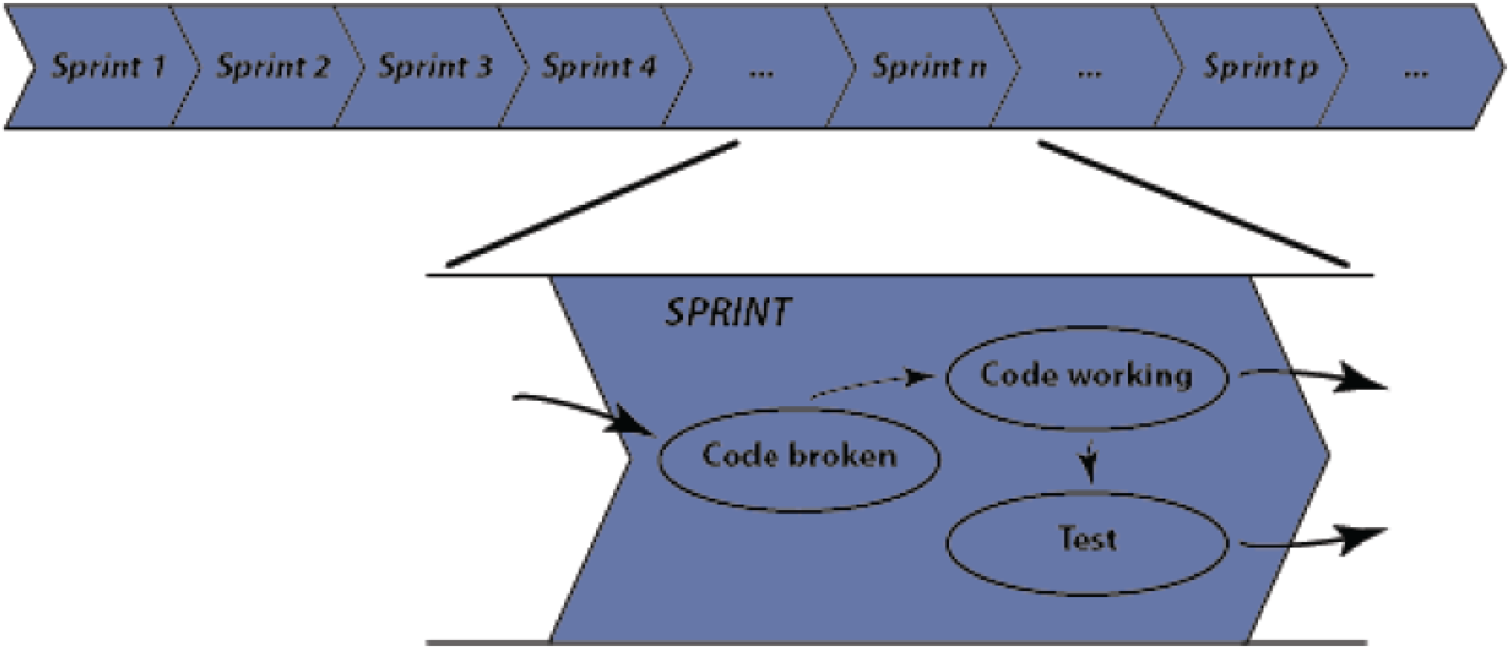
Barry W. Boehm. A Spiral Model of Software Development and Enhancement. IEEE Computer, 21(5), pp. 61-72, 1988.

Agile and Iterations



K. Schwaber and M. Beedle, Agile Software Development with SCRUM. Upper Saddle River, NJ: Prentice-Hall, 2002.

Agile Processes



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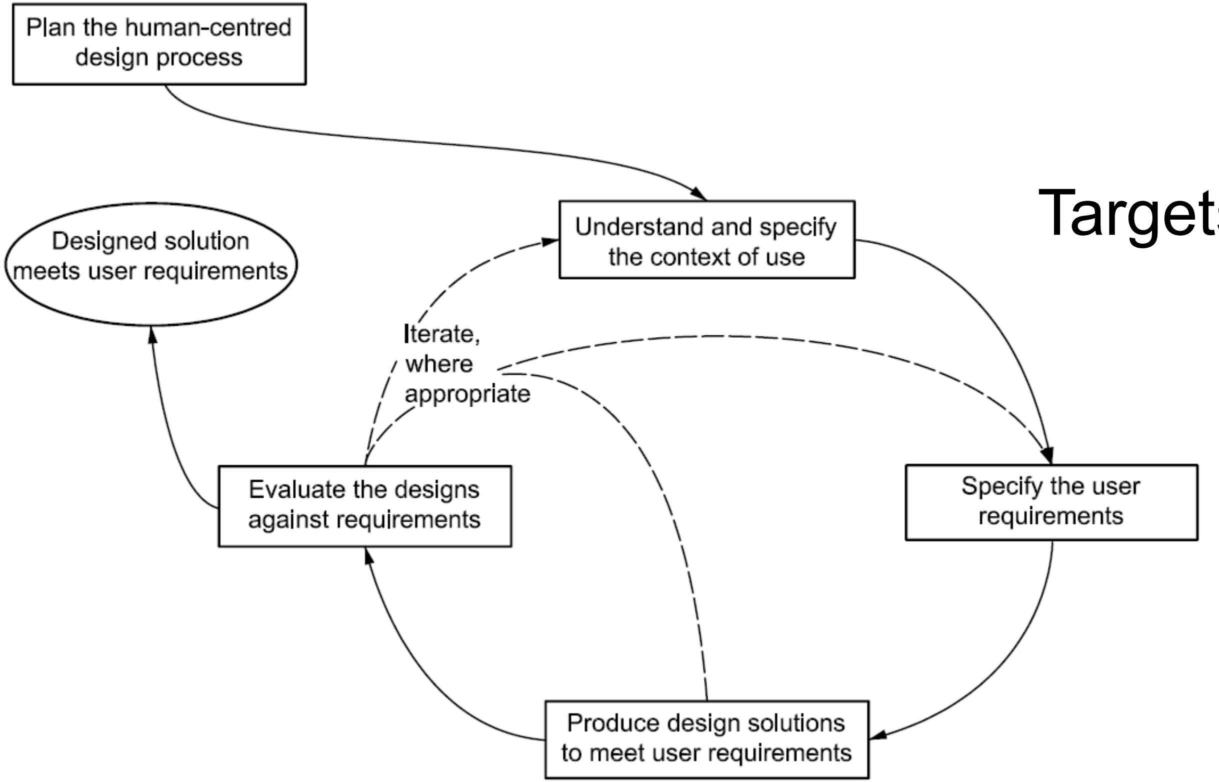
Designing for Usability

- Identifying elements that will improve
  - Performance
    - Support efficiently perception, cognition and action (commands and data entry)
    - Reduce errors
    - Gather commands into meta-commands
  - Effectiveness
    - Cover all the goals
    - Ensure that all the tasks can be performed with the system
  - Satisfaction
    - Don’t decrease too much satisfaction (usually follows effectiveness and efficiency evolution)
    - Reach an adequate level (above 70)
- Assess improvements: Iterate with real users (e.g. using prototypes)

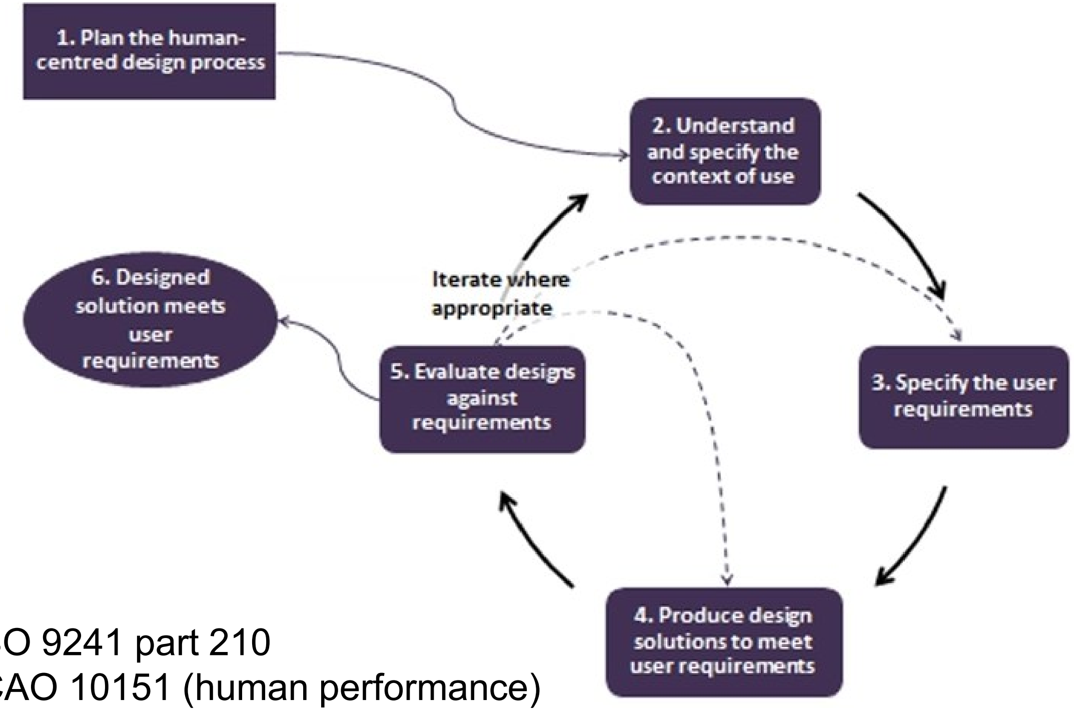
User Centered Design Process



Targets at Usability only

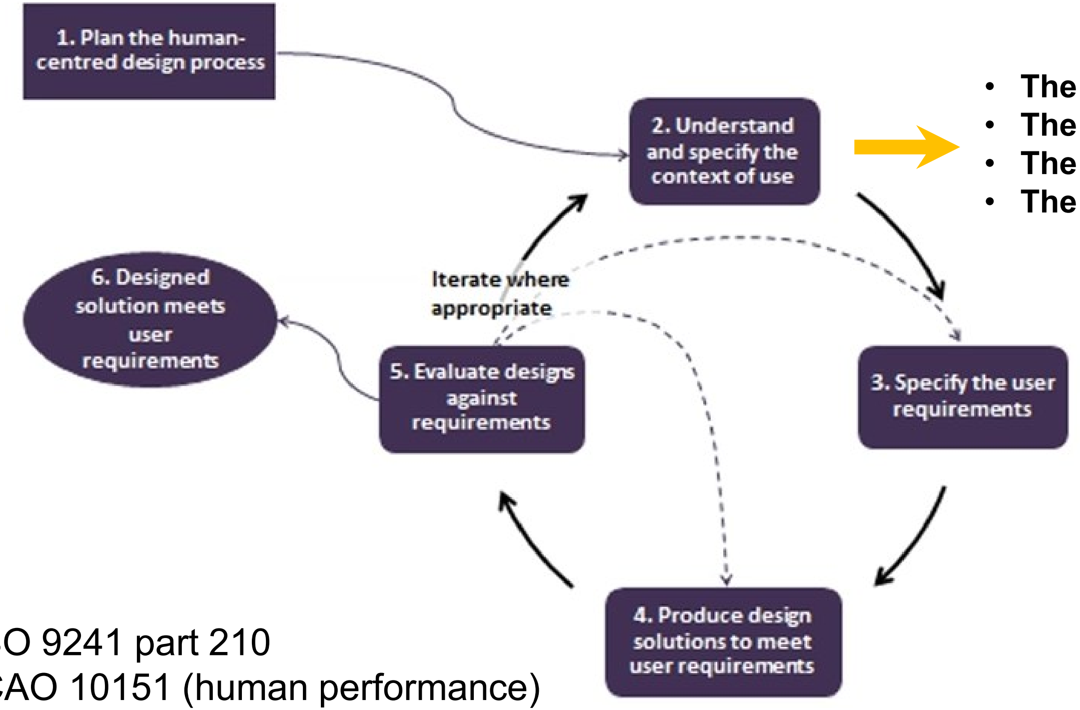
ISO 13407: Human-centred design process now in ISO 9241-210

User Centered Design Process



ISO 9241 part 210
ICAO 10151 (human performance)

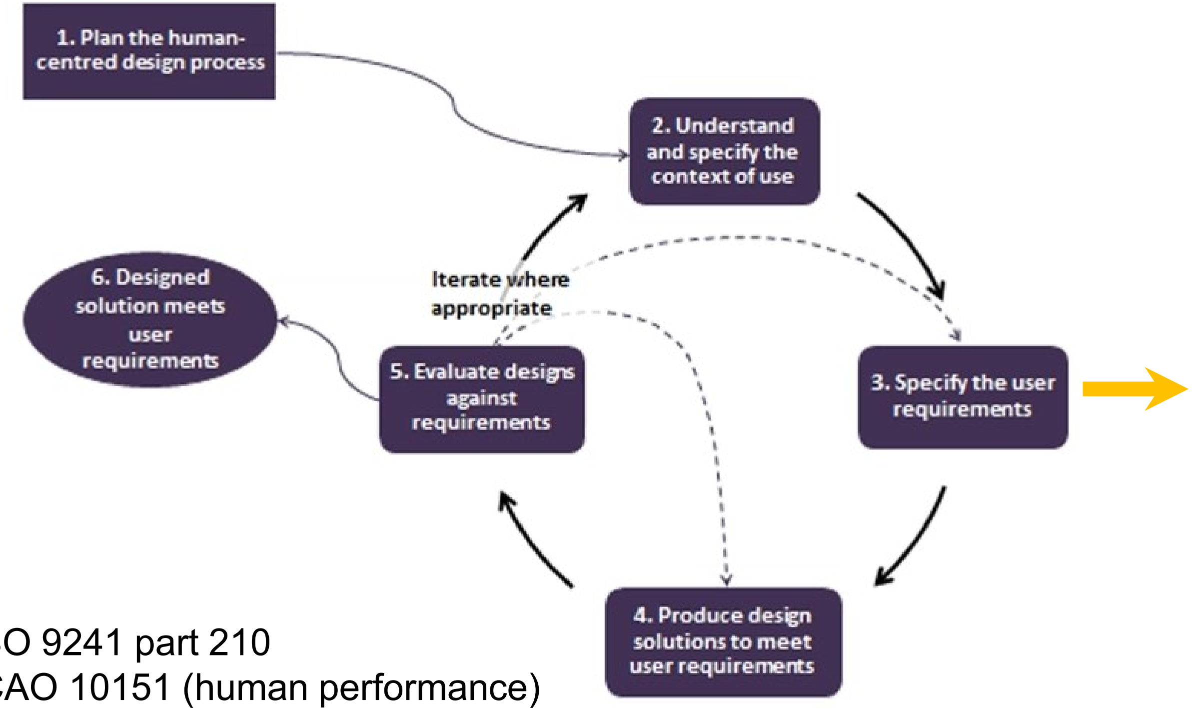
User Centered Design Process



- The users and other stakeholder groups
- The characteristics of the users or groups of users
- The goals and tasks of the users
- The environment(s) of the system

ISO 9241 part 210
ICAO 10151 (human performance)

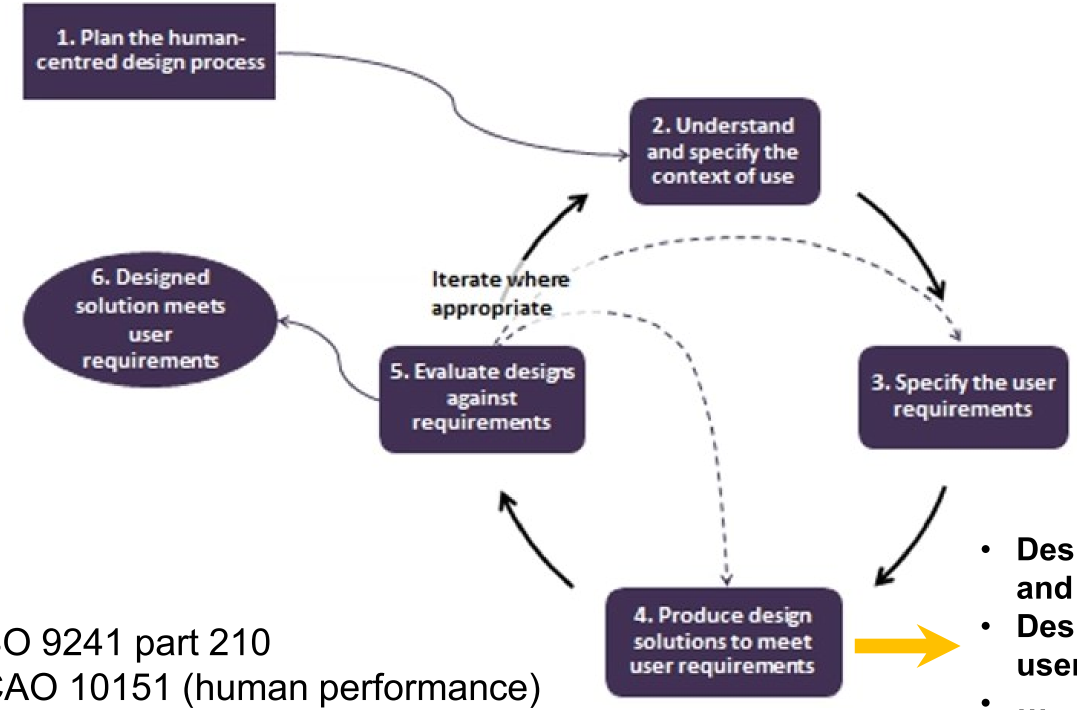
User Centered Design Process



- **Identifying user and other stakeholder needs**
- **Deriving user requirements**
- **Resolving trade-offs between user requirements**
- **Ensuring the quality of user requirements specifications**:

ISO 9241 part 210
ICAO 10151 (human performance)

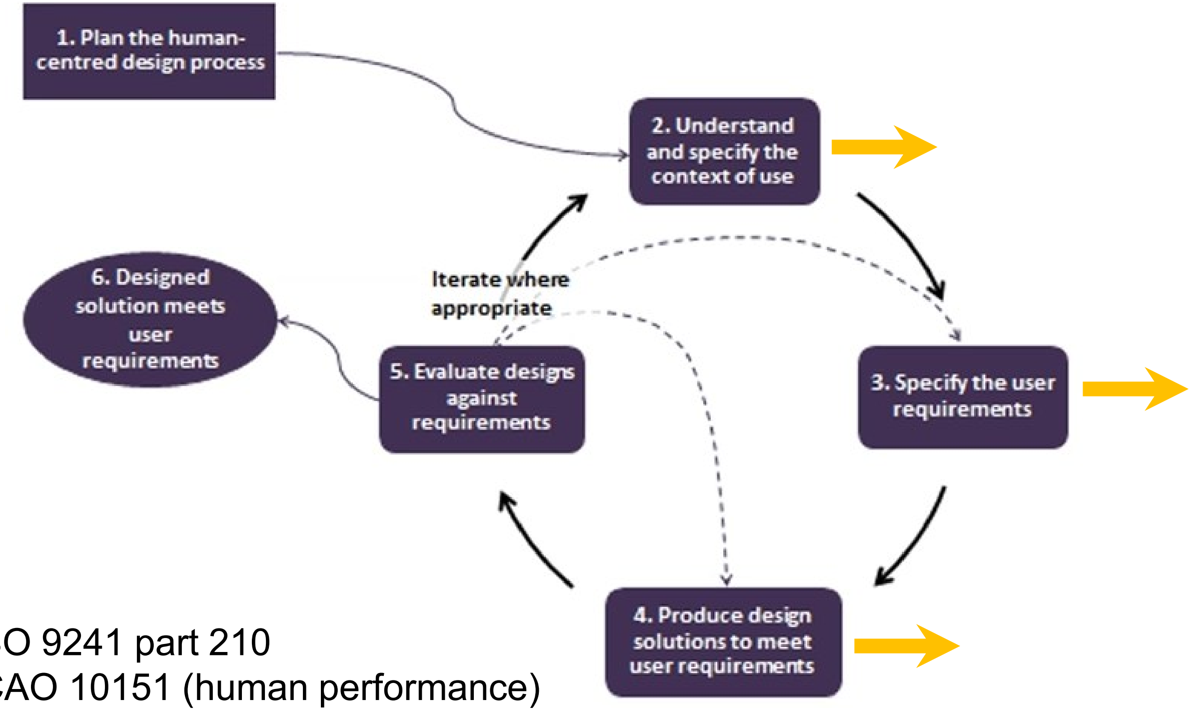
User Centered Design Process



ISO 9241 part 210
ICAO 10151 (human performance)

- **Designing user tasks, user–system interaction and user interface to meet user requirements …**
- **Designing the tasks and interaction between user and system**
- **…**

User Centered Design Process



ISO 9241 part 210
ICAO 10151 (human performance)

Analysis and understanding, modeling and exploitation of users tasks

as the corner stone

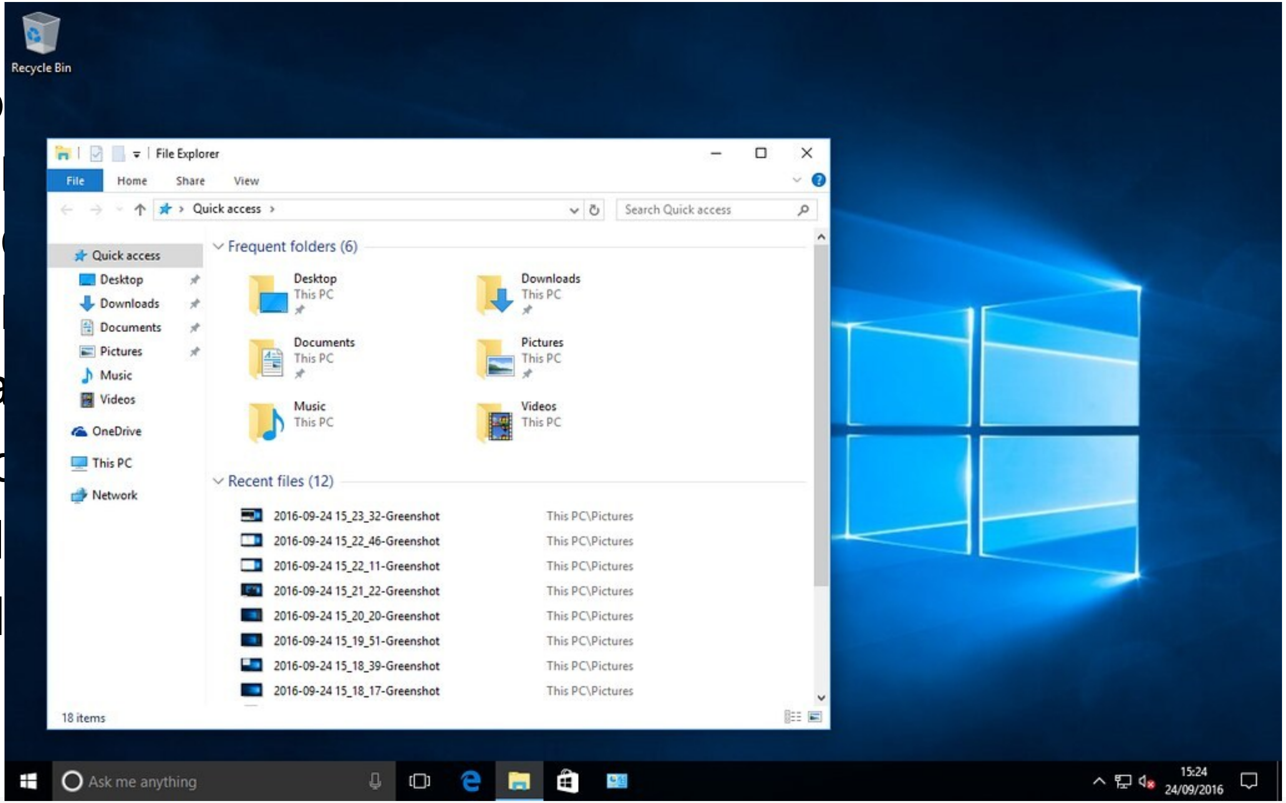
of the design, development and evaluation of socio-technical systems

Interactions on a file manager simple example

- Improve the way to interact with a file manager
  - Moving files between windows
  - Copying files in different directories
  - Keeping workspace as it is
- Analyze how it is currently done
- Find ways to do better
- Build a low precision prototype
- Build a high precision prototype

Interactions on a file manager simple example

- Imp
- Ana
- Fin
- Buil
- Buil



Interactions on a file manager

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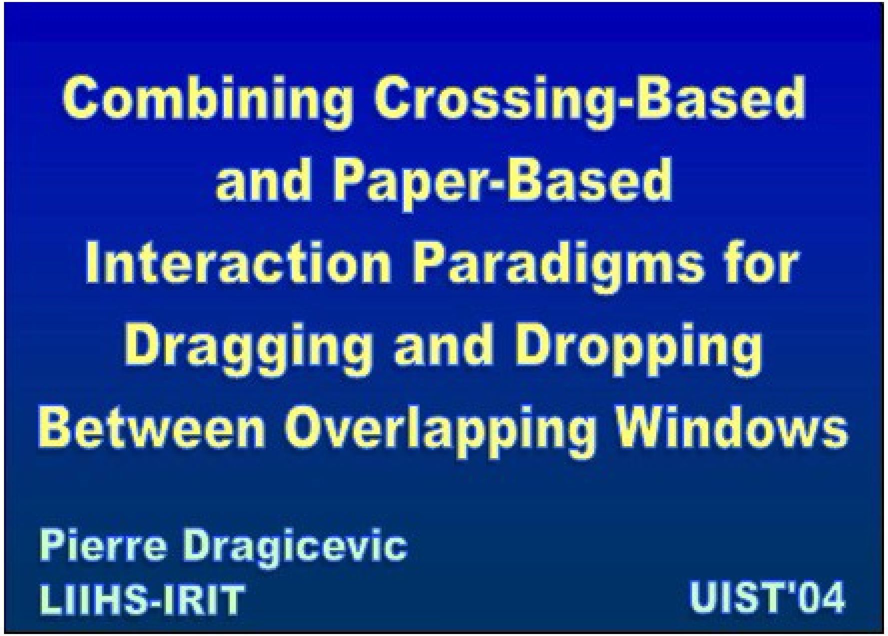
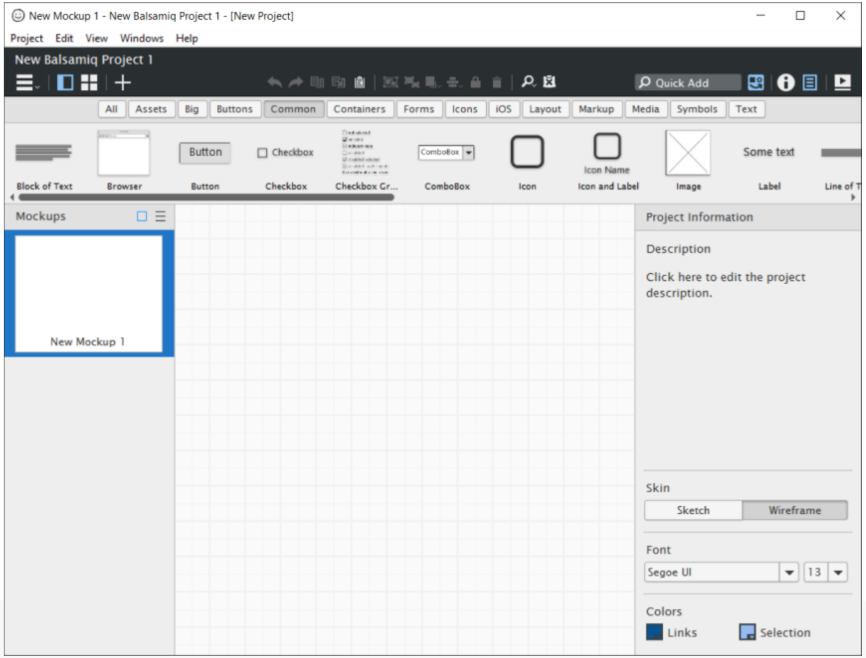
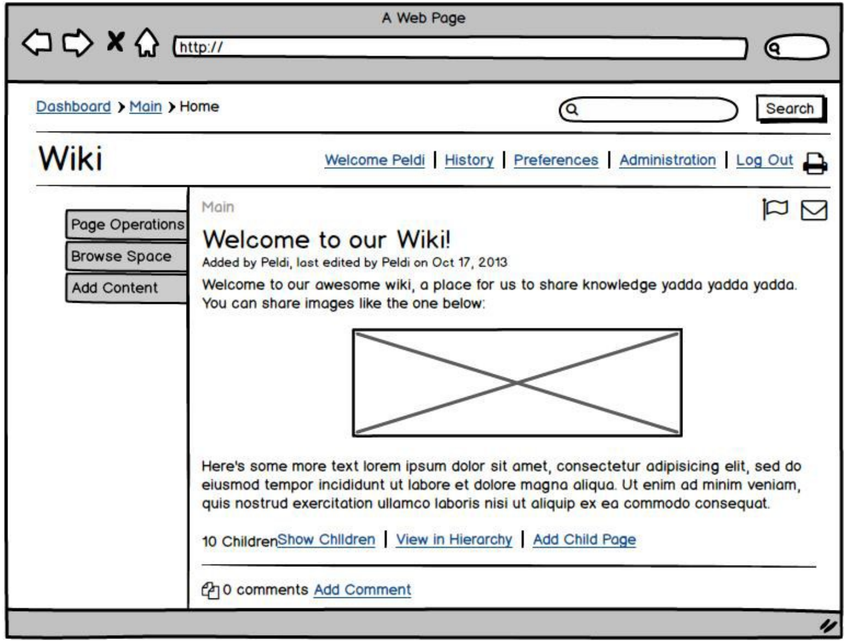


Example of prototyping tools

- Silk
- Balsamiq: http://www.balsamiq.com/products/mockups





Efficiency

- Input devices
- Interaction techniques (e.g. MS Word draw a table)
- Work
  - Minimal tasks
  - Structure of the user interface aligned to the tasks
  - Automation (tasks migrations)
  - Macro commands
- People
  - Selection
  - Training

Effectiveness

European Aviation Safety Agency

Certification Specifications
for
Large Aeroplanes

CS-25

Amendment 4
27 December 2007

CS 25.1302

GENERAL

CS 25.1301 Function and installation
(See AMC 25.1301)

Each item of installed equipment must –

(a) Be of a kind and design appropriate to its intended function;

(b) Be labelled as to its identification, function, or operating limitations, or any applicable combination of these factors. (See AMC 25.1301(b).)

(c) Be installed according to limitations specified for that equipment.

[Amdt. No.:25/2]

CS 25.1302 Installed systems and equipment for use by the flight crew
(See AMC 25.1302)

This paragraph applies to installed equipment intended for flight-crew members' use in the operation of the aeroplane from their normally seated positions on the flight deck. This installed equipment must be shown, individually and in combination with other such equipment, to be designed so that qualified flight-crew members trained in its use can safely perform their tasks associated with its intended function by meeting the following requirements:

(a) Flight deck controls must be installed to allow accomplishment of these tasks and information necessary to accomplish these tasks must be provided.

(b) Flight deck controls and information intended for flight crew use must:

(1) Be presented in a clear and unambiguous form, at resolution and precision appropriate to the task.

(2) Be accessible and usable by the flight crew in a manner consistent with the urgency, frequency, and duration of their tasks, and

(3) Enable flight crew awareness, if awareness is required for safe operation, of the effects on the aeroplane or systems resulting from flight crew actions.

(c) Operationally-relevant behaviour of the installed equipment must be:

(1) Predictable and unambiguous, and

(2) Designed to enable the flight crew to intervene in a manner appropriate to the task.

(d) To the extent practicable, installed equipment must enable the flight crew to manage errors resulting from the kinds of flight crew interactions with the equipment that can be reasonably expected in service, assuming the flight crew is acting in good faith. This sub-paragraph (d) does not apply to skill-related errors associated with manual control of the aeroplane.

[Amdt. No.:25/3]

CS 25.1303 Flight and navigation instruments

(a) The following flight and navigation instruments must be installed so that the instrument is visible from each pilot station:

(1) A free-air temperature indicator or an air-temperature indicator which provides indications that are convertible to free-air temperature.

(2) A clock displaying hours, minutes, and seconds with a sweep-second pointer or digital presentation.

(3) A direction indicator (non-stabilised magnetic compass).

(b) The following flight and navigation instruments must be installed at each pilot station:

(1) An airspeed indicator. If airspeed limitations vary with altitude, the indicator must have a maximum allowable airspeed indicator showing the variation of V_{MO} with altitude.

(2) An altimeter (sensitive).

(3) A rate-of-climb indicator (vertical speed).

(4) A gyroscopic rate of turn indicator combined with an integral slip-skid indicator (turn-and-bank indicator) except that only a slip-skid indicator is required on aeroplanes with a third attitude instrument system usable through flight attitudes of 360° of pitch and roll, which is powered from a source independent of the electrical generating system and continues reliable operation for a minimum of 30 minutes after total failure of the electrical generating system, and is installed in accordance with CS 25.1321 (a).

CS–25 BOOK 1

SUBPART F – EQUIPMENT

GENERAL

CS 25.1301 Function and installation
(See AMC 25.1301)

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CS 25.1302 Installed systems and equipment for use by the flight crew
(See AMC 25.1302)

This paragraph applies to installed equipment intended for flight-crew members' use in the operation of the aeroplane from their normally seated positions on the flight deck. This installed equipment must be shown, individually and in combination with other such equipment, to be designed so that qualified flight-crew members trained in its use can safely perform their tasks associated with its intended function by meeting the following requirements:

(a) Flight deck controls must be installed to allow accomplishment of these tasks and information necessary to accomplish these tasks must be provided.

(b) Flight deck controls and information intended for flight crew use must:

(1) Be presented in a clear and unambiguous form, at resolution and precision appropriate to the task.

(2) Be accessible and usable by the flight crew in a manner consistent with the urgency, frequency, and duration of their tasks, and

(3) Enable flight crew awareness, if awareness is required for safe operation, of the effects on the aeroplane or systems resulting from flight crew actions.

(c) Operationally-relevant behaviour of the installed equipment must be:

(1) Predictable and unambiguous, and

(2) Designed to enable the flight crew to intervene in a manner appropriate to the task.

(d) To the extent practicable, installed equipment must enable the flight crew to manage errors resulting from the kinds of flight crew interactions with the equipment that can be reasonably expected in service, assuming the flight

indications that are convertible to free-air temperature.

(2) A clock displaying hours, minutes, and seconds with a sweep-second pointer or digital presentation.

(3) A directio[n] magnetic compass).

(b) The following instruments must be insta[lled]

(1) An airspe[ed] limitations vary with [...] have a maximum all[...] showing the variation [...]

(2) An altimet[er]

(3) A rate-of-[...] speed).

(4) A gyrosco[pic] combined with an i[...] (turn-and-bank indicat[or]) skid indicator is requ[ired] third attitude instrum[ent] flight attitudes of 360° powered from a so[urce] electrical generating [...] reliable operation for a minimum of 30 minutes after total failure of the electrical generating system, and is installed in accordance with CS 25.1321 (a).

(a) Flight deck controls must be installed to allow accomplishment of these tasks and information necessary to accomplish these tasks must be provided.

(d) To the extent practicable, installed equipment must enable the flight crew to manage errors resulting from the kinds of flight crew interactions with the equipment that can be reasonably expected in service, assuming the flight crew is acting in good faith. This sub-paragraph (d) does not apply to skill-related errors associated with manual control of the aeroplane.

A350 Flight Crew Operating Manual

AIRBUS

A350

AIRBUS
FOR TRAINING ONLY

FLIGHT CREW
OPERATING MANUAL

Standard Operating Procedures

AIRBUS FOR TRAINING ONLY **A350** FLIGHT CREW OPERATING MANUAL	**PROCEDURES** **NORMAL PROCEDURES** STANDARD OPERATING PROCEDURES - GENERAL INFORMATION

STANDARD OPERATING PROCEDURES (SOP)

Ident.: PRO-NOR-SOP-10-00005563.0001001 / 04 NOV 13
Applicable to: ALL

SOPs consist of inspections, preparations, and normal procedures. All items of a given procedure are listed in a sequence that follows a standardized scan of the cockpit panels, unless that sequence goes against the action priority logic, to ensure that the flight crew performs all actions in the most efficient way.
SOPs are divided into flight phases. The flight crew should perform the SOPs actions by memory.
A panel scan sequence and flow pattern at the beginning of some flight phases indicate where the actions are performed. The flight crewmembers must apply the flow patterns to ensure that they perform the necessary actions for a specific flight phase, before completing the applicable checklist.
The SOPs assume that all systems are operating normally, and that all automatic functions are used normally.
The FCOM also contains normal procedures that are non-routine procedures in the Supplementary Procedures chapter and in the Special Operations chapter.

Preliminary Cockpit Preparation

The preliminary cockpit preparation includes:
- Aircraft power-up
- APU start
- ECAM/Logbook check
- OIS initialization
- OIS preparation.

AIRCRAFT POWER-UP

Ident.: PRO-NOR-SOP-40-A-PG02169
Applicable to: ALL

Ident.: PRO-NOR-SOP-40-A-00018118.0001001 / 03 DEC 15

1 **GENERAL**

WARNING	Do not pressurize the hydraulic systems until clearance is obtained from ground personnel.

Ident.: PRO-NOR-SOP-40-A-00018120.0001001 / 25 APR 14

ENG

ENG 1, 2 MASTER LEVERS....................OFF | CM2
ENG START selector....................NORM | CM2

Ident.: PRO-NOR-SOP-40-A-00018121.0001001 / 25 APR 14

L/G

L/G lever....................DOWN | CM2

Ident.: PRO-NOR-SOP-40-A-00018122.0001001 / 25 APR 14

WIPERS

Both WIPER selectors....................OFF | CM2

Preliminary Cockpit Preparation

Detailed textual task description

CM1= Crew Member 1
CM2 = Crew Member 2

Actor 1 = Captain (the big boss)
Actor 2 = First officer (the assistant)

Role 1 = Pilot Flying
Role 2 = Pilot Monitoring

The preliminary cockpit preparation includes:
- Aircraft power-up
- APU start
- ECAM/Logbook check
- OIS initialization
- OIS preparation.

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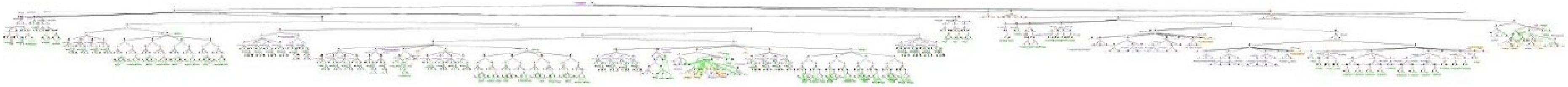
L/G lever....DOWN | CM2

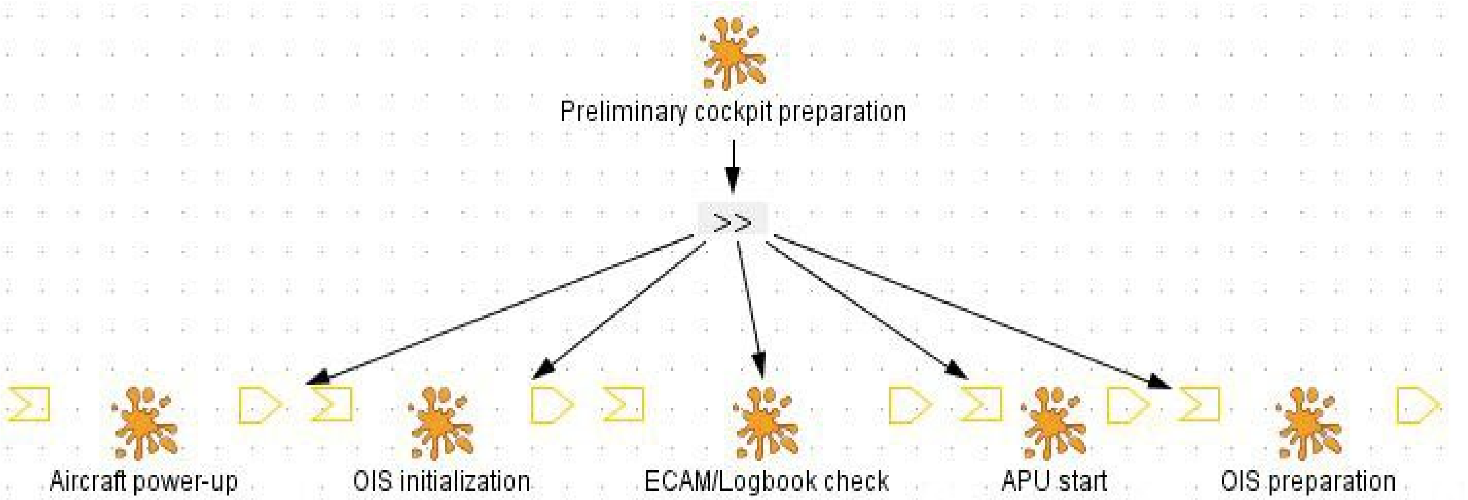
Ident.: PRO-NOR-SOP-40-A-00018122.0001001 / 25 APR 14

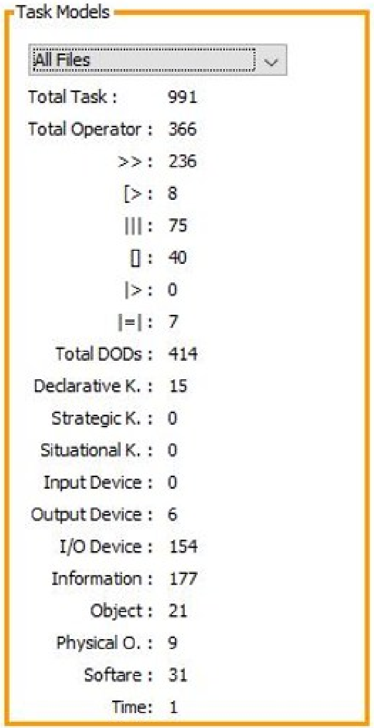
WIPERS

Both WIPER selectors....OFF | CM2

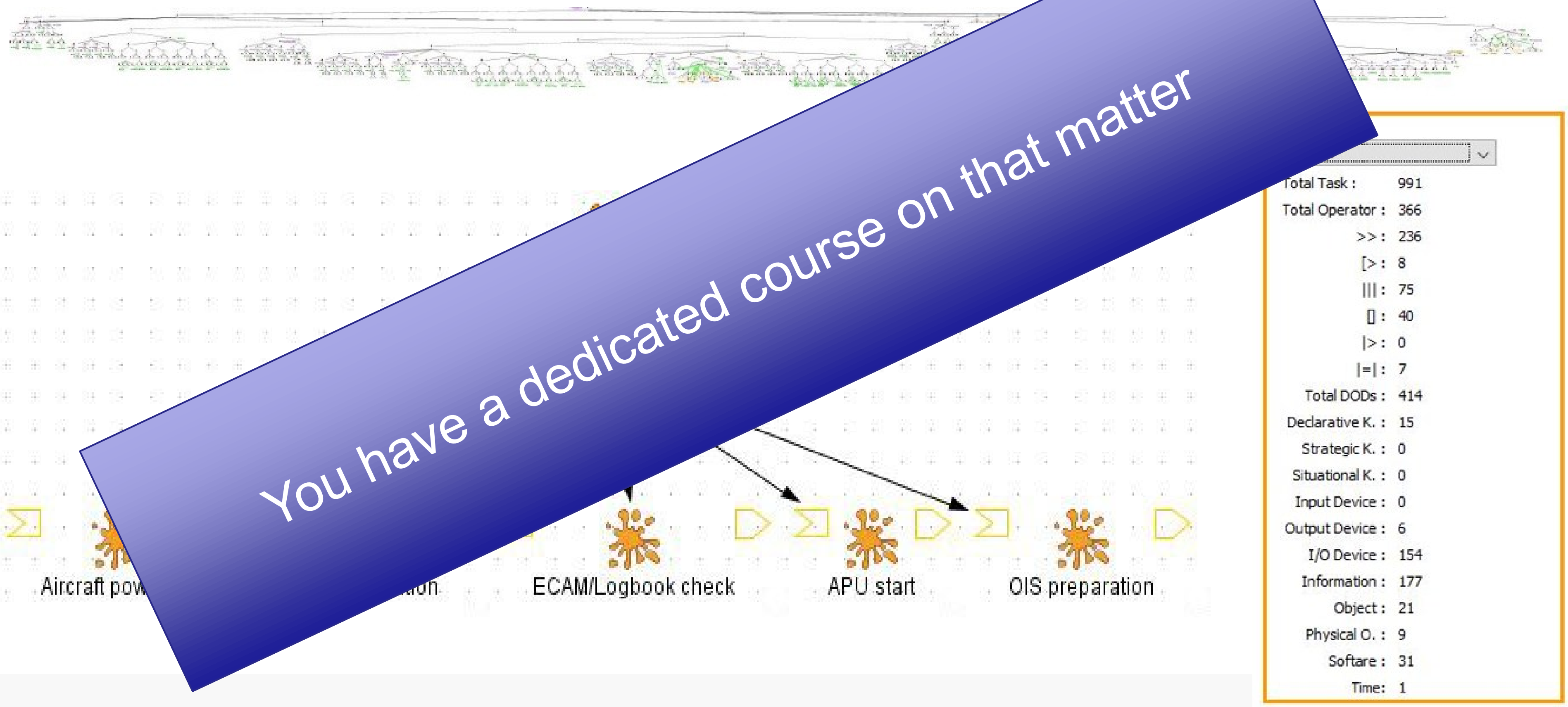
Preliminary Cockpit Preparation







Preliminary Cockpit Preparation



You have a dedicated course on that matter

Taxi

Complexity of designing tasks allocation
Checking related workload and feasibility
Training is key

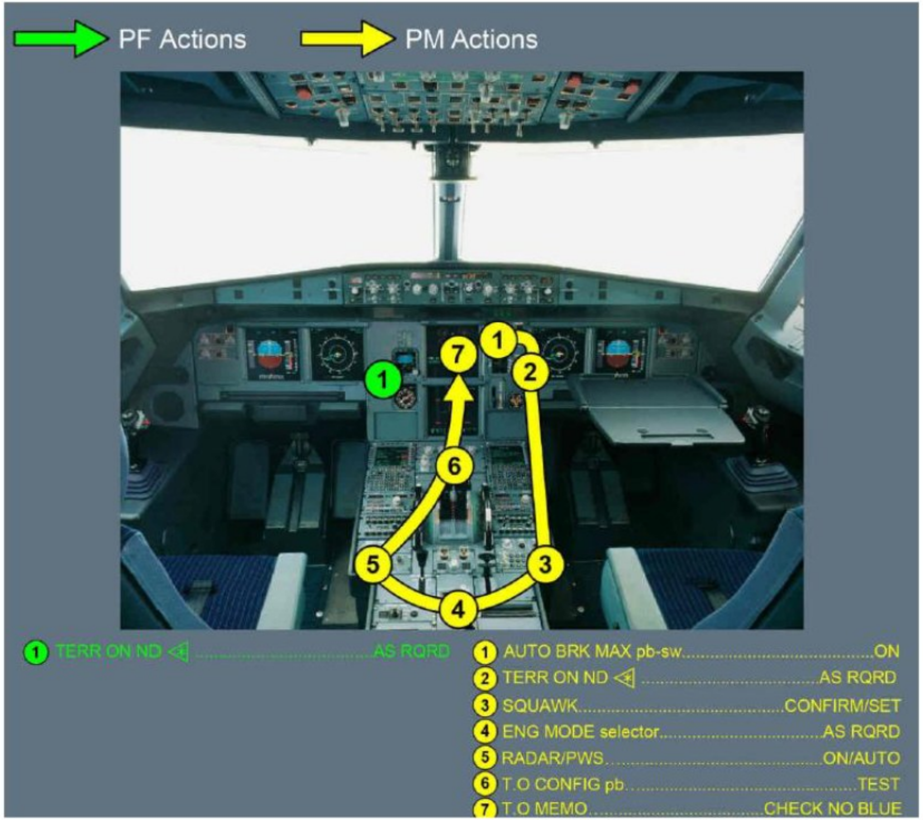
Role 1 = Pilot Flying
Role 2 = Pilot Monitoring

TAXI [1]

		PF	PM
PRE-REQ	**Flight Controls**	Check Flight Controls Before or During Taxi	
	Taxi Clearance		Obtain
	Taxi / Turn Off Light	ON [2]	
	Area Clearance	Call: "Clear Left Side"	Call: "Clear Right Side"
	Parking Brake	OFF – Brake Pressure Zero [3]	
	Brake Pedals	Press & Call: "BRAKE CHECK"	Call: "PRESSURE ZERO" [4]
CLEARANCE	**ATC Clearance**		Confirm
	AFS / Flight Instruments		**FMS**: F-PLN (SID, TRANS); Initial CLB SPD & SPD Limit **FCU**: Cleared Altitude; HDG Preset **FD**: Both ON
		PFD / ND – Check (Airspeeds, Altitude, Heading, FMA, SID)	
	Departure Briefing	Confirm	
D[illegible]	**Auto Brakes**		AUTO BRK MAX ON
	ATC Code / Mode		Confirm / Set
	Engine Mode Selector		As Required [5]
	Weather Radar		Radar – ON / ALL [6] Predictive Windshear – Auto / ON
	Terrain on ND	As Required – However, consider Radar on PF side & Terrain on PM side	
	Final Check		TO Config – Test TO Memo – Check No Blue
		"CABIN SECURED FOR TAKEOFF" – Report Receive	
	Taxi Checklist	**TAXI** FLIGHT CONTROLS.................. CHECKED (BOTH) FLAPS SETTING..................... CONF____ (BOTH) RADAR & PRED W/S.............................ON & AUTO ENG MODE SEL..____ ECAM MEMO..T.O NO BLUE - *AUTO BRK MAX* - *SIGNS ON* - *CABIN READY* - *SPLRS ARM* - *FLAPS TO* - *TO CONFIG NORM* CABIN..READY	

Connection between tasks and the interactive system

TAXI FLOW PATTERN



After Start Normal Procedure

AFTER START

	PF	PM
Engine Mode Selector	Normal [1]	
APU Bleed	OFF [2]	
Engine Anti-ice	As Required [3]	
Wing Anti-ice [4]	As Required	
APU Master Switch	OFF (if not required)	
Ground Spoilers		Arm
Rudder Trim		Neutral
Flaps		Takeoff Position [5]
Pitch Trim Handwheel		Set
Status Reminder	Check Not Displayed – If displayed then check ECAM status	
N/W STEER DISC MEMO	Check – Not Displayed	
Ground Crew	Announce: “Clear to Disconnect” “Hand Signal on the Left / Right”	
Checklist [6]	**AFTER START** ANTI ICE........................ ____ ECAM STATUS........................CHECKED PITCH TRIM........................ ____% RUDDER TRIM........................NEUTRAL	

After Start Normal Procedure (cues associated to tasks)

1. This is a cue to do "After Start" procedure.

2. This action enables to avoid ingestion of engine exhaust gases. If APU is necessary for performance purpose then bleed can be selected ON before takeoff.

3. Must be ON during all ground operation, when icing conditions (OAT/TAT < 10°C with visible moisture) exist or are anticipated. During ground operation in icing conditions with OAT +3°C or less, carry out ice shedding procedure from PRO-NOR-SUP-ADVWXR – ENGINE OPERATIONS ON GROUND IN ICING CONDITIONS.

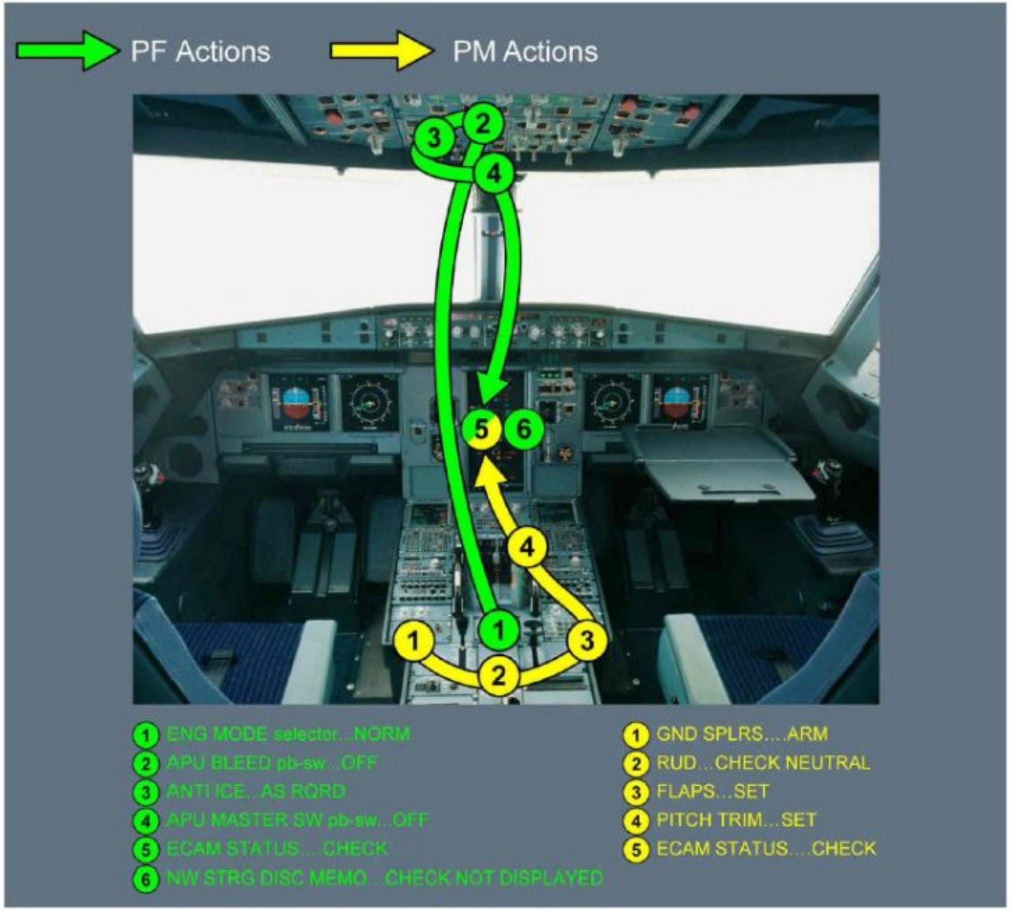
4. APU bleed not authorized for using wing anti ice. In icing conditions, wing anti-ice ***may be*** *turned on to prevent ice accretion on the wing leading edge. It* ***must be*** *turned on if there is evidence of ice accretion, such as ice on the visual indicator, or on the wipers, or with the SEVERE ICE DETECTED alert. Ice accretion is considered severe when the ice accumulation on the airframe reaches approximately 5mm thick or more.*

5. In icing conditions with rain, slush or snow, maintain flaps retracted until takeoff point.

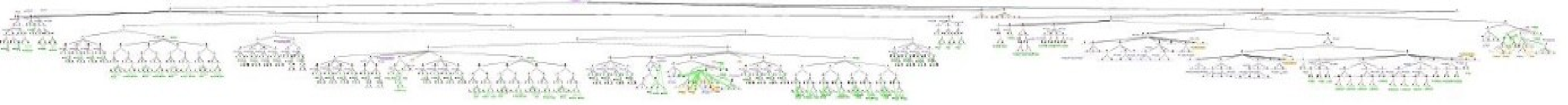
6. After receiving the hand signal from the ground crew, CM1/CM2 will call "HAND SIGNAL RECEIVED AND BYPASS PIN SIGHTED". Then CM1 will ask for "AFTER START CHECKLIST".

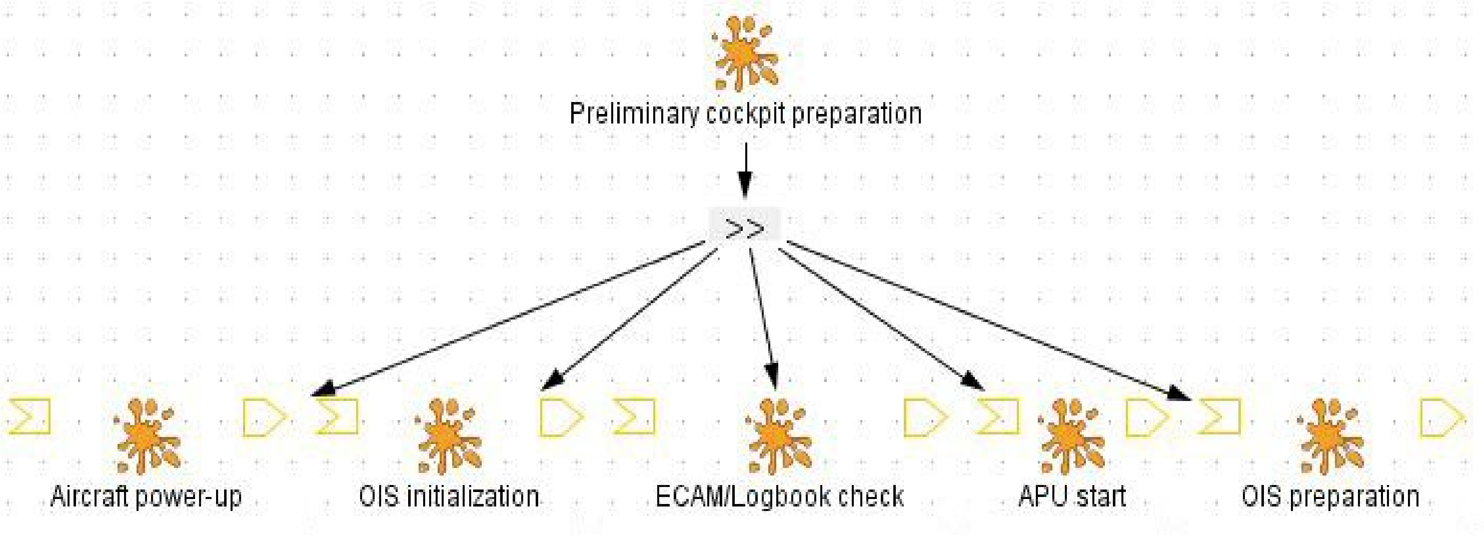
After Start Normal Procedure (connection to the interactive system (cockpit)

AFTER START FLOW PATTERN



Preliminary Cockpit Preparation





Task Models

All Files

Total Task :	991
Total Operator :	366
>> :	236
[> :	8
||| :	75
[] :	40
|> :	0
|=| :	7
Total DODs :	414
Declarative K. :	15
Strategic K. :	0
Situational K. :	0
Input Device :	0
Output Device :	6
I/O Device :	154
Information :	177
Object :	21
Physical O. :	9
Softare :	31
Time:	1

Complexity of tasks and important of modelling: PCP = 991 tasks to be performed

Outline of the course

- Context and definitions
  - Usability
  - User Experience
- Designing for Usability
- Designing for User Experience
- Measuring Usability and User Experience
- Going further
  - Acceptance
  - Learnability
- Specificities of command and control systems
- Take away message

Designing for User Experience

- As for usability, innovation is critical
- Solutions are available
- Solutions must be invented to fit the specific needs
  - Environment effects of cockpits (sun, night, turbulences)
  - Specific hardware / software systems (flight warning system)
  - Specific type of users and allocation of work (organic & radarist, …)

Before assessing

- Designing for each contributing factor of UX
  - Aesthetics
  - Emotions
  - Identification
  - Stimulation
  - Meaning and Value
  - Social connectedness
- Identifying the relevant ones
- Complexity related to branding, current fashion and fashion trends, contexts of use (e.g. war), …

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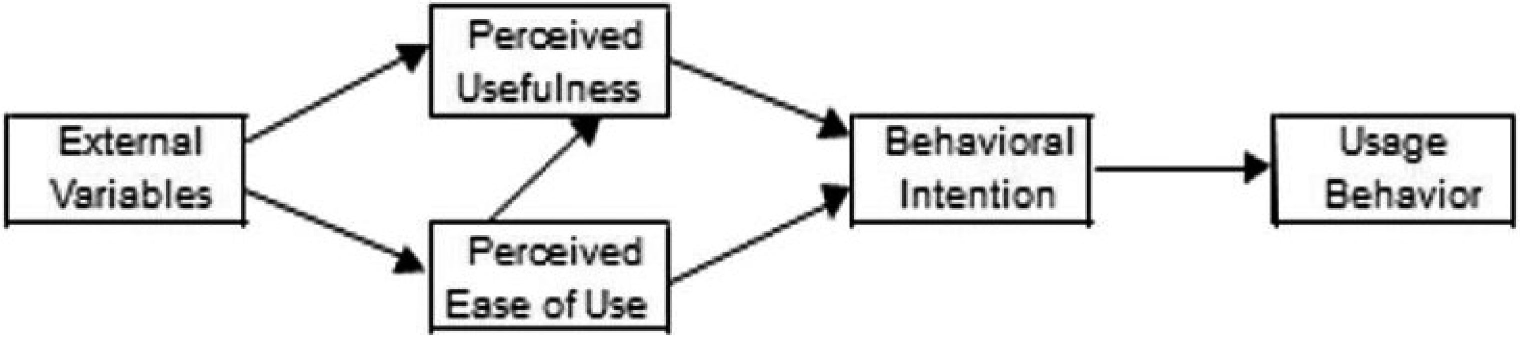
Technology Acceptance Model

- Technology Acceptance Model (TAM)
- Fred Davis (1986) – PhD Thesis at MIT
- Adaption of Fishbein and Ajzen’s Theory of Reasoned Action (TRA)
- Davis’ paper “Perceived usefulness, perceived ease of use, and user acceptance of information technology” (1989)

Citations on Google Scholar

19/05/10 - 5,403
06/03/13 - 15,092
07/03/13 - 15,103
31/01/14 - 18,410
08/07/20 - 52,259
20/09/21 – 62,216
20/09/23 – 80,916

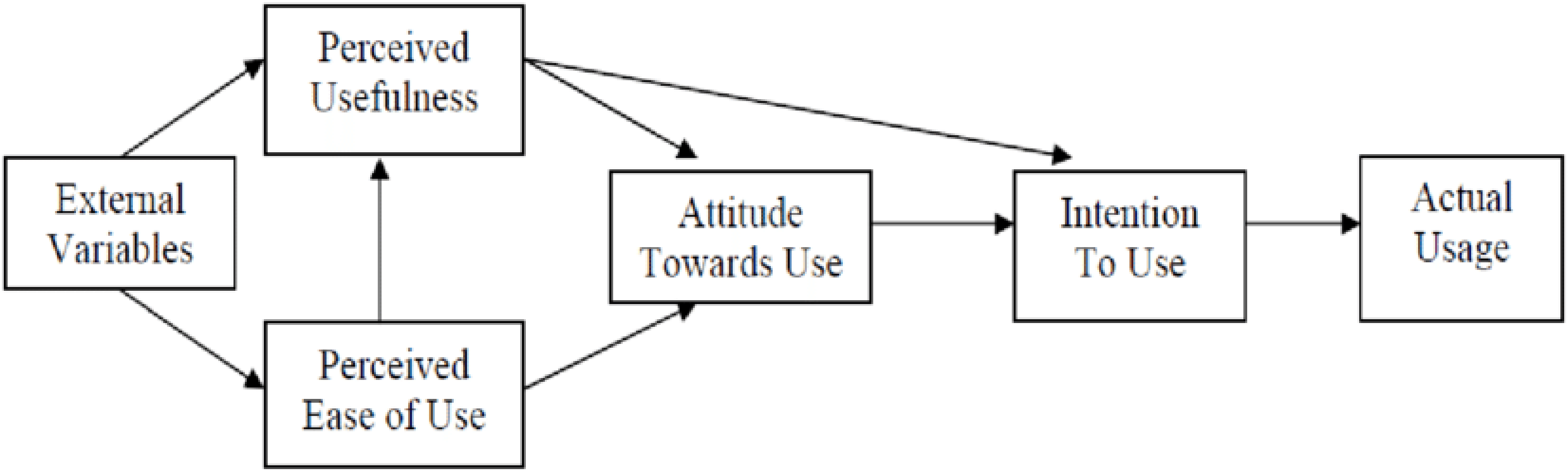
Technology Acceptance Model



- Technology Acceptance Model (TAM)
  - External Variables include “system characteristics, training, user involvement in design, and the nature of the implementation process” (Venkatesh & Davis, 1996)
  - These ‘[...] directly influence the perceived usefulness and perceived ease of use’, in turn mediating behavioural intention to use” (Davis, 1993)

Final Version of Technology Acceptance Model (TAM). Source: (Venkatesh & Davis, 1996)

Refined TAM from Davis



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  - User Experience
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  - Acceptance
  - Learnability
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- Take away message

What is a walk-up-and-use system

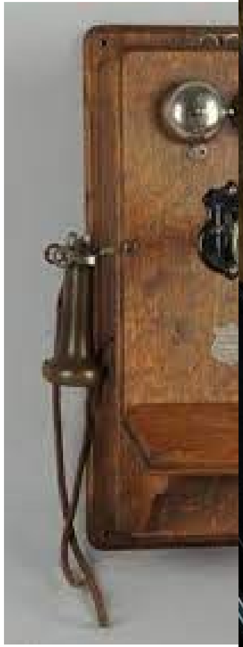
- We use a theory of exploratory learning developed by Polson and Lewis [19] to generate a list of theoretically motivated questions about the user interfaces of systems intended for use in applications with **minimal formal training requirements**, e.g. **advanced phone services** and other **consumer applications** of **computer technology**.

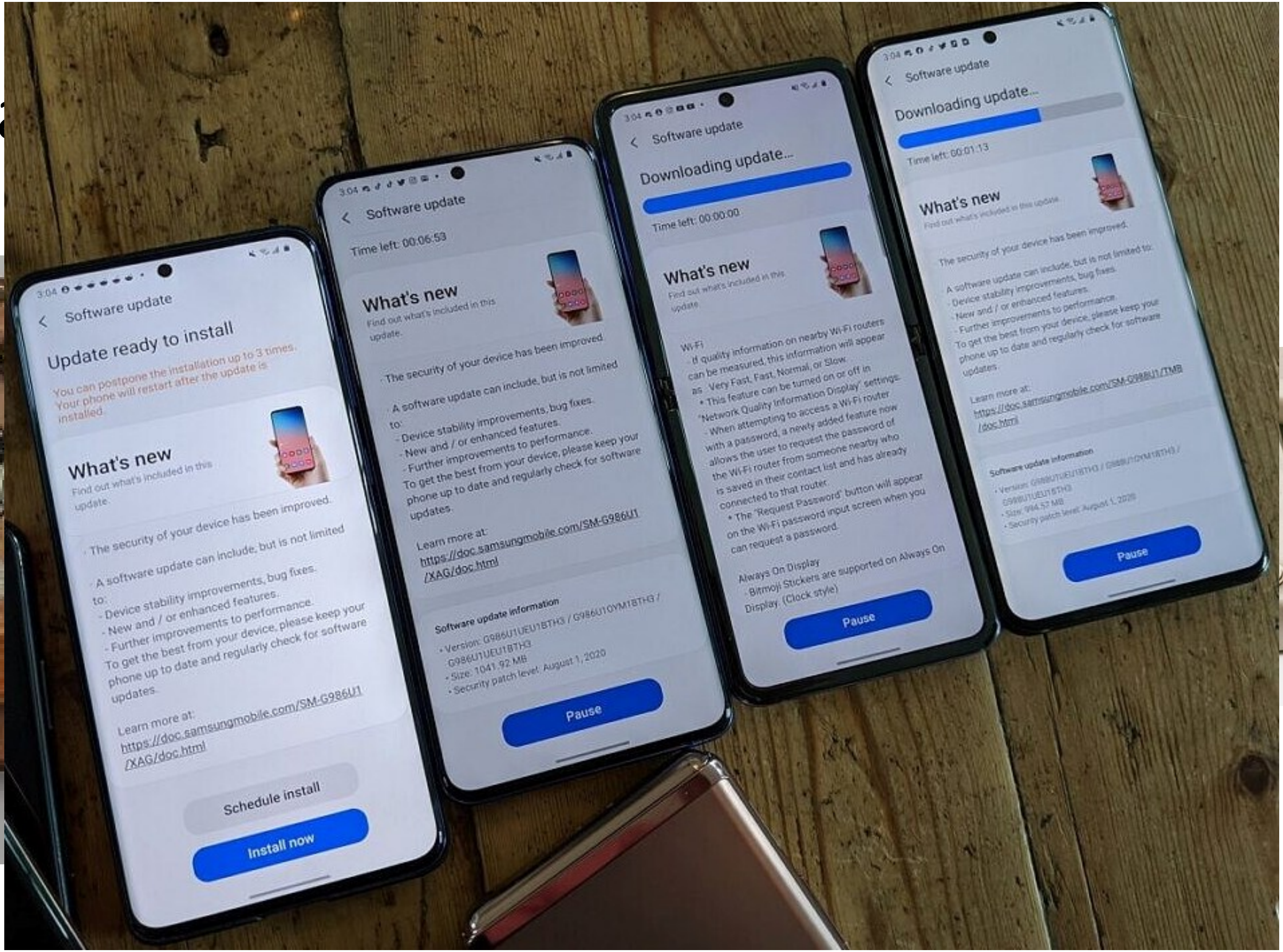
[19] Peter G. Polson and Clayton H. Lewis. 1990. **Theory-Based Design for Easily Learned Interfaces.** Hum.-Comput. Interact. 5, 2 (jun 1990), 191–220.

Clayton Lewis, Peter G. Polson, Cathleen Wharton, and John Rieman. 1990. **Testing a Walkthrough Methodology for Theory-Based Design of Walk-up-and-Use Interfaces**. In Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (Seattle, Washington, USA) (ACM CHI ’90).

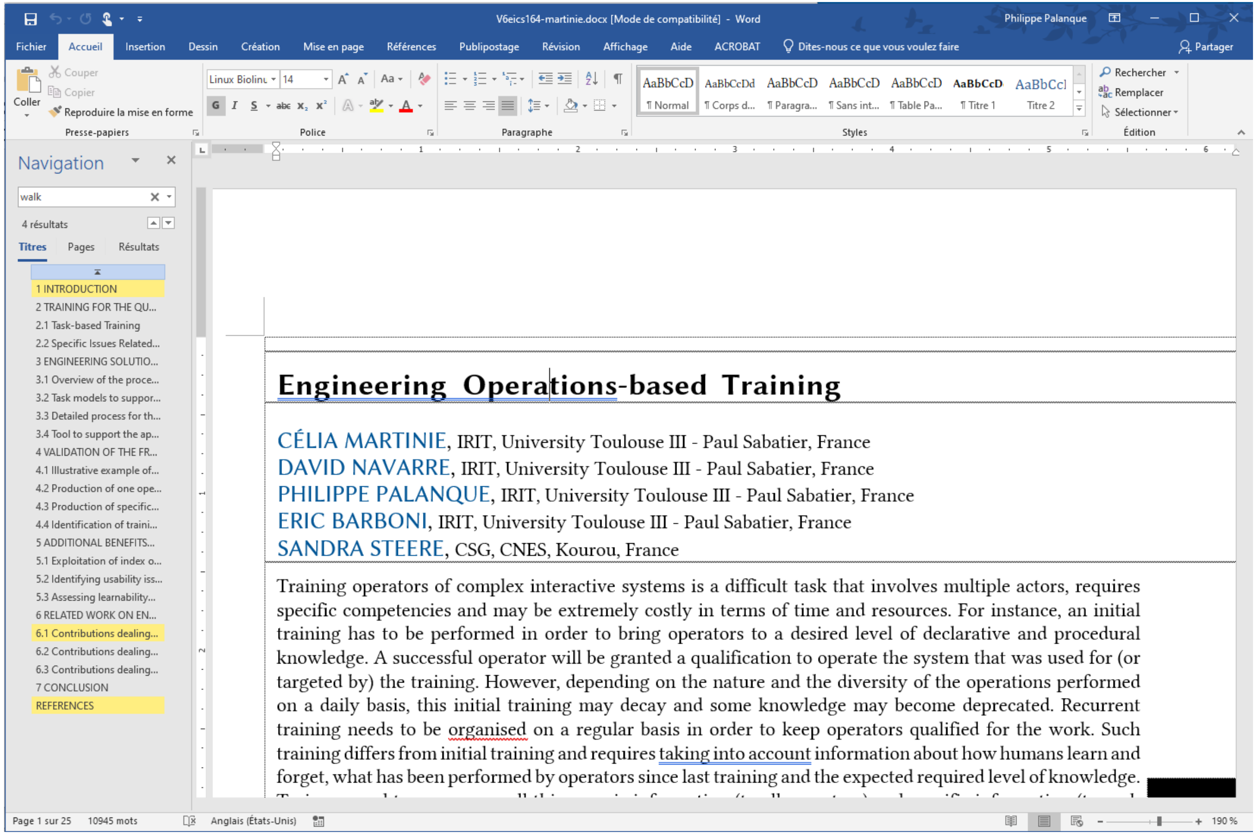
Advanced phone services

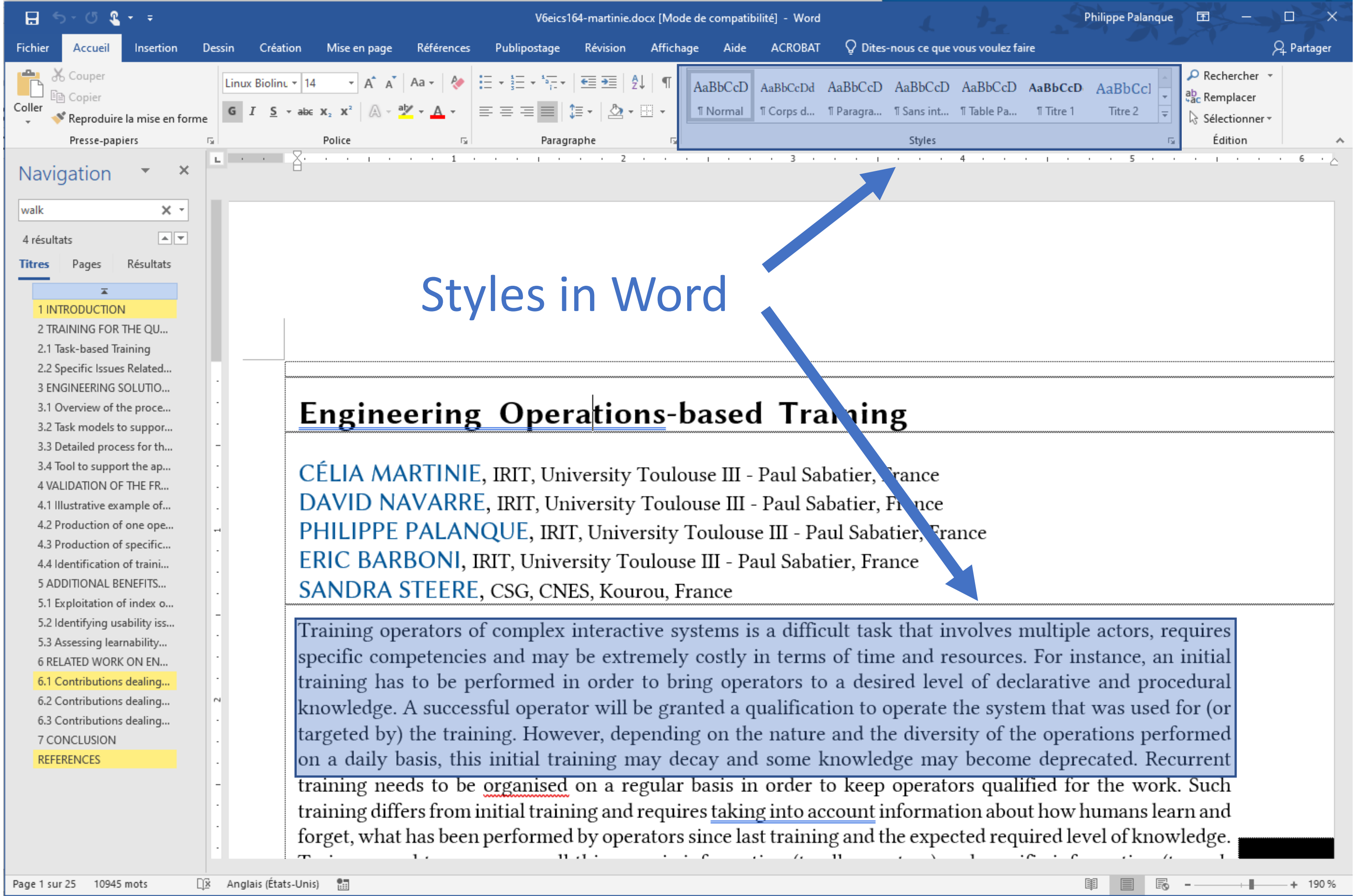
Adva




Software update
Update ready to install
What's new
Schedule install
Install now
Software update
Time left: 00:06:53
What's new
Pause
Software update
Downloading update...
Time left: 00:00:00
What's new
Pause
Software update
Downloading update...
Time left: 00:01:13
What's new
Pause

Computer Technology (e.g. MS Word)

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Philippe Palanque
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Mise en page
Références
Publipostage
Révision
Affichage
Aide
ACROBAT
Dites-nous ce que vous voulez faire
Partager
Couper
Copier
Coller
Reproduire la mise en forme
Presse-papiers
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¶ Titre 1
Titre 2
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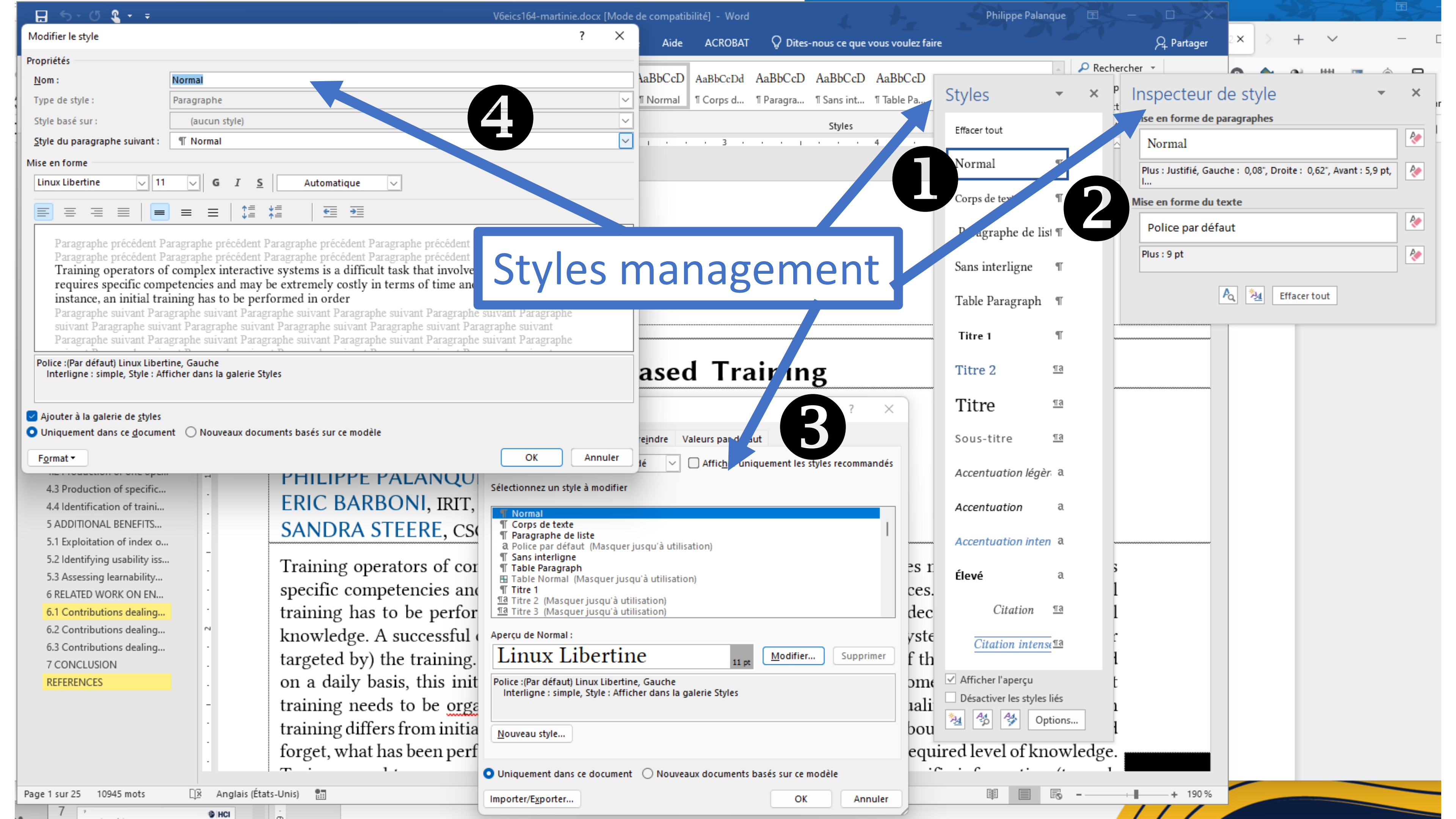
Résultats
1 INTRODUCTION
2 TRAINING FOR THE QU...
2.1 Task-based Training
2.2 Specific Issues Related...
3 ENGINEERING SOLUTIO...
3.1 Overview of the proce...
3.2 Task models to suppor...
3.3 Detailed process for th...
3.4 Tool to support the ap...
4 VALIDATION OF THE FR...
4.1 Illustrative example of...
4.2 Production of one ope...
4.3 Production of specific...
4.4 Identification of traini...
5 ADDITIONAL BENEFITS...
5.1 Exploitation of index o...
5.2 Identifying usability iss...
5.3 Assessing learnability...
6 RELATED WORK ON EN...
6.1 Contributions dealing...
6.2 Contributions dealing...
6.3 Contributions dealing...
7 CONCLUSION
REFERENCES
Styles in Word
Engineering Operations-based Training
CÉLIA MARTINIE, IRIT, University Toulouse III - Paul Sabatier, France
DAVID NAVARRE, IRIT, University Toulouse III - Paul Sabatier, France
PHILIPPE PALANQUE, IRIT, University Toulouse III - Paul Sabatier, France
ERIC BARBONI, IRIT, University Toulouse III - Paul Sabatier, France
SANDRA STEERE, CSG, CNES, Kourou, France
Training operators of complex interactive systems is a difficult task that involves multiple actors, requires specific competencies and may be extremely costly in terms of time and resources. For instance, an initial training has to be performed in order to bring operators to a desired level of declarative and procedural knowledge. A successful operator will be granted a qualification to operate the system that was used for (or targeted by) the training. However, depending on the nature and the diversity of the operations performed on a daily basis, this initial training may decay and some knowledge may become deprecated. Recurrent training needs to be organised on a regular basis in order to keep operators qualified for the work. Such training differs from initial training and requires taking into account information about how humans learn and forget, what has been performed by operators since last training and the expected required level of knowledge.
Page 1 sur 25
10945 mots
Anglais (États-Unis)
190 %



Styles management
4
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Philippe Palanque
Modifier le style
Propriétés
Nom : Normal
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Style basé sur : (aucun style)
Style du paragraphe suivant : ¶ Normal
Mise en forme
Linux Libertine
11
Automatique
Paragraphe précédent Paragraphe précédent Paragraphe précédent Paragraphe précédent
Training operators of complex interactive systems is a difficult task that involve
requires specific competencies and may be extremely costly in terms of time and
instance, an initial training has to be performed in order
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Ajouter à la galerie de styles
Uniquement dans ce document
Nouveaux documents basés sur ce modèle
Format
OK
Annuler
Aide
ACROBAT
Dites-nous ce que vous voulez faire
Partager
Rechercher
Styles
Effacer tout
Normal
Corps de texte
Paragraphe de liste
Sans interligne
Table Paragraph
Titre 1
Titre 2
Titre
Sous-titre
Accentuation légère
Accentuation
Accentuation intense
Élevé
Citation
Citation intense
Afficher l'aperçu
Désactiver les styles liés
Options...
Inspecteur de style
Mise en forme de paragraphes
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Police par défaut
Plus : 9 pt
Effacer tout
Valeurs par défaut
Afficher uniquement les styles recommandés
Sélectionnez un style à modifier
Normal
Corps de texte
Paragraphe de liste
Police par défaut (Masquer jusqu'à utilisation)
Sans interligne
Table Paragraph
Table Normal (Masquer jusqu'à utilisation)
Titre 1
Titre 2 (Masquer jusqu'à utilisation)
Titre 3 (Masquer jusqu'à utilisation)
Aperçu de Normal :
Linux Libertine
11 pt
Modifier...
Supprimer
Nouveau style...
Importer/Exporter...
4.3 Production of specific...
4.4 Identification of traini...
5 ADDITIONAL BENEFITS...
5.1 Exploitation of index o...
5.2 Identifying usability iss...
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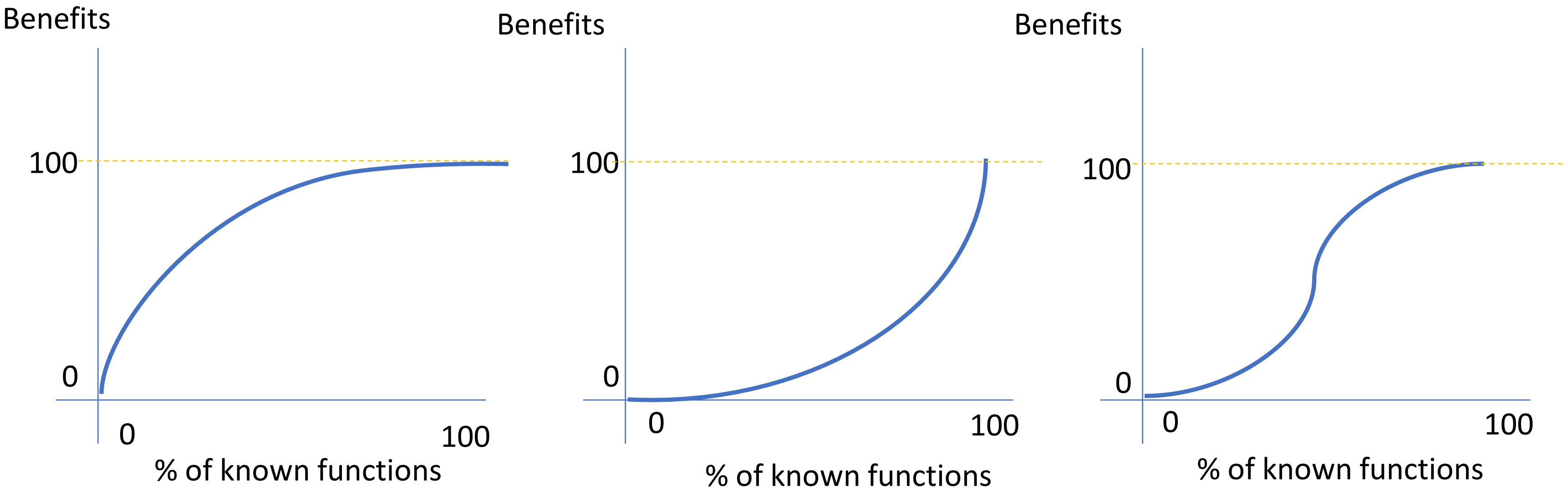
REFERENCES
PHILIPPE PALANQUE
ERIC BARBONI, IRIT,
SANDRA STEERE, CS
Page 1 sur 25
10945 mots
Anglais (États-Unis)
190 %

Styles functionality in MS Word

- A usability function that gathers a set of functions in a single command
  - Increases efficiency
  - Reduces error rates
  - Increases consistency
  - Increases modifiability
- Learning
  - Hard to learn 100% of the functions (e.g. styles inspector)
  - Learning is
    - Usually not done
    - Exploratory
    - By Proxy / osmosys

Styles functionality in MS Word

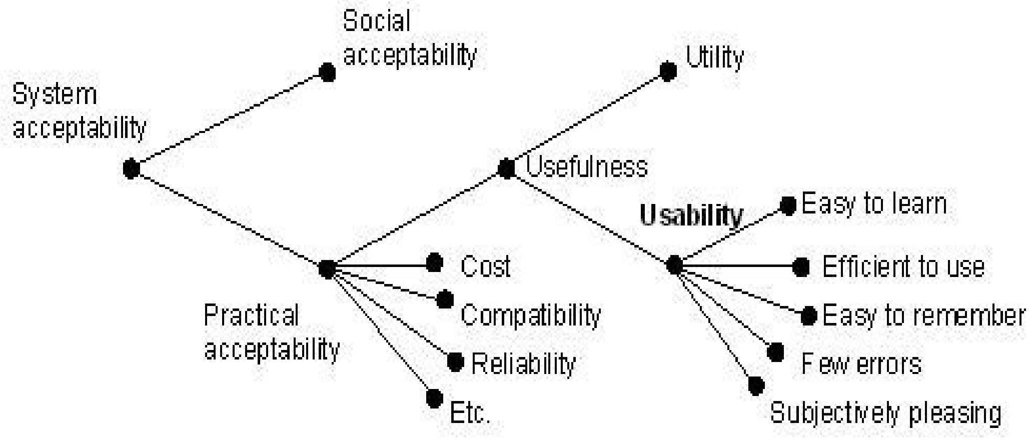


Outline of the presentation

- Short introduction to usability definitions and discrepencies
- The global POISE framework for interactive systems
- Learning and learnability
- Engineering training
  - Initial training
  - Recurring training
- Conclusions and research challenges

Different definitions of usability

- Usability from Jackob Nielsen (Nielsen 1992)



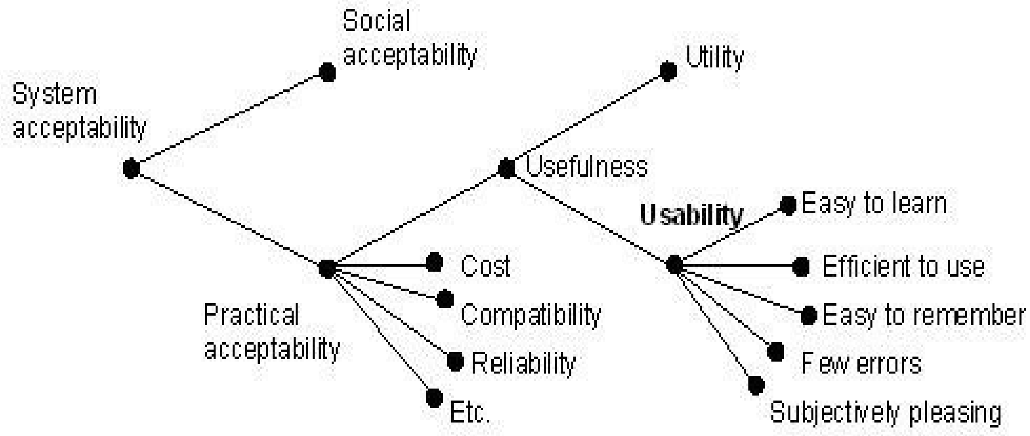
- Usability from ISO 9241 – part 11

ISO 9241 – Part 11 - *Guidance on usability*

ISO 9241 – Usability is The **effectiveness**, **efficiency** and **satisfaction** with which specified users achieve specified goals in particular environments

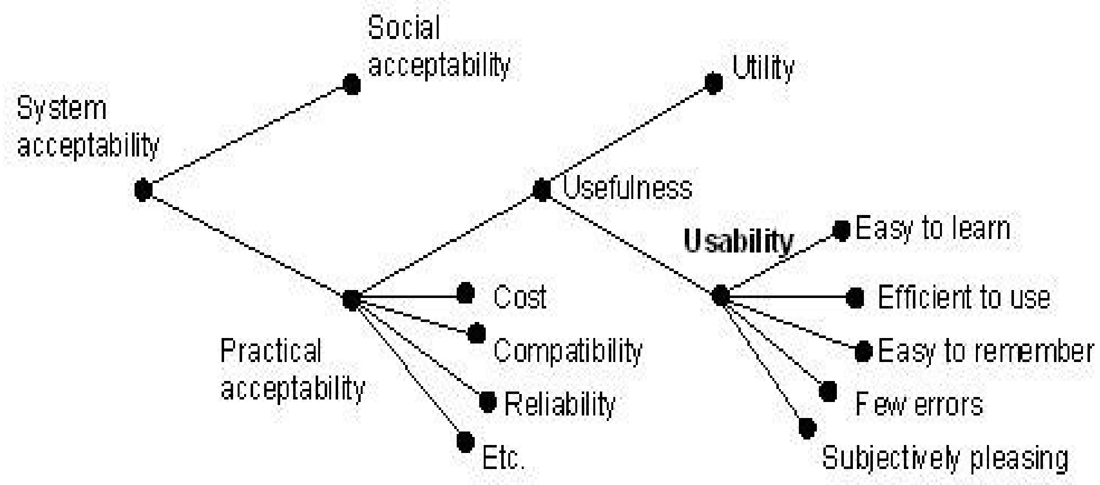
- **Effectiveness**: the accuracy and completeness with which specified users can achieve specified goals in particular environments
- **Efficiency**: the resources expended in relation to the accuracy and completeness of goals achieved (time, error rate, …)
- **Satisfaction**: the comfort and acceptability of the work system to its users and other people affected by its use

What is the difference?



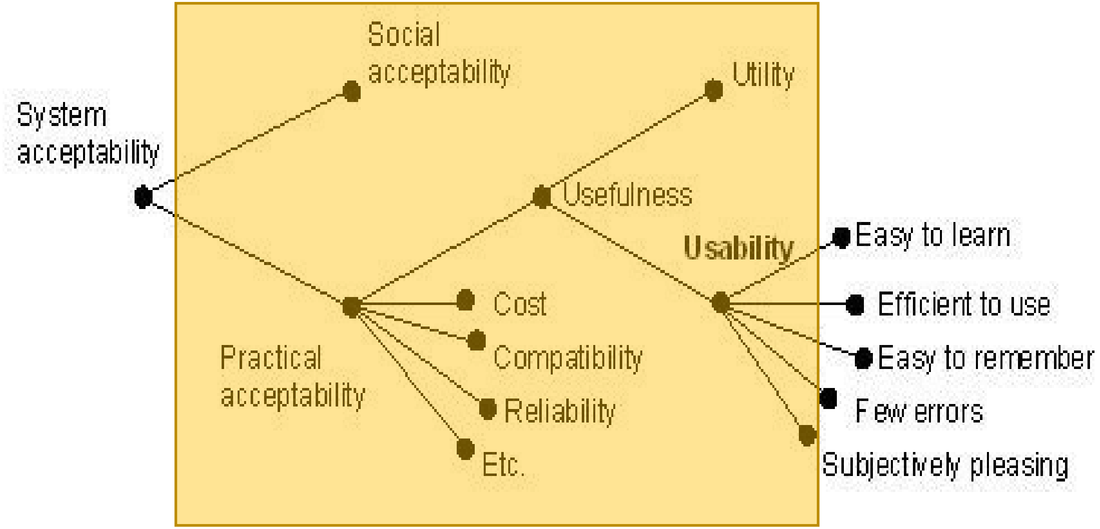
What is the difference?

- First: usability within a braoder context



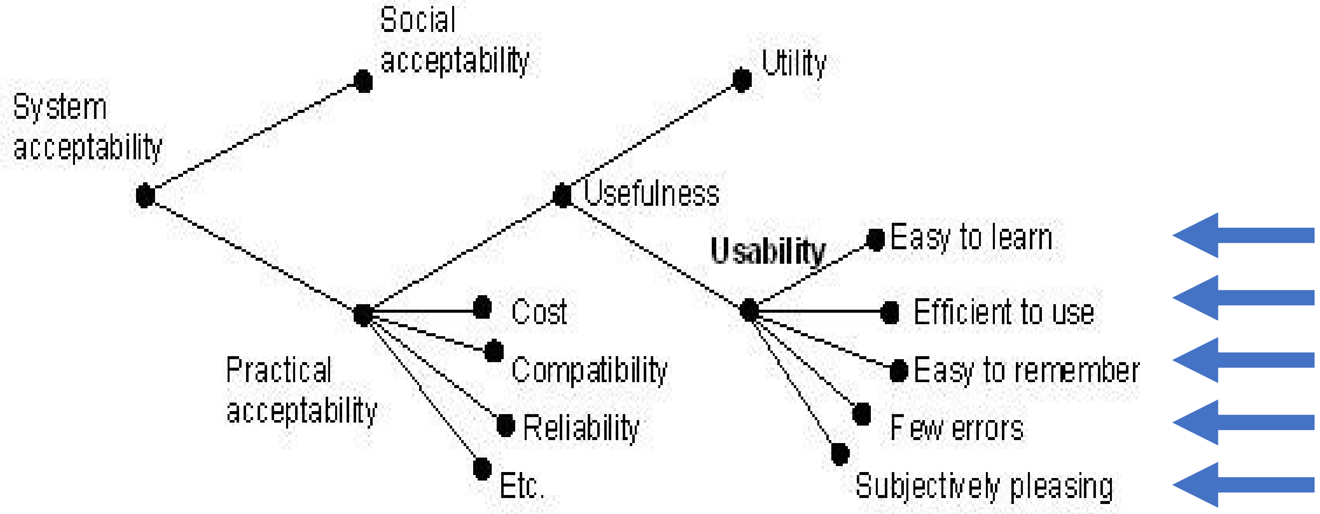
What is the difference?

- **First**: usability within a braoder context



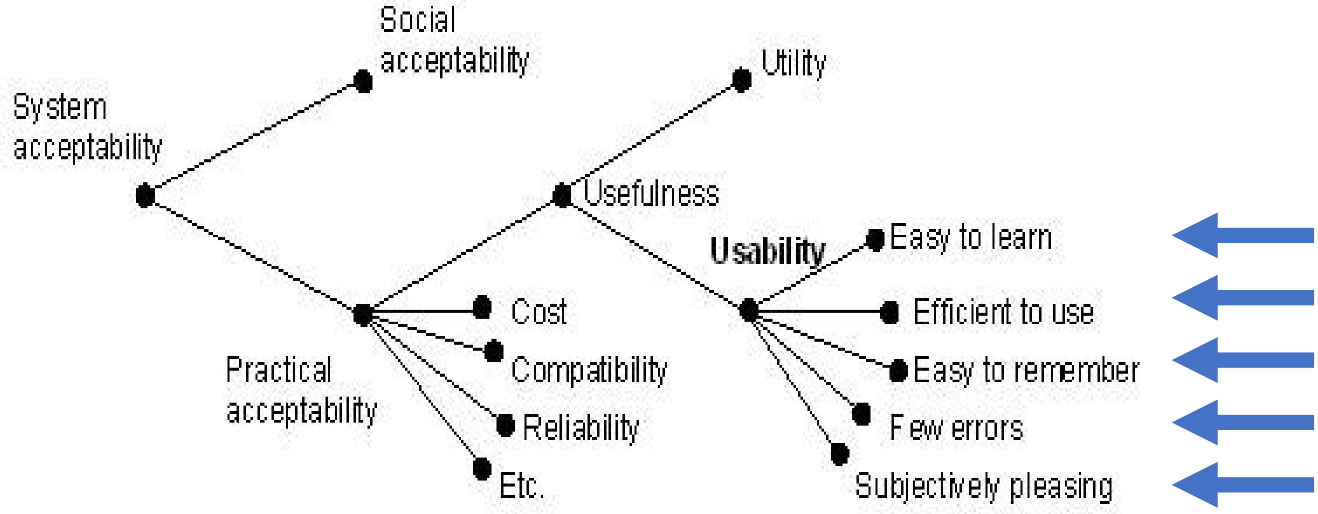
What is the difference?

Second: more contributing factors to usability



What is the difference?

Second: more contributing factors to usability

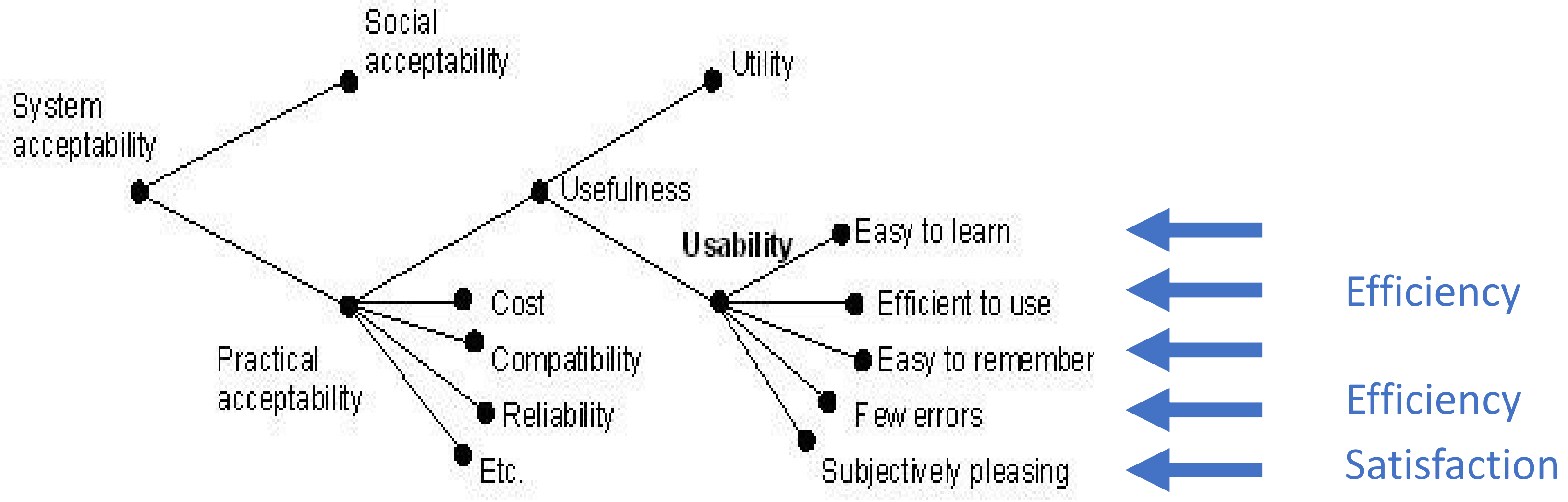


Efficiency

Efficiency

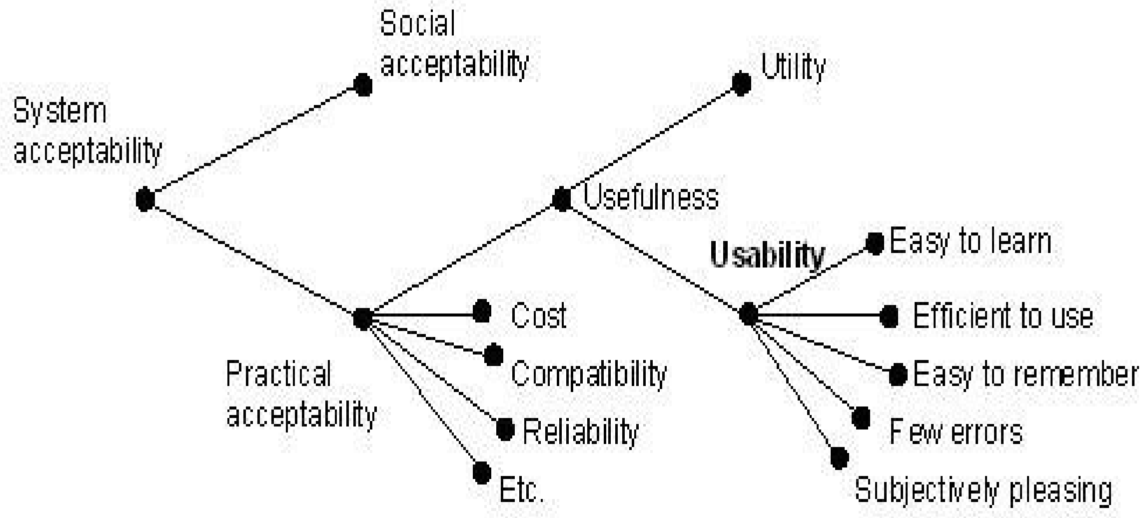
What is the difference?

Second: more contributing factors to usability



What is the difference?

Second: more contributing factors to usability



Effectiveness ?

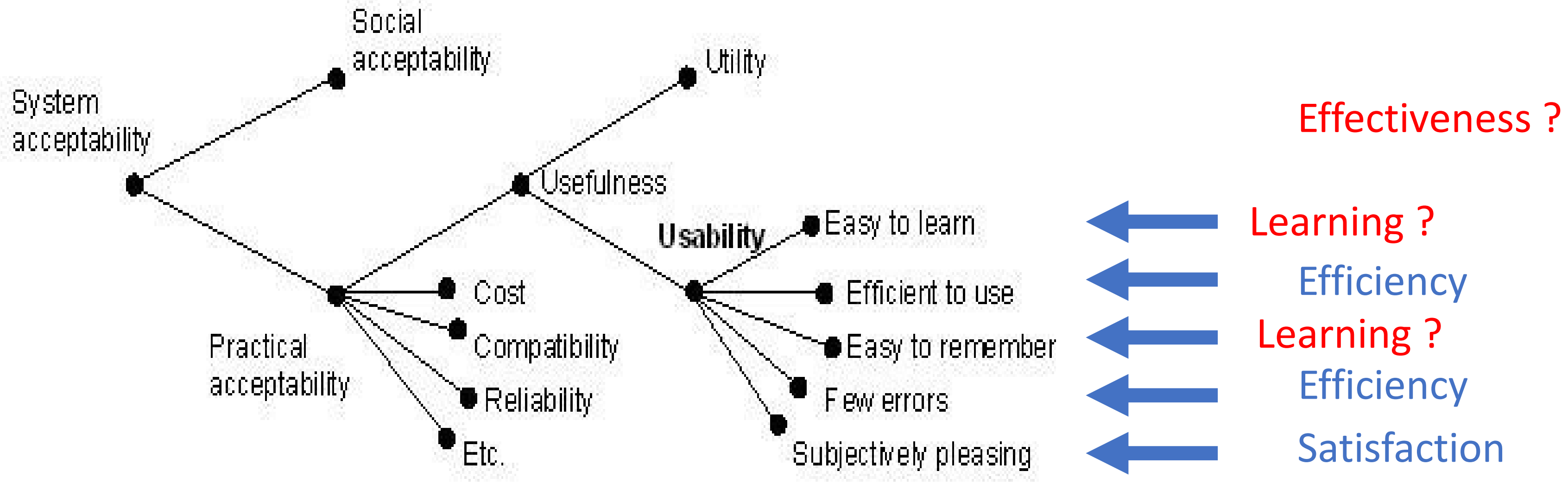
Efficiency

Efficiency

Satisfaction

What is the difference?

Second: more contributing factors to usability



Why did learning disappear?

- HCI research focusses on simple systems (web applications, …)
- HCI research moved away from system supporting people work (Computer Supported Cooperative Work discussing the removal of "work" in their title … more into Social Computing see ACM ToSC
- Dealing with work requires a more complex understanding of users

HCI researchers and practitioners believe in the Graal:

a well designed system should be usable without learning how to use it

Learnability property

- **Synthesizability** — the user can evaluate (and gather knowledge from) the effect of his/her past actions with the system
- **Predictability** — determinism of the user interface and visibility of available functions
- **Familiarity** — fit between system interface and expectations from user previous usage of systems (déjà vu)
- **Consistency** — probability that the behavior of the interface (input and output) is the same in similar contexts/situations
- **Generalizability** — possibility to extend what has been learnt to new situations/contexts/tasks

Learnability with a usability perspective

- **Effectiveness *of training***

  Percentage of system functions learnt

  Percentage of tasks learnt

  Percentage of knowledge and skills acquired

- **Efficiency *of training***

  Time spent in the activity of learning

  Number of re-training until qualification

- **Satisfaction *of training***

  Subjective evaluation of ease of learning

  Satisfaction during the training program execution

  Satisfaction from the training outcome (knowledge, skills, …)

Learnability

- **Effectiveness**

  Percentage of functions learnt

- **Efficiency**

  Time spent in the activity of learning

- **Satisfaction**

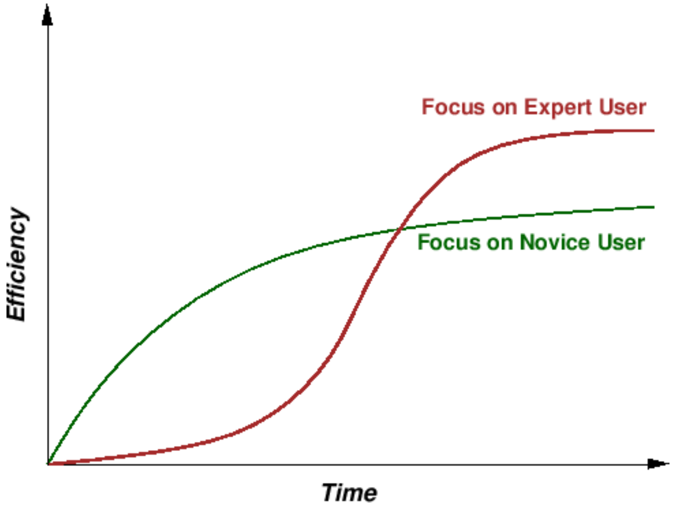
  Subjective evaluation of ease of learning

Continuous and non monotonous value

Quantity of memorization/recall after 1 day, 1 week, …

Learnability perspectives

- Focus on effectiveness (red curve)
- Focus on relative efficiency (green curve)



Learnability perspectives

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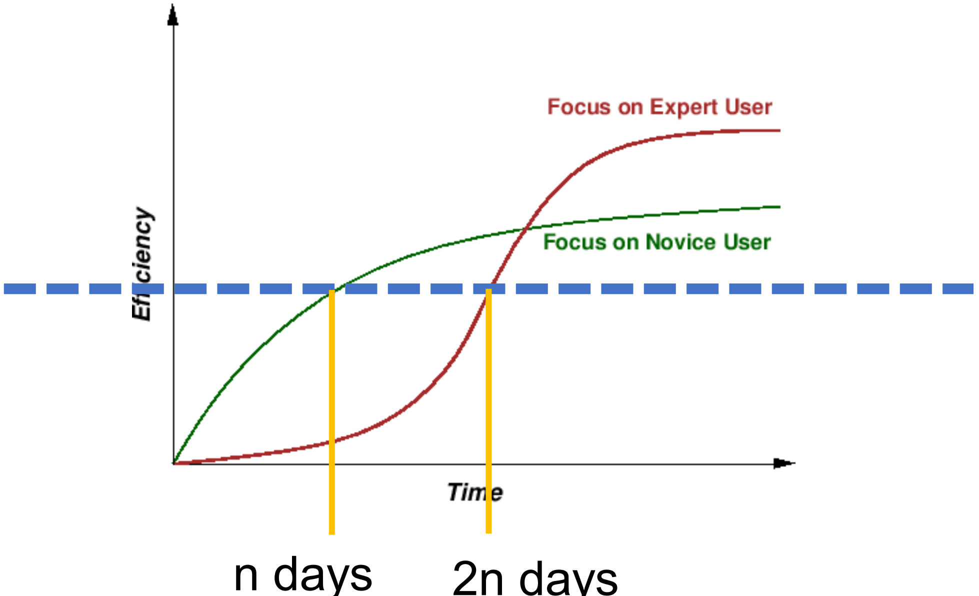
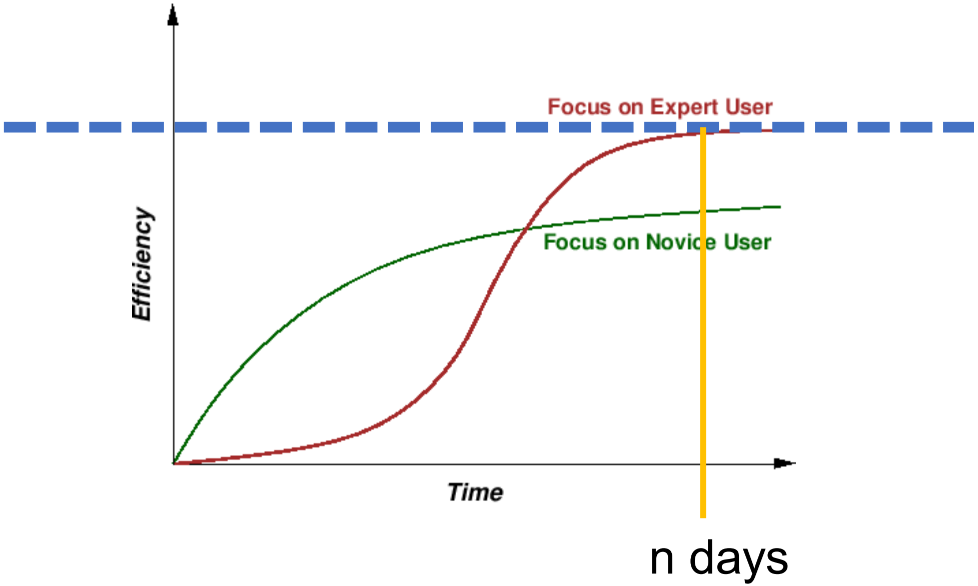


User is able to use part of the system

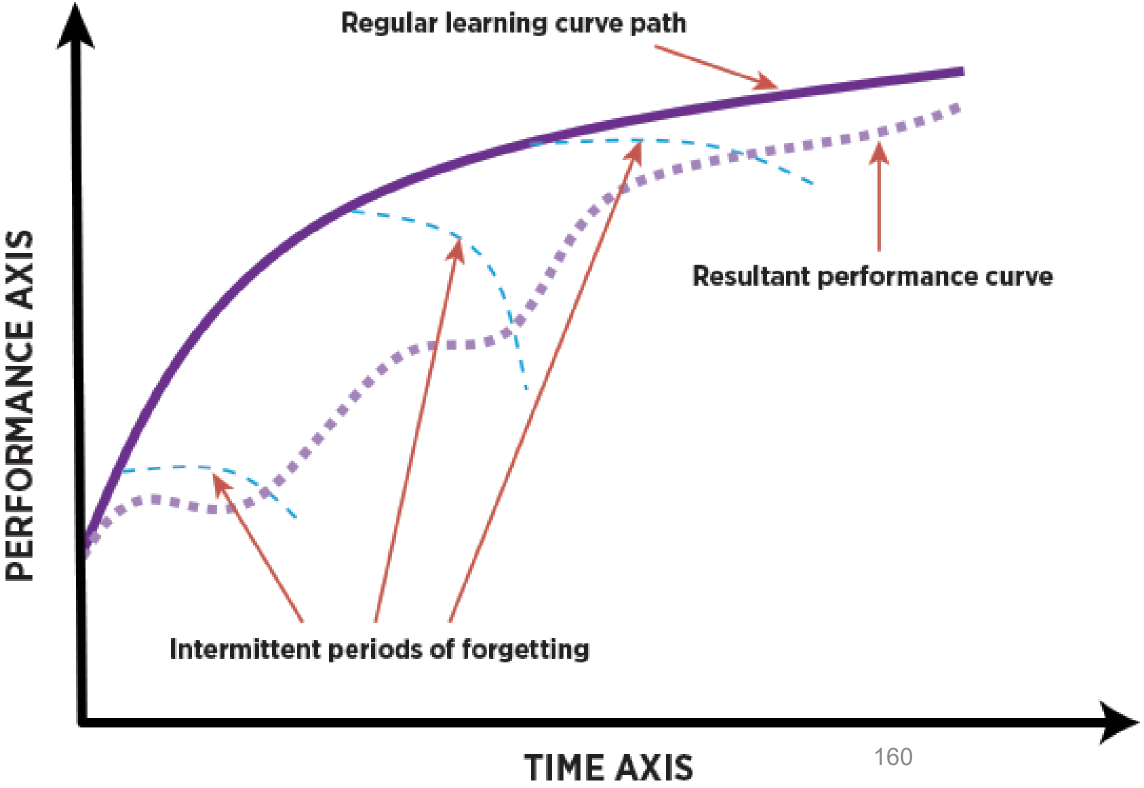
Learnability perspectives

- Focus on effectiveness (red curve)
- Focus on relative efficiency (green curve)



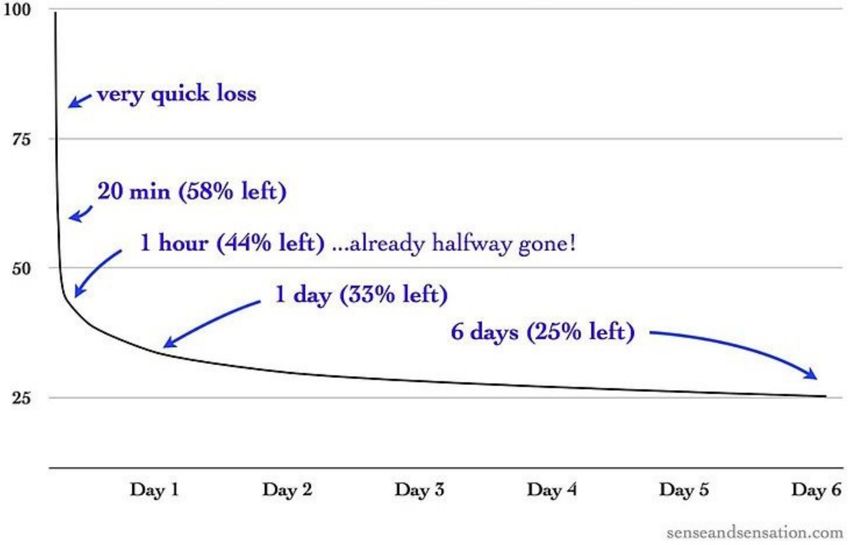
User is able to entirely use the system

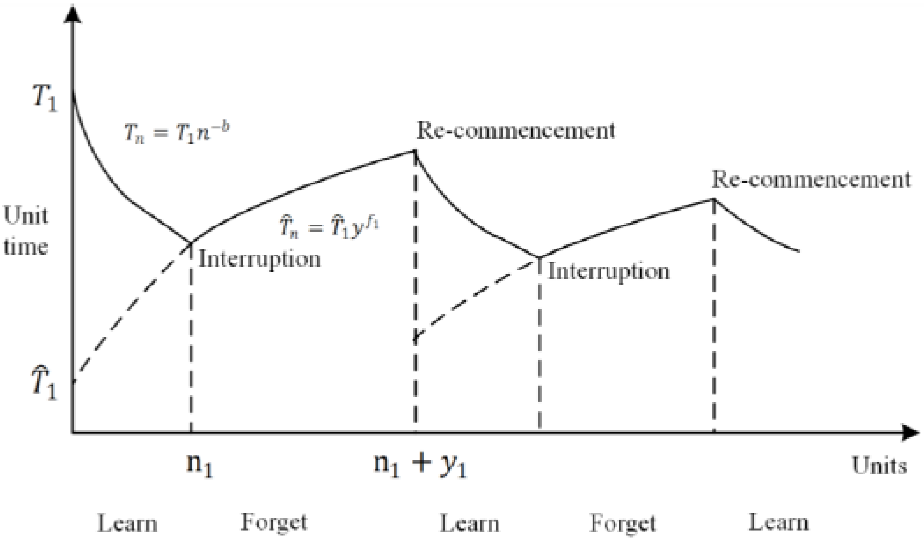
Reality of Learnability



Learning-forgetting curve

- Forgetting curve Ebbinghaus 1913
- Learning-forgetting curve Jaber & Bonney 1997





Ebbinghaus H. 1913. Memory: A contribution to experimental psychology. *H. A. Ruger & C. E. Bussenius, Trans.* (1913).
Mohamad Y. Jaber, Maurice Bonney, A comparative study of learning curves with forgetting, Applied Mathematical Modelling, Volume 21, Issue 8, 1997, Pages 523-531, ISSN 0307-904X,

Let’s Compare Two Systems

≠

different yes, but how?

different yes, but which one is better?

if 2 is better, how to design training?

Outline of the course

- Context and definitions
  - Usability
  - User Experience
- Designing for Usability
- Designing for User Experience
- Measuring Usability and User Experience
- Going further
  - Acceptance
  - Learnability
- Specificities of command and control systems (real systems)
- Take away message

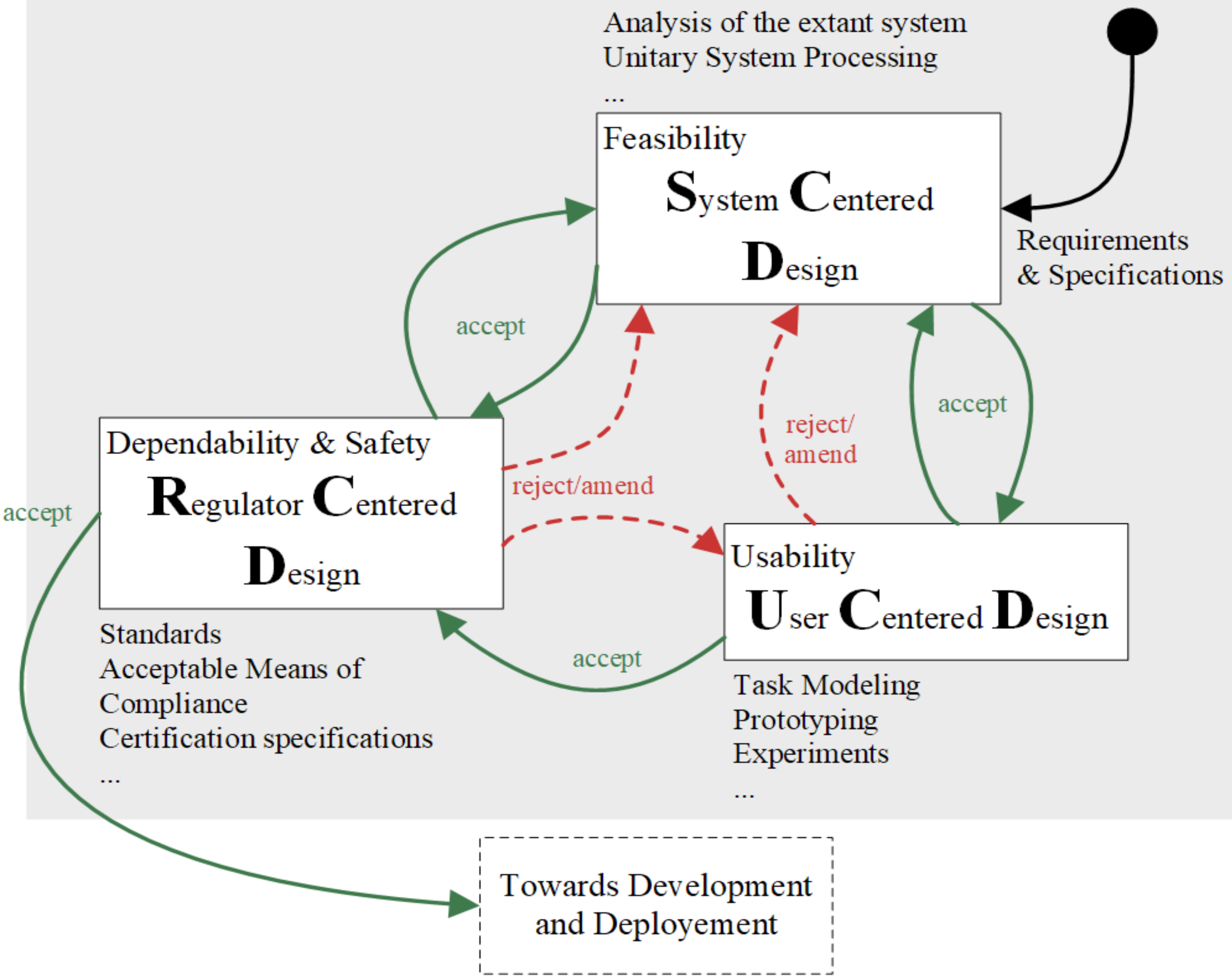
Specificities of command and control systems (1/2)

- Complexity of the system must be understood
- Complexity of the system must inform the user interface design
- Regulations must be accounted for
  - In term of certification (e.g. CS 25.1302)
  - In terms of interactive system development (e.g. DO 178C)

Elodie Bouzekri, Alexandre Canny, Célia Martinie, Philippe A. Palanque, Christine Gris: **Deep System Knowledge Required: Revisiting UCD Contribution in the Design of Complex Command and Control Systems**. INTERACT (1) 2019: 699-720

Specificities o

- Complexity o
- Complexity o
- Regulations r
  - In term of c
  - In terms of



Large variety of tasks structured around a mantra

Fly

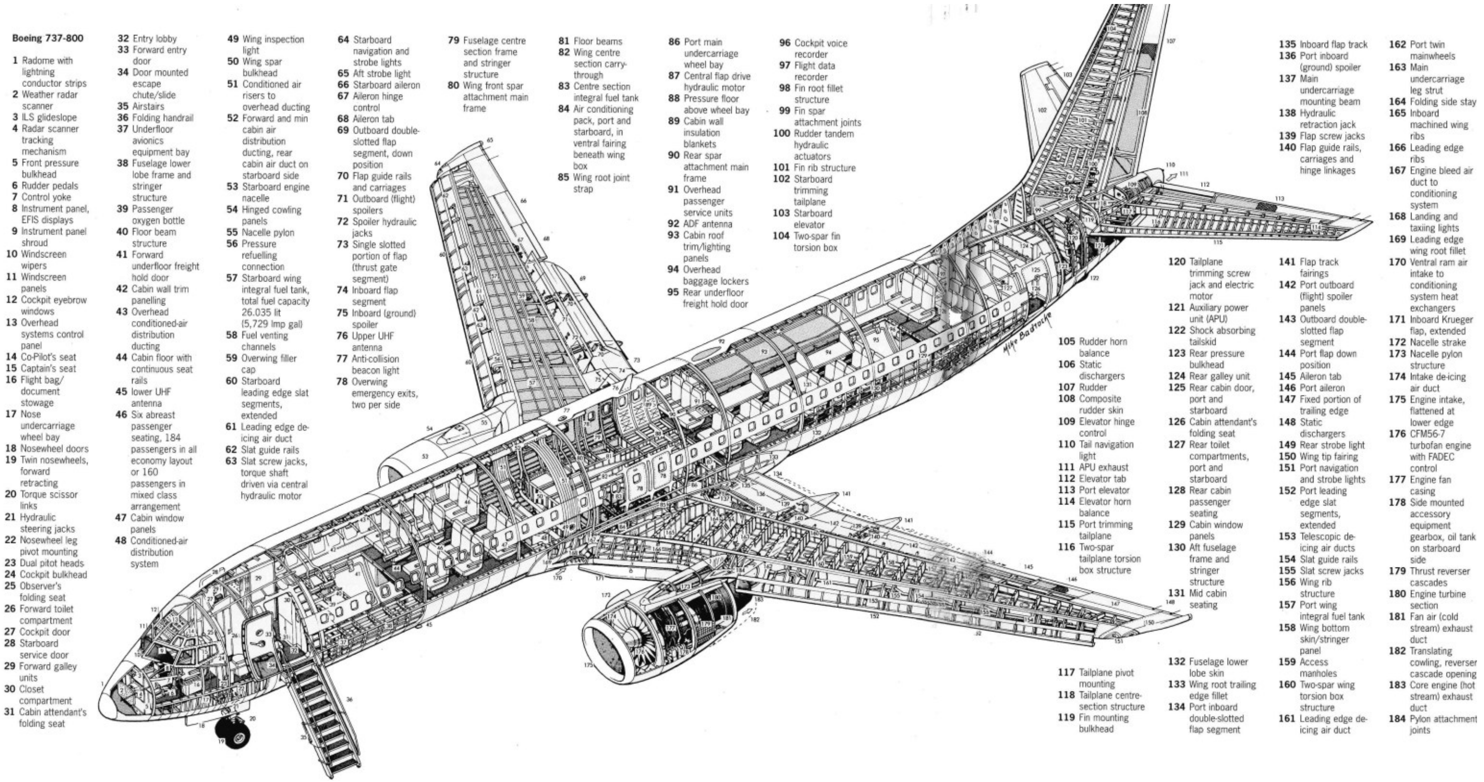
Flight crew deal with many activities

Navigate

Communicate

Manage Systems

Manage (a lot of) systems

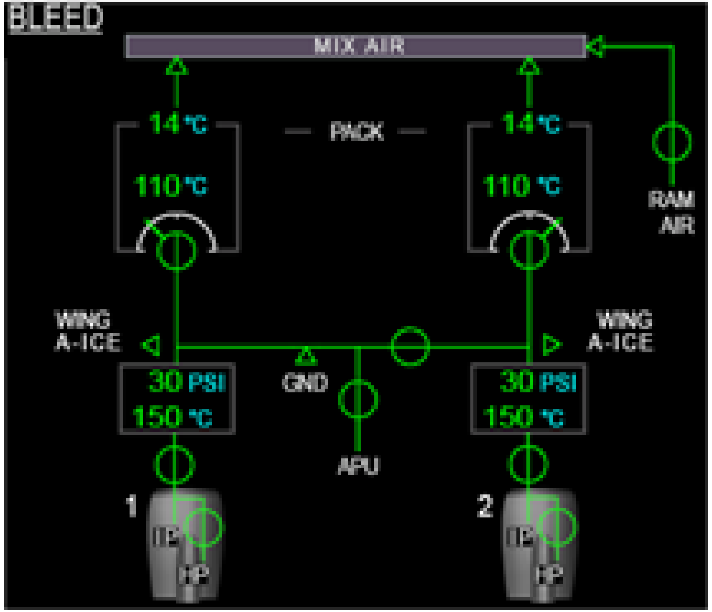


FWS in the cockpit

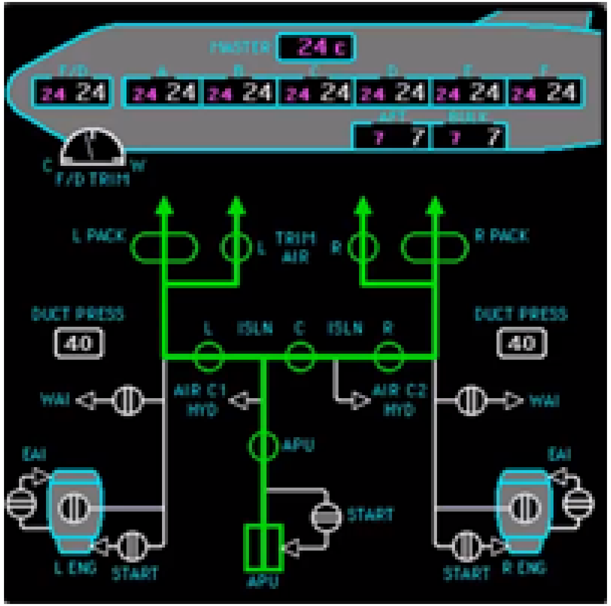

ON
FIRE
1
FAULT
OFF
ENG START
NORM
CRANK
IGN
START
ON
FIRE
2
FAULT
OFF
T.O
CONFIG
DEFRD
ABN
RCL
LAST
RCL
ALL
DISPCH
STS
MORE
SD
ENG
DOOR
WHEEL
FUEL
VIDEO
CLEAR
EL/AC
EL/DC
HYD
F/CTL
VIDEO
CLEAR
ALL
APU
BLEED
COND
PRESS

System-specific pages

Airbus A350 BLEED SD Page

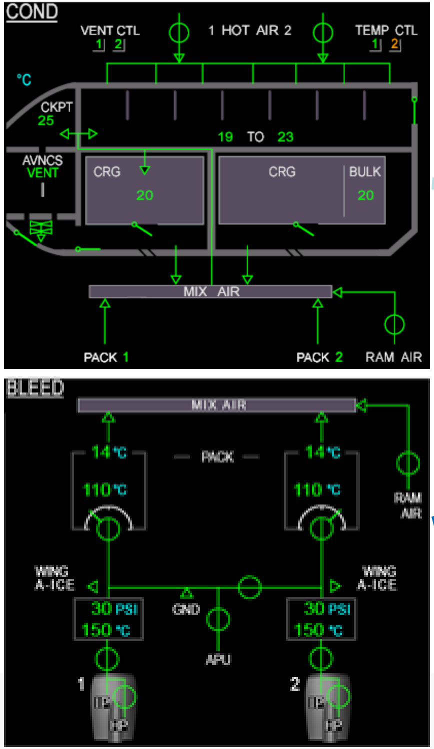


Boeing 777 AIR Synoptic

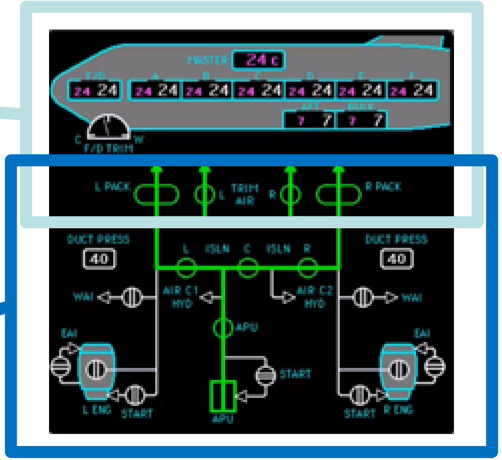


System-specific pages

Airbus A350
COND+BLEED SD Pages



Boeing 777 AIR Synoptic

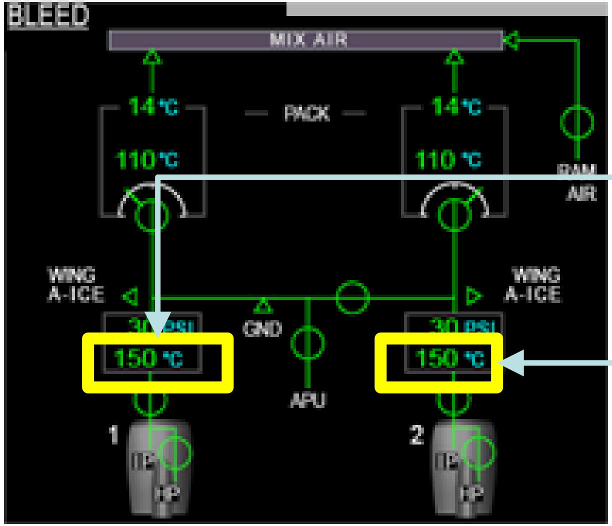


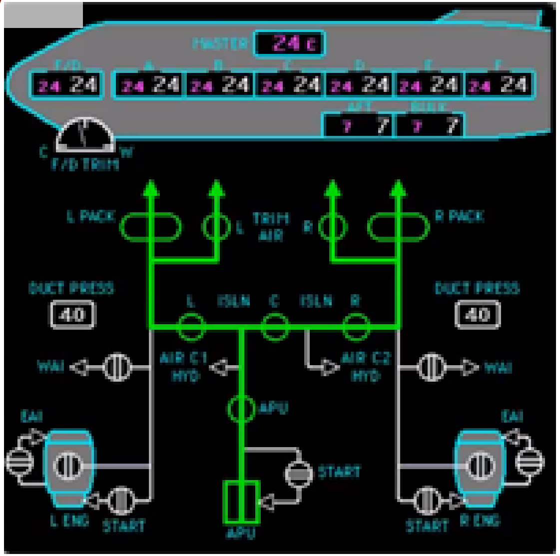
System-specific pages

Airbus A350 COND+BLEED SD Pages

Boeing 777 AIR Synoptic

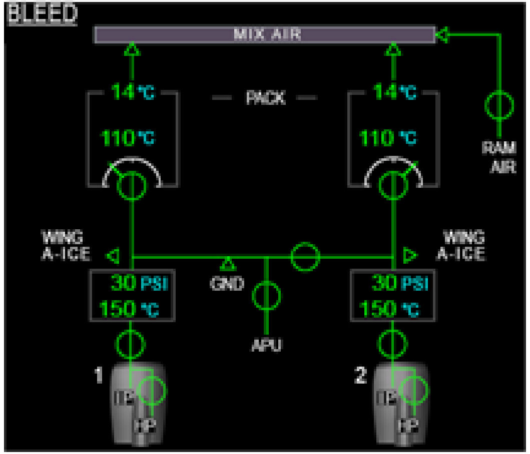
BLEED AIR temp on A350 Only



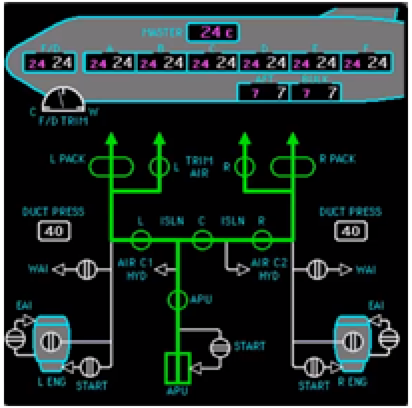


System-specific pages

Airbus A350
COND+BLEED SD Pages



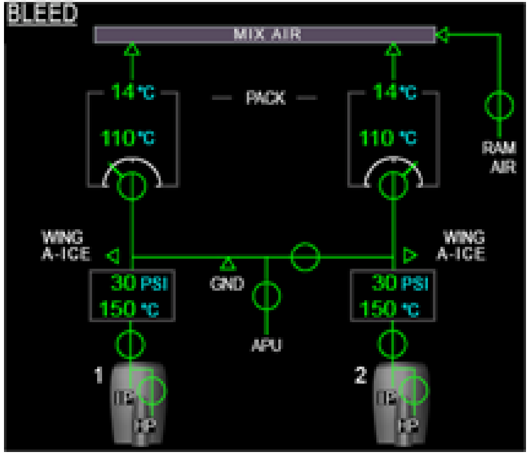
Boeing 777 AIR Synoptic



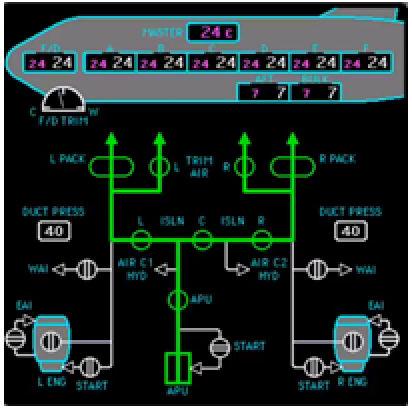
Is bleed air temp relevant to the flight crew? Is the cabin temp more relevant to the flight crew?

System-specific pages

Airbus A350
COND+BLEED SD Pages



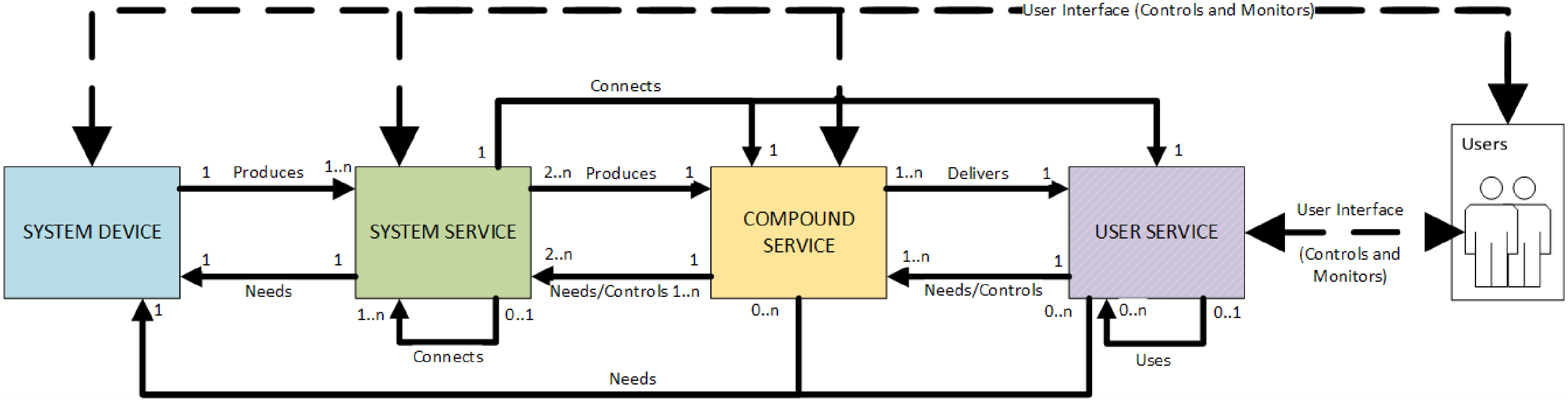
Boeing 777 AIR Synoptic



Is bleed air temp relevant to the flight crew? Is the cabin temp more relevant to the flight crew?

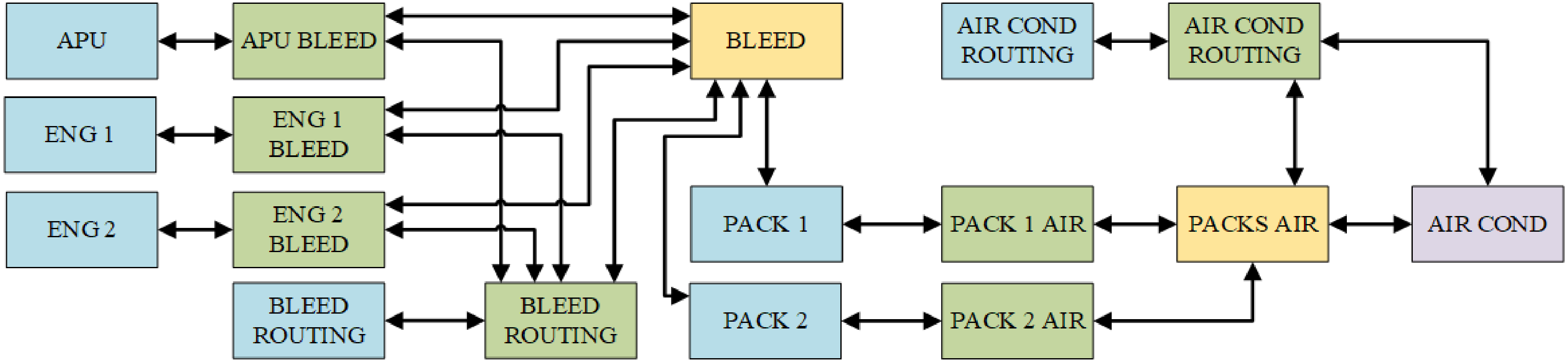
System Centered Design

- A generic architecture based on 4 main types of components



Outcome of System Centered Design

- Views on systems and their relationships



Outline of the course

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  - Acceptance
  - Learnability
- Specificities of command and control systems
- Take away message

The Problem: Limitations of UCD

- Learnability is out of the loop of the design process while
  - Learning is key for system deployment
  - Learnability has to be assessed during the system development (a hard to learn system might require redesign to make training more efficient) e.g. Boeing reported spending 17% of the training on 4% of the functions (auto-pilot) issues
  - Learning never comes from scratch but differential training is used
  - Airbus success come (in part) from cross qualification designs
- How to support the design of muliple UIs for supporting users when they are evolving e.g. from novice to intermediate to expert?
- How to assess that the same application on different platform is the same or at least how much it is different?
- How to design an application which looks like and Apple app but is still different from an Apple app (to avoid going to trial)

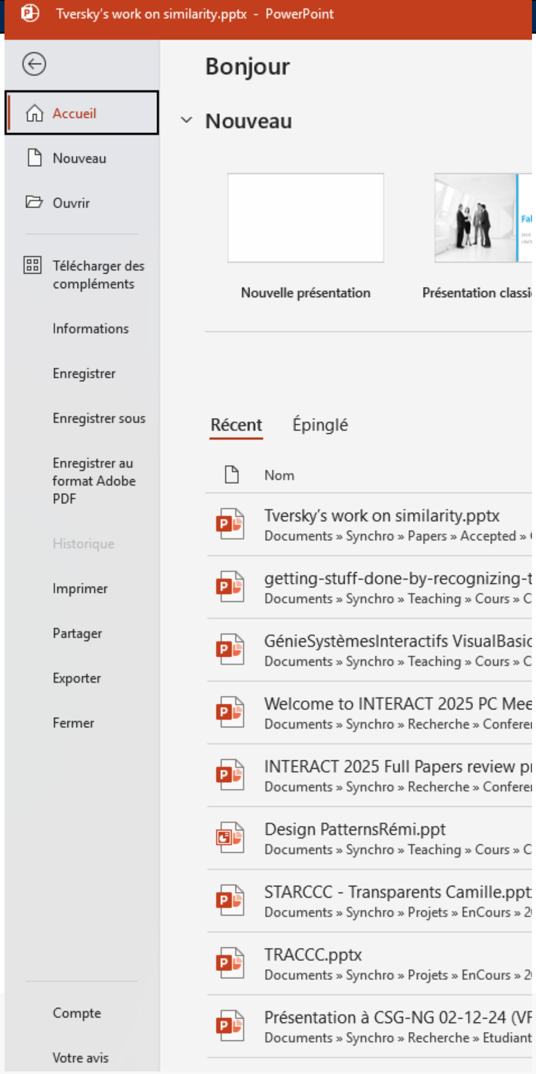
Differential training

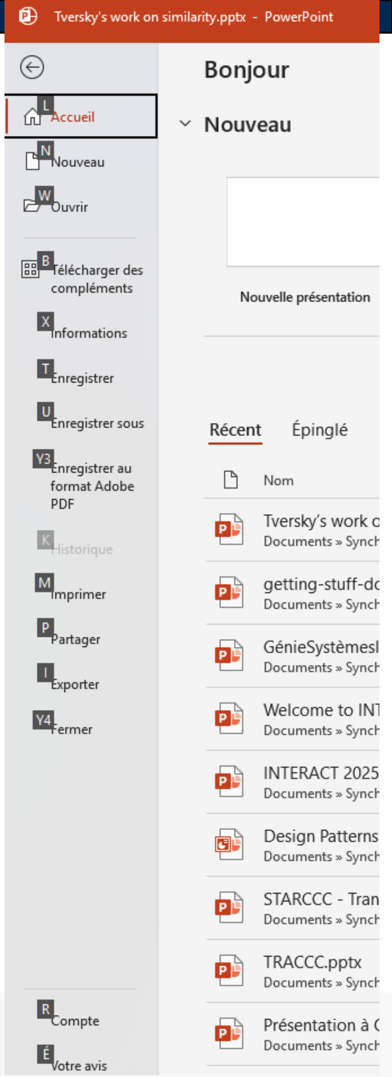
Cross-Crew Qualification (CCQ)

- CCQ is a type of transition training based on the principle of **commonality**, which is the high degree of similarity in cockpit layout, handling characteristics, and operational procedures across an aircraft family
- CCQ focuses only on the **differences** between the "base" aircraft (the one they're already qualified on) and the " candidate" aircraft (the new one they want to fly)

Support users evolution

Expert operator VS novice operator (interaction level)





Evolution of users

Differential training: Operator Difference Requirement (ODR) Table

4. A320ceo Family with Sharklet to A320 Family neo – Maneuvers – Abnormal Operations (per flight phase)

No difference in maneuvers/procedures in abnormal operations per flight phase.

5. A320ceo Family with Sharklet to A320 Family neo – Maneuvers – Abnormal Operations (per ATA chapter)

SYSTEMS	DIFFERENCES	FLT CHAR	PROC CHG	DIFFERENCE LEVELS TRNG	CHECK	CUR
27 – Flight Controls	- On the A319neo aircraft only: • For the malfunctions implying the loss of yaw damper (AUTO FLT FAC1 + 2 FAULT and AUTO FLT YAW DAMPER SYS), the ECAM displays "ALTN LAW: PROT LOST". However, the active law for both pitch and roll will remain NORM, the active law for yaw will be MECH. As a consequence, the flare mode will remain active for landing and the ECAM will not display the message "WHEN L/G DN: DIRECT LAW". Note: for the same feature on A320 and A321, refer to the ODR table Sharklet.	No	Yes	A	A	B 12 m
70 – Engines	The A320neo family aircraft have the following additional abnormal procedures: - All A320neo family aircraft: • ENG 1(2) CTL SYS FAULT • ENG 1(2) FUEL LEAK • ENG 1(2) HIGH VIBRATION. - Aircraft equipped with PW1100G engines: • ENG 1(2) AIR SYS FAULT • ENG 1(2) BOWED ROTOR PROTECTION FAULT.	No	Yes	A	A	B 36 m

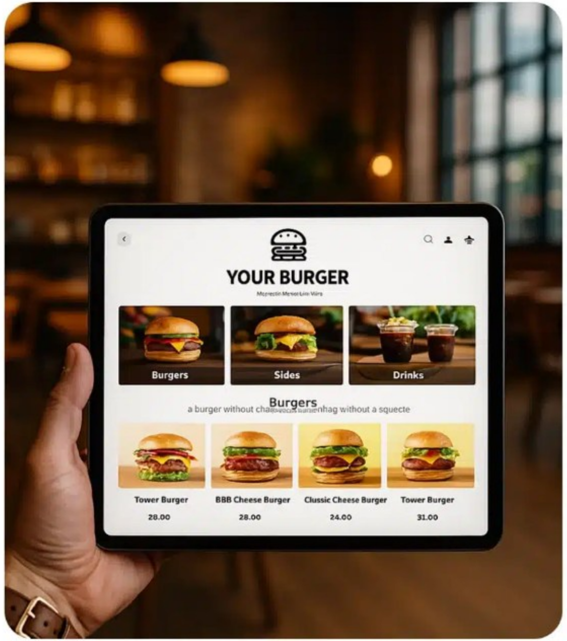
6. A320ceo Family with Sharklet to A320 Family neo – Training Areas of Special Emphasis (TASE)

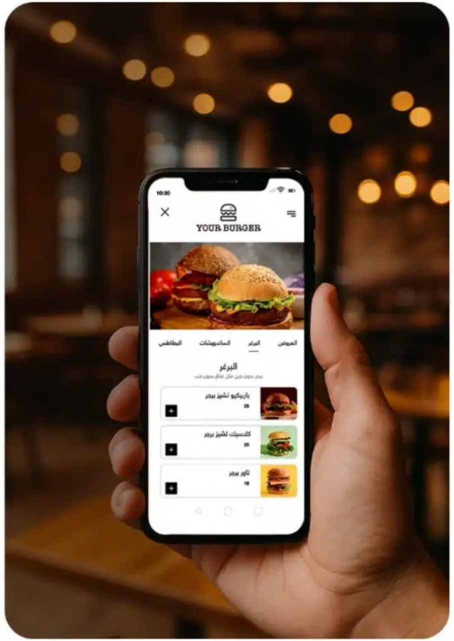
No specific Training Areas of Special Emphasis.

Difference of the same app on different platforms

Platform (visual appearance and layout level)

What about interaction, behaviour, structure, shortcuts … and taking benefits of the specificities of the platform (stylus)





Similar but different: counterfeit

Apple vs Samsung

Apple initially accused Samsung of infringing:

- `381 patent: touchscreen interactions
- `915 patent: using an API to scroll through documents
- `163 patent: tap-to-zoom
- `677 patent: general outline and ornamental design
- `087 patent: general outline and ornamental design
- `889 patent: ornamental design
- `305 patent: GUI for a display screen

Samsung countersued, accusing Apple of infringing:

- `941 patent: mobile phone 3G capabilities
- `516 patent: mobile phone 3G capabilities
- `711 patent: MP3 playback on a mobile device
- `460 patent: device that functions as a phone and a camera
- `893 patent: method for remembering user's place in a gallery

Samsung triumphs over Apple in US Supreme Court patent row

6 December 2016

The patent row between Apple and Samsung began in 2011

The US Supreme Court has sided with Samsung in the South Korean company's long-running patent fight with Apple.

The court rejected an earlier ruling that Samsung must pay $399m to Apple for copying some iPhone designs.

Objective of the work: understand similarities of interactive systems

- On the theoretical side have a clear understanding of the similarity property (a property which is accross systems)
- On the practical side understand the similarities in the context of interactive systems (generic architecture side)
- On the application side provide a generic description schema to capture the "relevant" features of an interactive application (UI objects, properties of the objects, …) to assess similarity between 2 or more Uis

Navarre, David, et al. Similarity as a design driver for user interfaces of dependable critical systems. IFIP Conference on Human-Computer Interaction. Workshop proceedings: Springer, 2017.

Bouzekri, Elodie, et al. Characterizing Sets of Systems: Representation and Analysis of Across-Systems Properties. INTERACT conf. on Human-Computer Interaction. Springer, 2019.

How we propose to address the problem

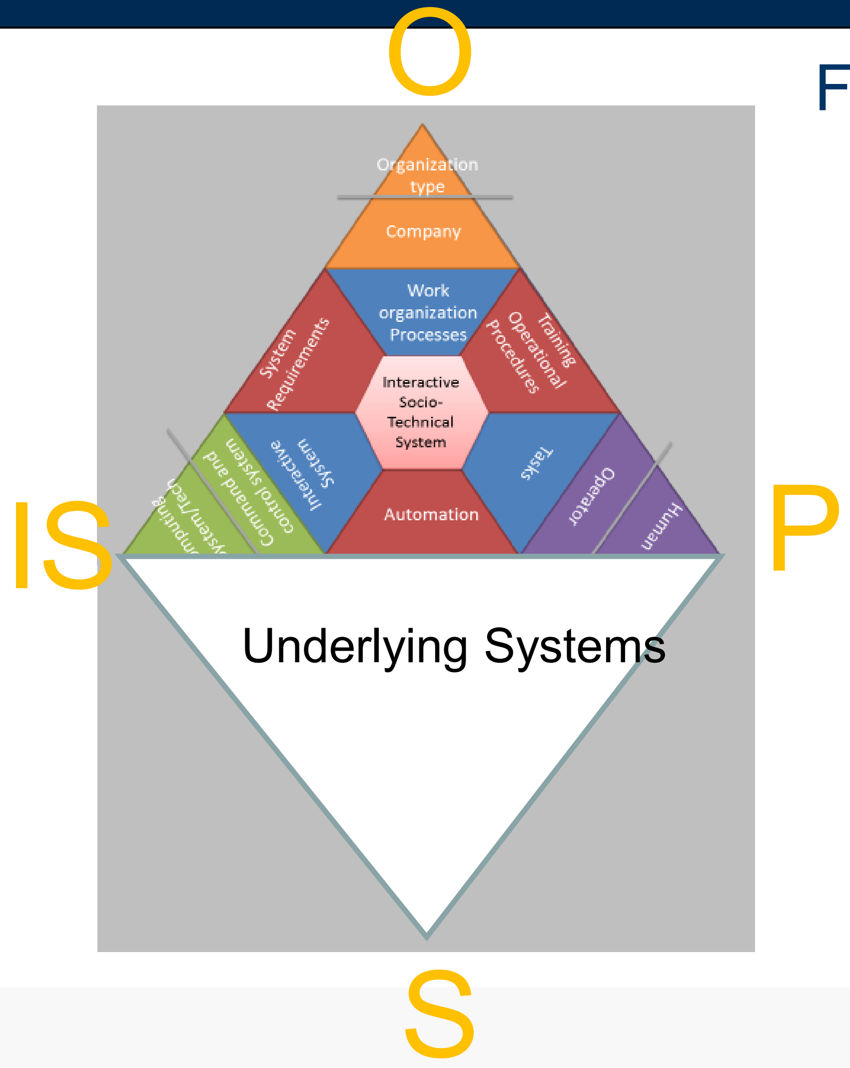
- Similarity is processed folowing two concepts
  - Features of an interactive system
  - "Features Matching" according to Twersky's work on similarity

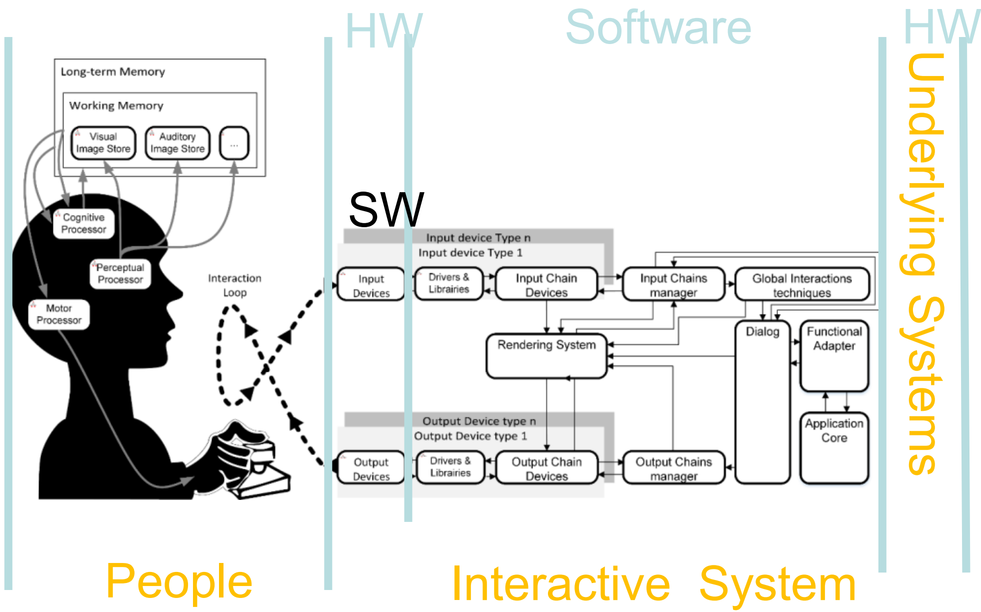
    Tversky, A. (1977). Features of similarity. *Psychological Review, 84*(4), 327–352.
- Features are split in two parts
  - POISES systemic framework for understanding socio-technical systems to decompose the problem into People aspects, Organization aspects, Interactive Systems aspects, Environmental aspects and underlying System aspects
  - Interactive Systems architecture (our target) including Input Devices, Output Devices, Interaction Techniques and the interactive application (dialog and functional core)

Palanque, P. POISE: a framework for designing perfect interactive systems with and for imperfect people. IFIP Conference on Human-Computer Interaction. Cham: Springer International Publishing, 2021.

Cronel, M. et al. MIODMIT: a generic architecture for dynamic multimodal interactive systems. International Conference on Human-Centred Software Engineering. Springer, 2018.

Features of Similarity





Outline of the presentation

- On the need for multiple interactive system variants (identified needs, existing related work and limitations)
- The design context of interactive system variants for the Launch Vehicle Flight Safety Operations (POISE framework application)
- Our generic solution : the Multi-variant UCD process (rapid presentation of the process)
- Application example of the Multi-variant UCD process : the Ariane 6 and CALLISTO UI prototypes
- Conclusion and lessons learnt

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- Conclusion and lessons learned
- Future work

The design context of Launch Vehicle Flight Safety Operations

CENTRE SPATIAL GUYANAIS

Port spatial de l'Europe
Europe's Spaceport

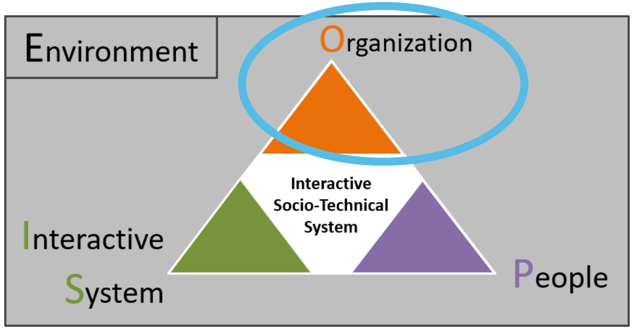
esa arianespace arianeGROUP cnes

- Europe Spaceport is undergoing a significant renewal, notably of its Core Launch Range system and services

- In particular the **interactive socio-technical** Flight Safety Command and Control System (CCS) :
  - It is essential for:
    - Ensuring the safety of people, property, and the environment from damage caused by in-flight launcher operations.
  - It must be adapted to:
    - Accommodate increased launch frequency
    - Support new types of launch vehicles, such as reusable launchers (e.g. CALLISTO)

- Due to the complexity of this CCS, we can decompose it using the POISE-S framework in …

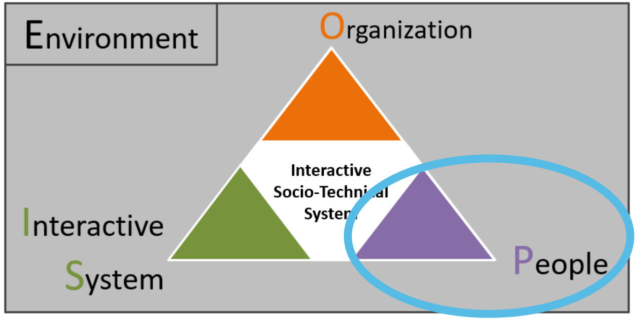
The design context of Launch Vehicle Flight Safety Operations



Organization :

- Several organizations involved on the launch of a space vehicle
- Spaceport leading organization (CNES) is responsible, among others, for the safety of operations
- CNES ensures compliance of the French Space Operations act, describing , among others, the flight safety operations

The design context of Launch Vehicle Flight Safety Operations



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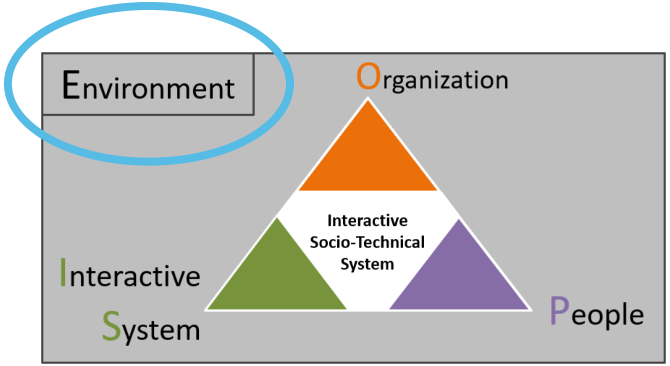
People :

- Flight Safety Operations Team (several operators) -> **focus only on the Flight Safety Operator (FSO)**
- French Space Operations act defines that the FSO must :
  - Monitor the launch vehicle to asses its dangerous nature
  - Intervene if necessary, by sending an order on board
  - FSO is trained for 3 years to be qualified

The design context of Launch Vehicle Flight Safety Operations



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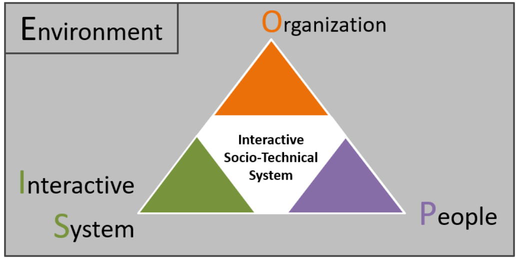
Environment :

- Dedicated room with limited light source

The design context of Launch Vehicle Flight Safety Operations



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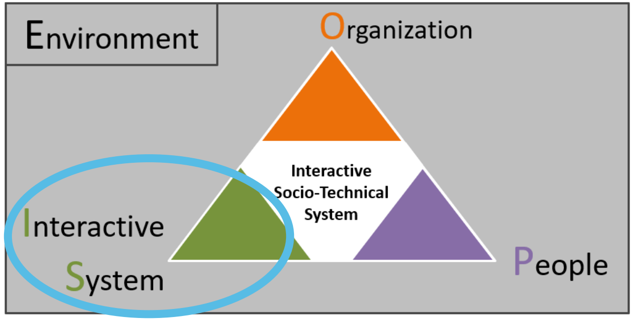
➔ Spaceport leading organization (**O**), FSO operators (**P**), and operational environment (**E**) remain constant regardless of the mission or launch vehicle

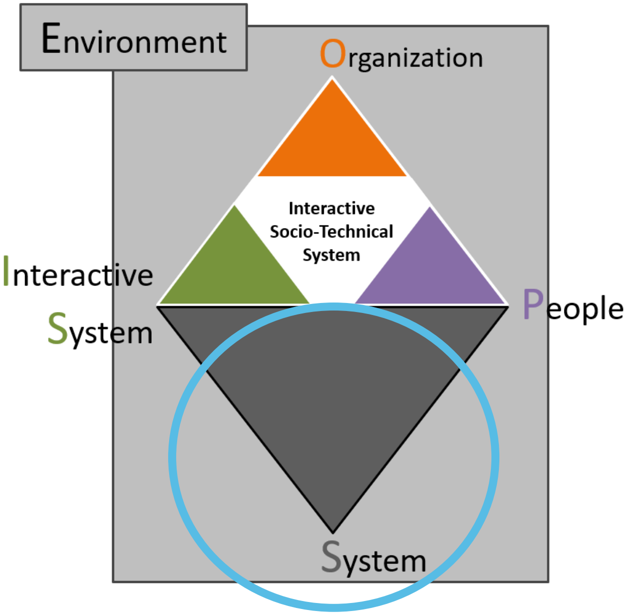
The design context of Launch Vehicle Flight Safety Operations



Interactive System :

- Ground segment interactive CCS for FSO offers the following main interaction means :
  - Monitoring User Interface
  - Physical control panel

The design context of Launch Vehicle Flight Safety Operations



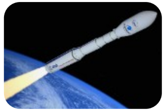
Interactive System :

- Ground segment interactive CCS for FSO offers the following main interaction means :
  - Monitoring User Interface
  - Physical control panel

Systems (under control) :

- Several ground segment systems (localization means, ground station antennas, …)
- Different types of launchers (with specific characteristics but shared a goal, to put satellites in orbit)

Ariane 6

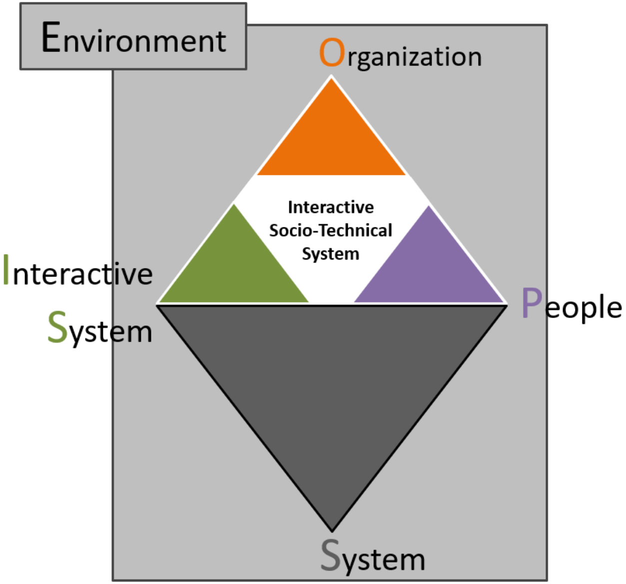


Vega

CALLISTO

Others …

The design context of Launch Vehicle Flight Safety Operations



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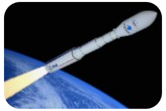


Vega

CALLISTO

Others …

➔ Monitoring UI (**IS**) can vary with the launch vehicle type and the ground segment systems (**S**) chosen for a particular mission

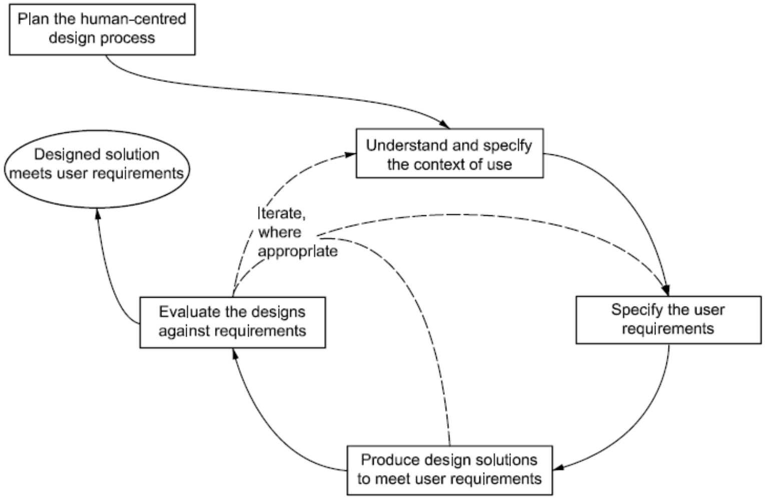
/!\ Content of the IS is imposed by the underlying system (S)

Outline of the presentation

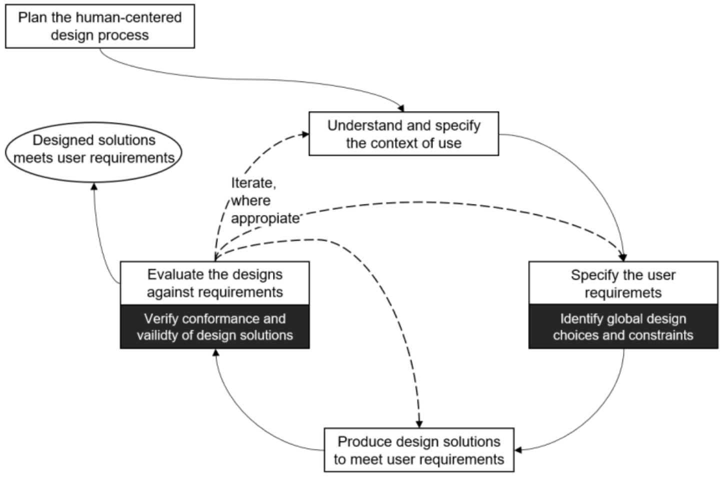
- On the need for multiple interactive system variants (identified needs, existing related work and limitations)
- The design context of interactive system variants for the Launch Vehicle Flight Safety Operations (POISE framework application)
- Our generic solution : the Multi-variant UCD process (rapid presentation of the process)
- Application example of the Multi-variant UCD process : the Ariane 6 and CALLISTO UI prototypes
- Conclusion and lessons learnt
- Future work

The Multi-variant UCD process

Classic UCD process of ISO 9241 Part 210

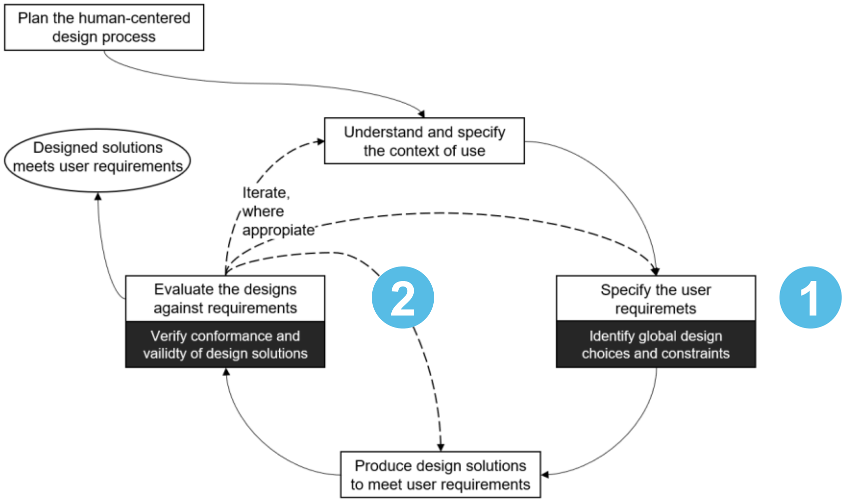


Multi variant UCD process



Process to create **usable** interactive system variants, with a high degree of similarity for the same user or user group, designed by the same design team

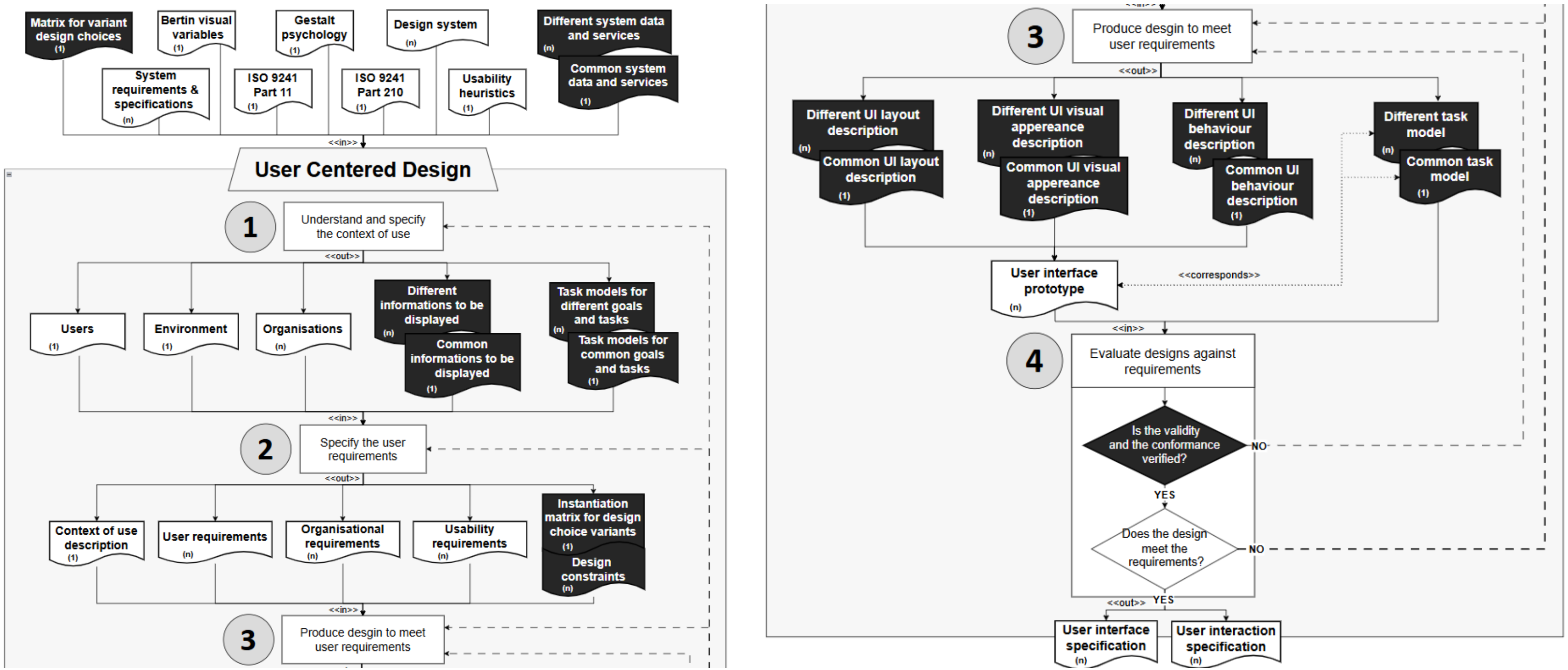
The Multi-variant UCD process (simplified version)



Two main extensions of the UCD process :

- Refinement of **1** by adding «Identify global design and constraints » for the:
  - Instantiation of the matrix of design choices -> to determine which combinations of POISE components are accepted or not in the design solutions
  - Definition of design constraints-> to limit accepted variability of design solutions (ex. similarity constraint -> common information must be encoded using the same visual variable)
- Refinement of **2** by adding « Verify conformance and validity of design solutions » for the:
  - Verification that all common user-system tasks are feasible with the design solutions, and that specific tasks are also feasible
  - Validation of the adherence of design solutions to established design choices and constraints

The Multi-variant UCD process (detailed version)



/!\ Common and different parts of all POISE components that can vary must be identified
This identification extends to the user-system tasks, directly related to the capabilities of the IS component

Rodriguez-Hernando, D., Martinie, C., Palanque, P., & Steere, S. (2025, September). **Multi-Variant UCD-A Process for the Design of Interactive System Variants: Application to Launch Vehicles Flight Safety Operations.** In *IFIP INTERACT conference* (pp. 371-396).

Outline of the presentation

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Application example of the Multi-variant UCD process : the Ariane 6 and CALLISTO UI prototypes

Design of two UI prototypes of the Flight Safety CCS

Ariane 6

Two different flight phases:

- Close range
- Far range

Ballistic trajectory with a static flight corridor

CALLISTO

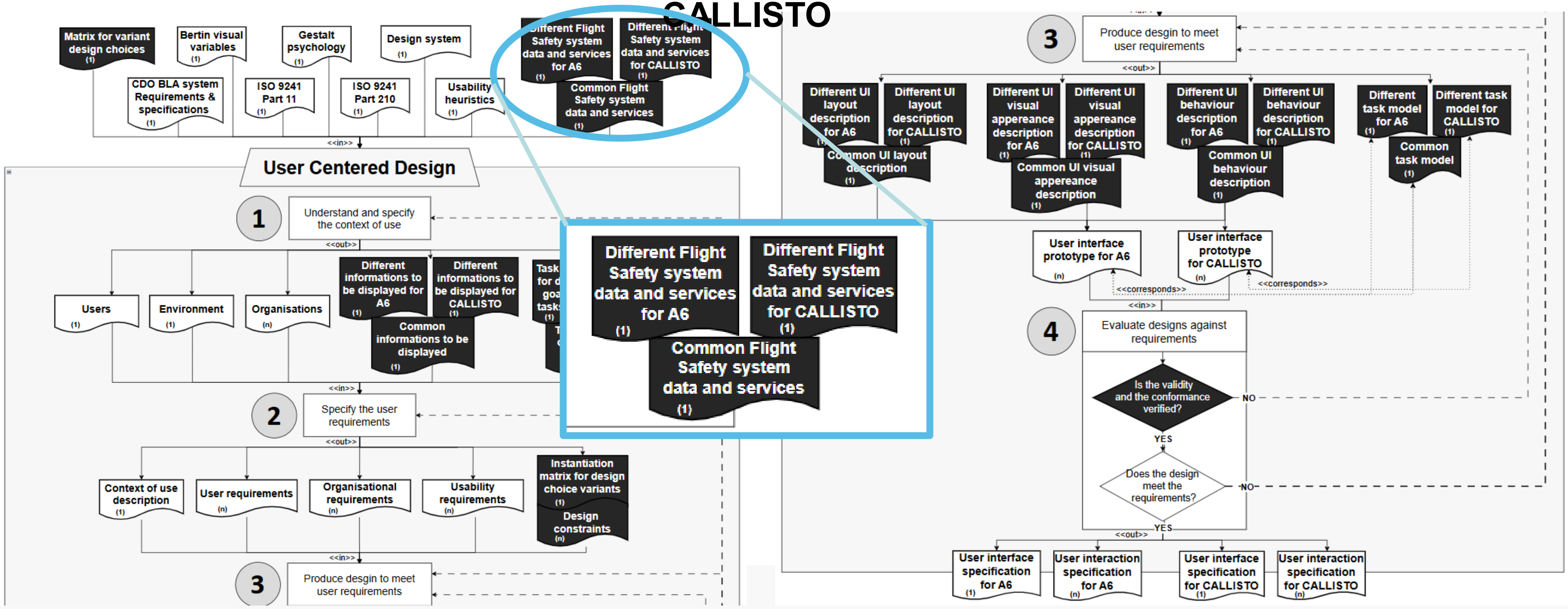
Three different flight phases:

- Climb
- Culmination
- Descent

Vertical take-off/vertical landing trajectory with a dynamic flight corridor

Application example of the Multi-variant UCD process : the Ariane 6 and CALLISTO UI prototypes

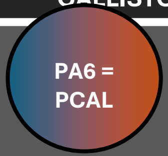
Instantiated Multi-variant UCD process for Ariane 6 and CALLISTO



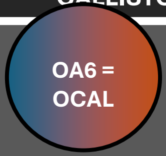
Application example of the Multi-variant UCD process : the Ariane 6 and CALLISTO UI prototypes

Set based decomposition of all POISE-S components for Ariane 6 and CALLISTO

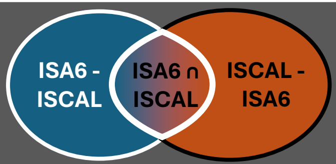
P (PEOPLE)
- PA6 (P Ariane 6)
- PCAL (P CALLISTO)



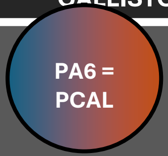
O (ORGANIZATION)
- OA6 (O Ariane 6)
- OCAL (O CALLISTO)



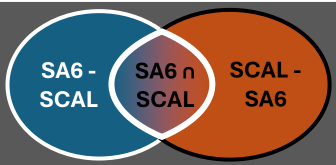
IS (INTERACTIVE SYSTEM)
- ISA6 (IS Ariane 6)
- ISCAL (IS CALLISTO)



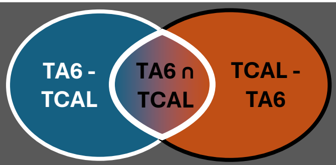
E (ENVIRONMENT)
- EA6 (E Ariane 6)
- ECAL (E CALLISTO)



S (SYSTEM UNDER CONTROL)
- SA6 (S Ariane 6)
- SCAL (S CALLISTO)

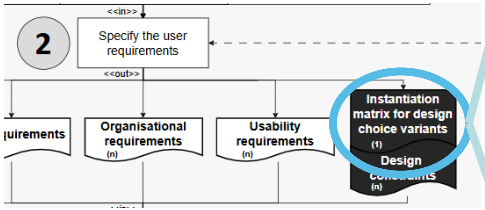


T (USER-SYSTEM TASKS)
- TA6 (T Ariane 6)
- TCAL (T CALLISTO)



Application example of the Multi-variant UCD process : the Ariane 6 and CALLISTO UI prototypes

Process Step 2



	P (PEOPLE) - PA6 (P of Ariane 6) - PCAL(P of CALLISTO)	S (SYSTEM) - SA6 (S of Ariane 6) - SCAL (S of CALLISTO)	IS (INTERACTIVE SYSTEM) - ISA6 (IS of Ariane6) - ISCAL (IS of CALLISTO)	T (TASKS) - TA6 (T of Ariane6) - TCAL (T of CALLISTO)	ACCEPTATION FOR CASE STUDY
#DesignChoice1	PA6 = PCAL	SA6 - SCAL SCAL - SA6	ISA6 - ISCAL ISCAL - ISA6	TA6 - TCAL TCAL - TA6	ACCEPTED
#DesignChoice2	PA6 = PCAL	SA6 - SCAL SCAL - SA6	ISA6 - ISCAL ISCAL - ISA6	TA6 ∩ TCAL	ACCEPTED
#DesignChoice3	PA6 = PCAL	SA6 - SCAL SCAL - SA6	ISA6 ∩ ISCAL	TA6 - TCAL TCAL - TA6	NOT ACCEPTED
#DesignChoice4	PA6 = PCAL	SA6 - SCAL SCAL - SA6	ISA6 ∩ ISCAL	TA6 ∩ TCAL	ACCEPTED
#DesignChoice5	PA6 = PCAL	SA6 ∩ SCAL	ISA6 - ISCAL ISCAL - ISA6	TA6 - TCAL TCAL - TA6	NOT ACCEPTED
#DesignChoice6	PA6 = PCAL	SA6 ∩ SCAL	ISA6 - ISCAL ISCAL - ISA6	TA6 ∩ TCAL	ACCEPTED
#DesignChoice7	PA6 = PCAL	SA6 ∩ SCAL	ISA6 ∩ ISCAL	TA6 - TCAL TCAL - TA6	NOT ACCEPTED
#DesignChoice8	PA6 = PCAL	SA6 ∩ SCAL	ISA6 ∩ ISCAL	TA6 ∩ TCAL	ACCEPTED

Instantiation **done by the design team**

Application example of the Multi-variant UCD process : the Ariane 6 and CALLISTO UI prototypes

#DesignChoice5	PA6 = PCAL	SA6 ∩ SCAL	ISA6 - ISCAL ISCAL - ISA6	TA6 - TCAL TCAL - TA6	NOT ACCEPTED

#Design choice 5

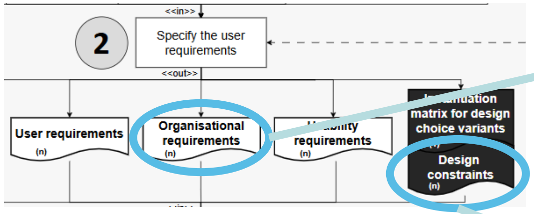
We dont want to adress common System parts with different Interactive System parts and its associated different Tasks

Airbus vs Boeing

- System : Fly-by-wire (S Airbus ∩ S Boeing)
- Interacive System : Airbus side-stick (IS Airbus – IS Boeing) different from Boeing yoke (IS Boeing – IS Airbus)
- Tasks : Airbus side-stick tasks (T Airbus – T Boeing) different from Boeing yoke tasks (T Boeing – T Airbus)

Application example of the Multi-variant UCD process : the Ariane 6 and CALLISTO UI prototypes

Process Step 2



Requirement example

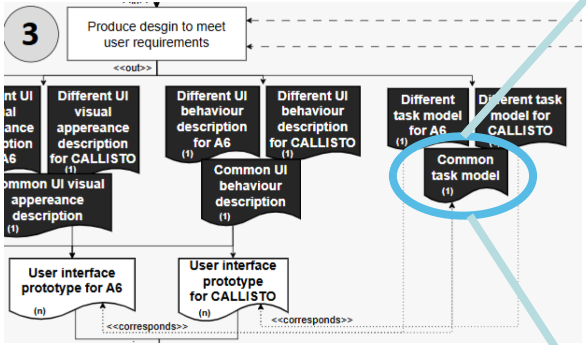
#REQ_1 : CCS UIs must cover all the FSO activities defined by the French Space Operations act

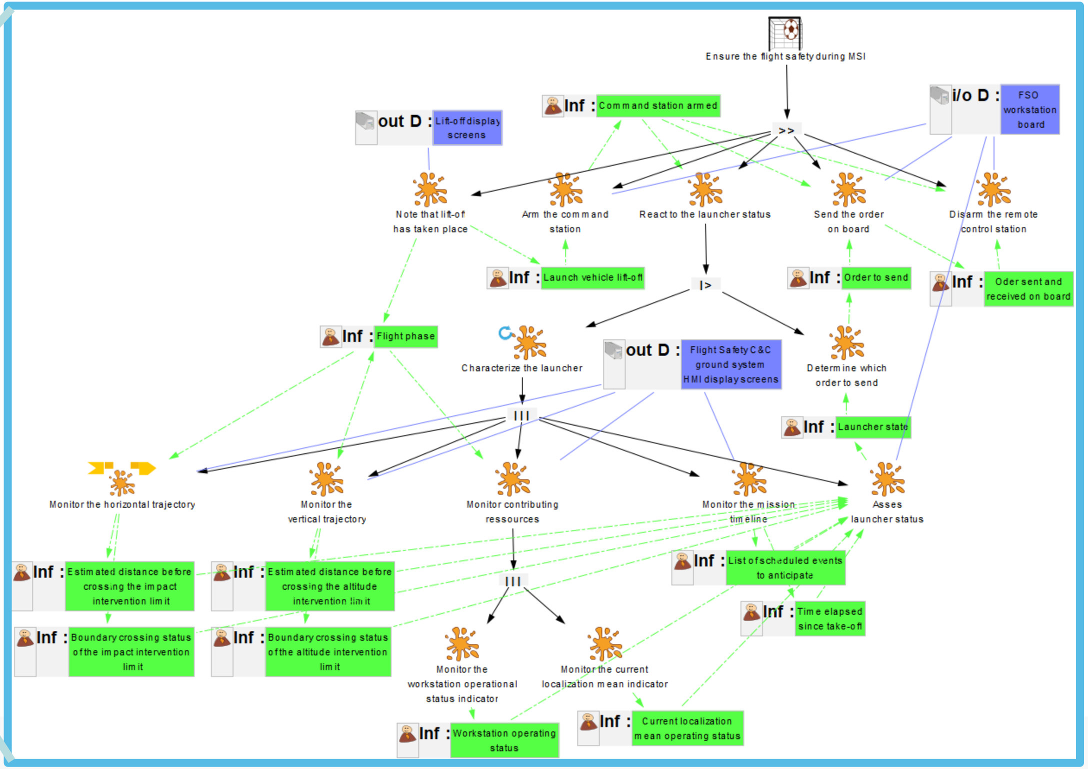
Design constraint example

"If UI element aspects Visual Apparence or Behaviour are disimilar for a common information displayed on UI, then the whole UI element is consider as dissimilar"

Application example of the Multi-variant UCD process : the Ariane 6 and CALLISTO UI prototypes

Process Step 3





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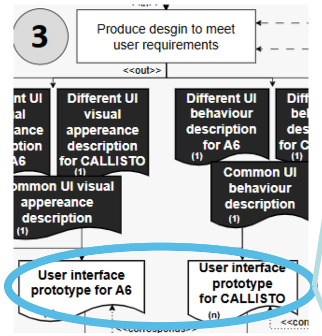
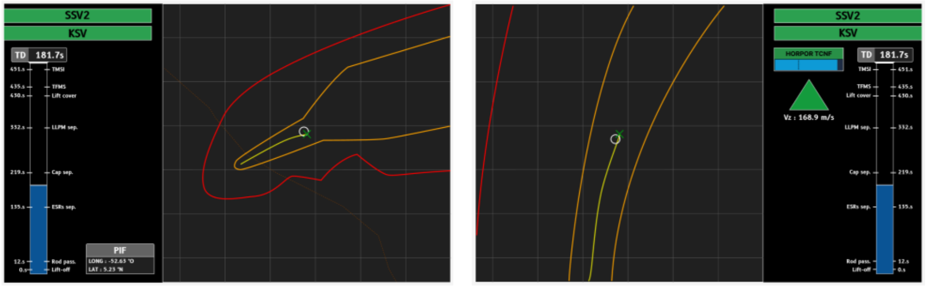
Application example of the Multi-variant UCD process : the Ariane 6 and CALLISTO UI prototypes

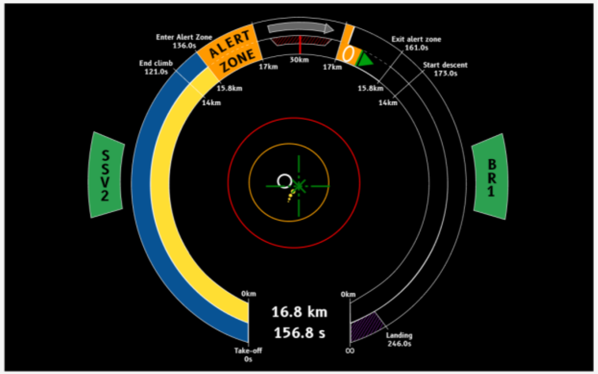
Process Step 3



Ariane 6 UI Prototype :
- Two display screens
- Information arrangement inherited from Ariane 5 UI

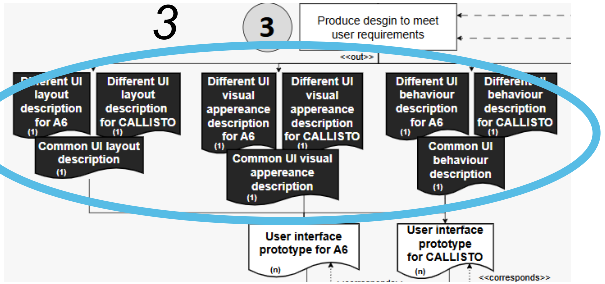


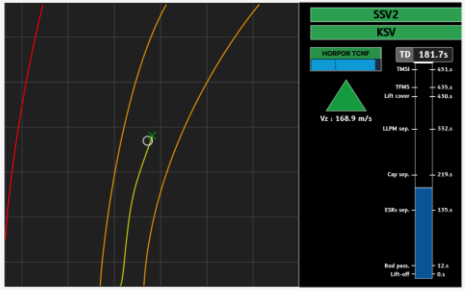
CALLISTO UI Prototype :
- One display screen
- Information displayed in small area of focus

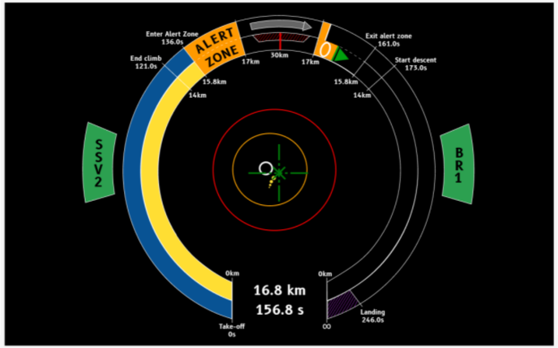


Application example of the Multi-variant UCD process : the Ariane 6 and CALLISTO UI prototypes

Process Step 3





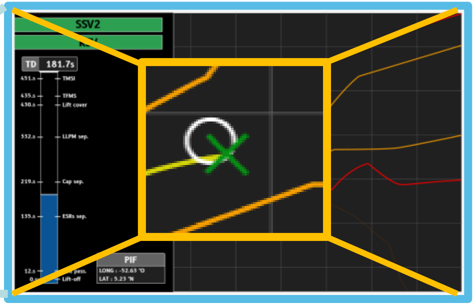


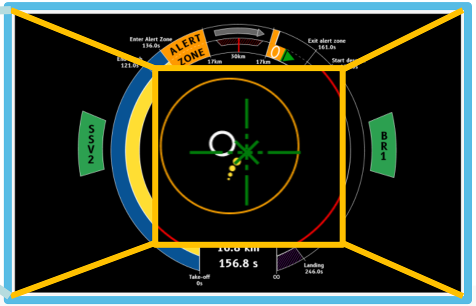
Both UI prototypes should be decomposed in :

- **Information displayed** (ex. Launcher's horizontal position)
- **Layout** (UI elements arrangement, order, …)
- **Visual appearance** (UI elements color, shape, etc.)
- **Behavior** (UI elements states, states transitions, etc.)

Application example of the Multi-variant UCD process : the Ariane 6 and CALLISTO UI prototypes

INTERACTIVE C&C SYSTEM (*Only the UI part*) UIPA6 (for the Ariane 6 variant) UIPCAL (for the CALLISTO variant)			
UI PROTOTYPE INFORMATION UIPIA6 (for the Ariane 6 variant) UIPICAL (for the CALLISTO variant)	**UI PROTOTYPE LAYOUT** UIPLA6 (for the Ariane 6 variant) UIPLCAL (for the CALLISTO variant)	**UI PROTOTYPE VISUAL APPEREANCE** UIPVAA6 (for the Ariane 6 variant) UIPVACAL (for the CALLISTO variant)	**UI PROTOTYPE BEHAVIOUR** UIPBA6 (for the Ariane 6 variant) UIPBCAL (for the CALLISTO variant)
ISA6 - ISCAL			
ISCAL - ISA6			
UIPIA6 ∩ UIPICAL Current position	**UIPLA6 - UIPLCAL** Horizontality graph (**current position**): Order = (0), Arrangement = (x=598, y=0), Size = (width=1085, height=1077)	**UIPVAA6 - UIPVACAL** Color = Green, Shape = Cross	**UIPBA6 ∩ UIPBCAL** Always in the center of the horizontality graph. It correspond to the center of the viewport.
	UIPLCAL - UIPLA6 Horizontality graph (**current position**) : Order = (0), Arrangement = (x=465, y=145), Size = (width=745, height=760)	**UIPVACAL - UIPVAA6** Color = Green, Shape = Crosshair	





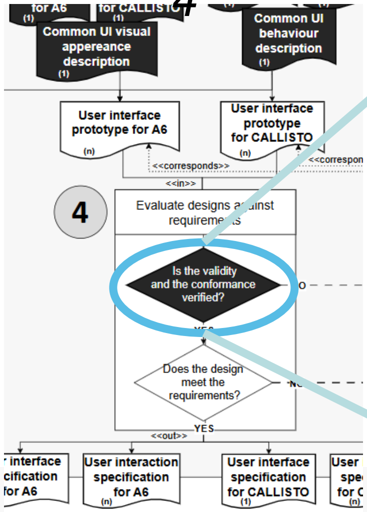
Design choice

"If UI element Visual Appearance or Behavior are dissimilar for a common information displayed on UI, then the whole UI element is consider as dissimilar"

Application example of the Multi-variant UCD process : the Ariane 6 and CALLISTO UI prototypes

#DesignChoice6	PA6 = PCAL	SA6 ∩ SCAL	ISA6 - ISCAL ISCAL - ISA6	TA6 ∩ TCAL	ACCEPTED

Process Step 4



SYSTEM SA6 (for the Ariane 6 variant) SCAL (for the CALLISTO variant)		INTERACTIVE C&C SYSTEM (*Only the UI part*) UIPA6 (for the Ariane 6 variant) UIPCAL (for the CALLISTO variant)				TASKS TA6 (for the Ariane 6 variant) TCAL (for the CALLISTO variant)	ASSOCIATED DESGIN CHOICE
LAUNCHER SYSTEM LSA6 (for the Ariane 6 variant) LSCAL (for the CALLISTO variant)	GROUND SEGMENT SYSTEM GSSA6 (for the Ariane 6 variant) GSSCAL (for the CALLISTO variant)	UI PROTOTYPE INFORMATION UIPIA6 (for the Ariane 6 variant) UIPICAL (for the CALLISTO variant)	UI PROTOTYPE LAYOUT UIPLA6 (for the Ariane 6 variant) UIPLCAL (for the CALLISTO variant)	UI PROTOTYPE VISUAL APPEREANCE UIPVAA6 (for the Ariane 6 variant) UIPVACAL (for the CALLISTO variant)	UI PROTOTYPE BEHAVIOUR UIPBA6 (for the Ariane 6 variant) UIPBCAL (for the CALLISTO variant)		
SA6 ∩ SCAL		ISA6 - ISCAL ISCAL - ISA6				TA6 ∩ TCAL	#DesginChoice6
LSA6 ∩ LSCAL System responsible for cummunicaiting launch vehicle position to radars	GSSA6 ∩ GSSCAL System responsible for retrieving the launch vehicle's position from radars and converting it into coordinates	UIPIA6 ∩ UIPICAL Current position	UIPLA6 - UIPLCAL Horizontality graph (**current position**): Order = (0), Arrangement = (x=598, y=0), Size = (width=1085, height=1077) UIPLCAL - UIPLA6 Horizontality graph (**current position**) : Order = (0), Arrangement = (x=465, y=145), Size = (width=745, height=760)	UIPVAA6 - UIPVACAL Color = Green, Shape = Cross UIPVACAL - UIPVAA6 Color = Green, Shape = Crosshair	UIPBA6 ∩ UIPBCAL Always in the center of the horizontality graph. It correspond to the center of the viewport.	TA6 ∩ TCAL Look at the current position	

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Conclusions and Lessons Learnt

- This is one of the few research work on properties accross interactive systems thus going beyond the classical usability, UX, dependability, safety …
- We proposed a framework for defining similary between two systems A and B (features and features matching)
- We proposed a framework for assessing similarity between two interactive systems A and B (exploitig IS architecture)
- We demonstrated the multiple benefits in designing and building similar and disimilar IS (consistency, interference errors reduction, training costs reductions, early adoption, …)

Outline of the course

- Context and definitions
  - Usability
  - User Experience
- Designing for Usability
- Designing for User Experience
- Measuring Usability and User Experience
- Going further
  - Acceptance
  - Learnability
- Specificities of command and control systems
- Take away message

Take away message

- User eXperience and Usability are different properties
- UX and Usability contribute to Interactive Systems quality
  - They encompass subjective and objective attributes
  - They may be in conflict
  - They require strong investment in design (User Interface, Interaction, graphical aspects, sound, haptic, …)
- UX and Usability are non-functional requirements
  - But they are embedded as functions (e.g. undo, social connectedness)
  - They require early planning and resource allocation
  - It is hard to find skilled people/designers/developers in UX (usually they believe it is the same as usability) and they don't know how to build what they designed
- Learnability requires complex engineering of Interactive Systems

Take away message – no need to add AI

- POISE framework brings together 4 perspectives (P, O, IS & E)
- There is a need to address multiple properties together
  - Usability and User Experience (to remove human-made operational faults)
  - Learnability (otherwise Usability and UX are useless)
  - Reliability (to remove human-made development faults)
  - Dependability (to remove natural inside system and inside human faults)
  - And many other ones (Acceptance, Security, …)
- There is a need to address conflicting properties at design and maintenance times

All clear ...? Anything to add ...?